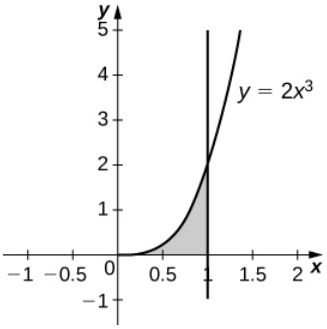
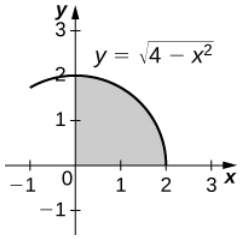


$V = \frac{4\pi}{5}$ units^3

27) $y = 3x^2$, $x = 0$, and $y = 3$

28) $y = \sqrt{4 - x^2}$, $y = 0$, and $x = 0$

Answer

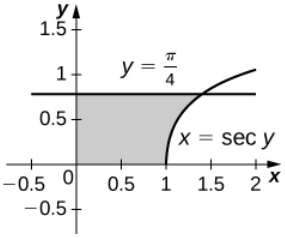


$V = \frac{16\pi}{3}$ units^3

29) $y = \dfrac{1}{\sqrt{x+1}}$, $x = 0$, and $x = 3$

30) $x = \sec(y)$ and $y = \dfrac{\pi}{4}$, $y = 0$ and $x = 0$

Answer

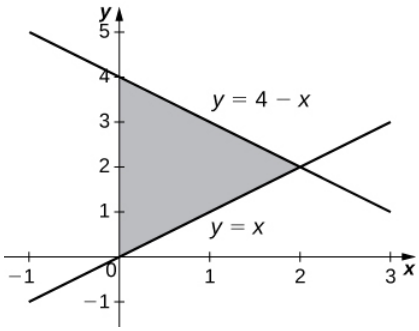


$V = \pi$ units^3

31) $y = \dfrac{1}{x+1}$, $x = 0$, and $x = 2$

32) $y = 4 - x$, $y = x$, and $x = 0$

Answer



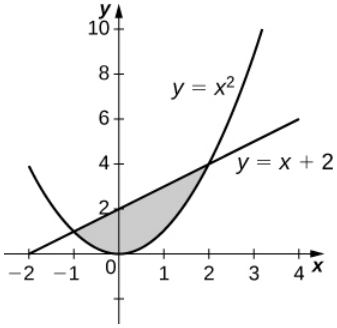
$V = \frac{16\pi}{3}$ units^3

For exercises 33 - 40, draw the region bounded by the curves. Then, find the volume when the region is rotated around the x -axis.

33) $y = x + 2, \quad y = x + 6, \quad x = 0$, and $x = 5$

34) $y = x^2$ and $y = x + 2$

Answer

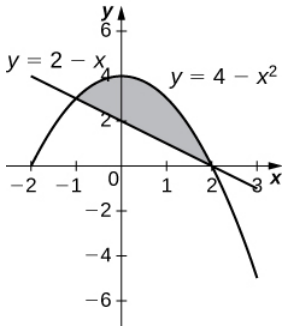


$V = \frac{72\pi}{5}$ units^3

35) $x^2 = y^3$ and $x^3 = y^2$

36) $y = 4 - x^2$ and $y = 2 - x$

Answer

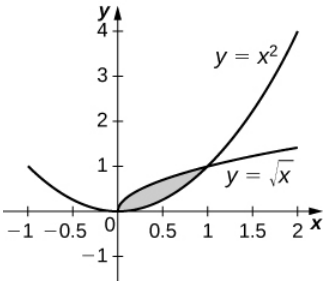


$V = \frac{108\pi}{5}$ units^3

37) [T] $y = \cos x, \quad y = e^{-x}, \quad x = 0$, and $x = 1.2927$

38) $y = \sqrt{x}$ and $y = x^2$

Answer

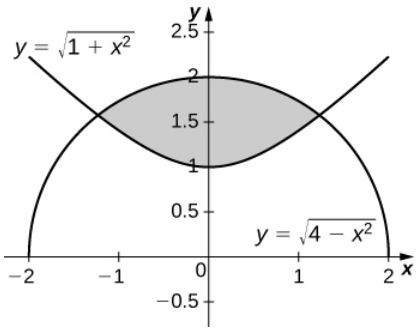


$V = \frac{3\pi}{10}$ units^3

39) $y = \sin x, \quad y = 5 \sin x, \quad x = 0$ and $x = \pi$

40) $y = \sqrt{1 + x^2}$ and $y = \sqrt{4 - x^2}$

Answer



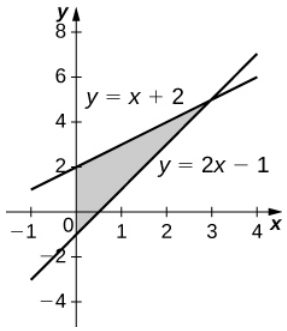
$V = 2\sqrt{6}\pi$ units^3

For exercises 41 - 45, draw the region bounded by the curves. Then, use the washer method to find the volume when the region is revolved around the y-axis.

41) $y = \sqrt{x}, \quad x = 4$, and $y = 0$

42) $y = x + 2, \quad y = 2x - 1$, and $x = 0$

Answer

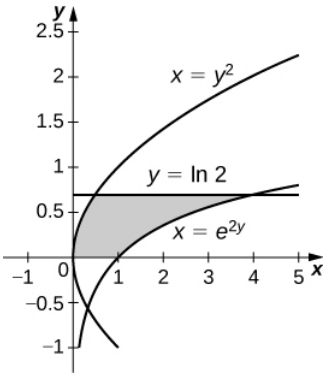


$V = 9\pi$ units^3

43) $y = \dfrac{3}{x}$ and $y = x^3$

44) $x = e^{2y}, \quad x = y^2, \quad y = 0$, and $y = \ln(2)$

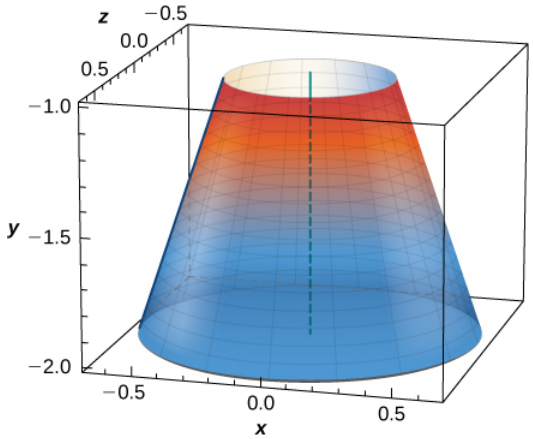
Answer



$V = \frac{\pi}{20}(75 - 4\ln^5(2))$ units^3

45) $x = \sqrt{9 - y^2}$, $x = e^{-y}$, $y = 0$, and $y = 3$

46) Yogurt containers can be shaped like frustums. Rotate the line $y = (\frac{1}{m})x$ around the y-axis to find the volume between $y = a$ and $y = b$.

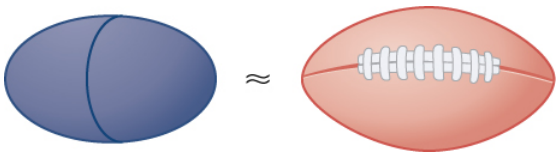




Answer

$V = \frac{m^2\pi}{3}(b^3 - a^3)$ units^3

47) Rotate the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ around the x-axis to approximate the volume of a football, as seen here.



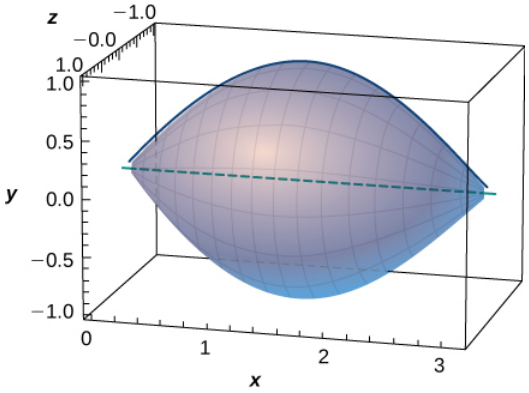
https://math.libretexts.org/@go/page/69809

48) Rotate the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ around the y-axis to approximate the volume of a football.

Answer

$V=\frac{4a^2b\pi}{3}$ units^3

49) A better approximation of the volume of a football is given by the solid that comes from rotating $y=\sin x$ around the x-axis from $x=0$ to $x=\pi$. What is the volume of this football approximation, as seen here?



For exercises 51 - 56, find the volume of the solid described.

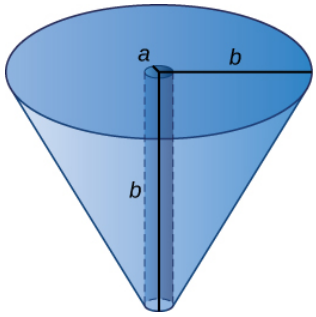
51) The base is the region between $y=x$ and $y=x^2$. Slices perpendicular to the x-axis are semicircles.

52) The base is the region enclosed by the generic ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Slices perpendicular to the x-axis are semicircles.

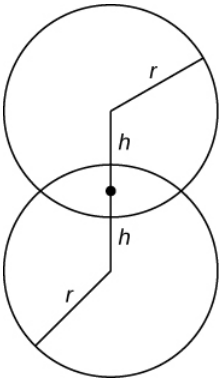
Answer

$V=\frac{2ab^2\pi}{3}$ units^3

53) Bore a hole of radius a down the axis of a right cone and through the base of radius b, as seen here.



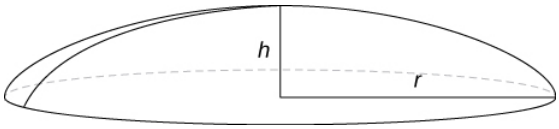
54) Find the volume common to two spheres of radius r with centers that are $2h$ apart, as shown here.



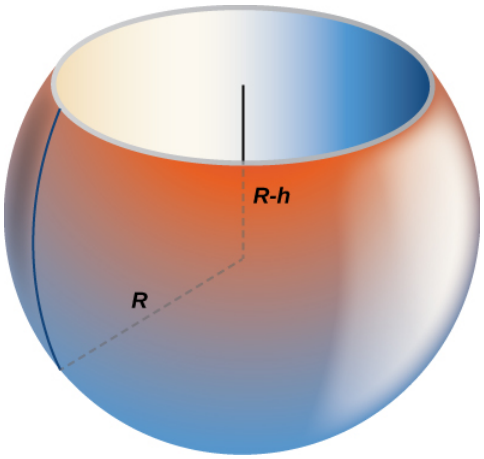
Answer

$V = \frac{\pi}{12}(r+h)^2(6r-h)$ units^3

55) Find the volume of a spherical cap of height h and radius r where $h < r$, as seen here.



56) Find the volume of a sphere of radius R with a cap of height h removed from the top, as seen here.



Answer

$V = \frac{\pi}{3}(h+R)(h-2R)^2$ units^3

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 6.2E: Exercises for Section 6.2 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.3: Volumes of Revolution - Cylindrical Shells

Learning Objectives

- Calculate the volume of a solid of revolution by using the method of cylindrical shells.
- Compare the different methods for calculating a volume of revolution.

In this section, we examine the method of cylindrical shells, the final method for finding the volume of a solid of revolution. We can use this method on the same kinds of solids as the disk method or the washer method; however, with the disk and washer methods, we integrate along the coordinate axis parallel to the axis of revolution. With the method of cylindrical shells, we integrate along the coordinate *axis perpendicular* to the axis of revolution. The ability to choose which variable of integration we want to use can be a significant advantage with more complicated functions. Also, the specific geometry of the solid sometimes makes the method of using cylindrical shells more appealing than using the washer method. In the last part of this section, we review all the methods for finding volume that we have studied and lay out some guidelines to help you determine which method to use in a given situation.

The Method of Cylindrical Shells

Again, we are working with a solid of revolution. As before, we define a region R, bounded above by the graph of a function $y = f(x)$, below by the x**-axis**, and on the left and right by the lines $x = a$ and $x = b$, respectively, as shown in Figure 6.3.1a. We then revolve this region around the y-axis, as shown in Figure 6.3.1b. Note that this is different from what we have done before. Previously, regions defined in terms of functions of x were revolved around the x**-axis** or a line parallel to it.

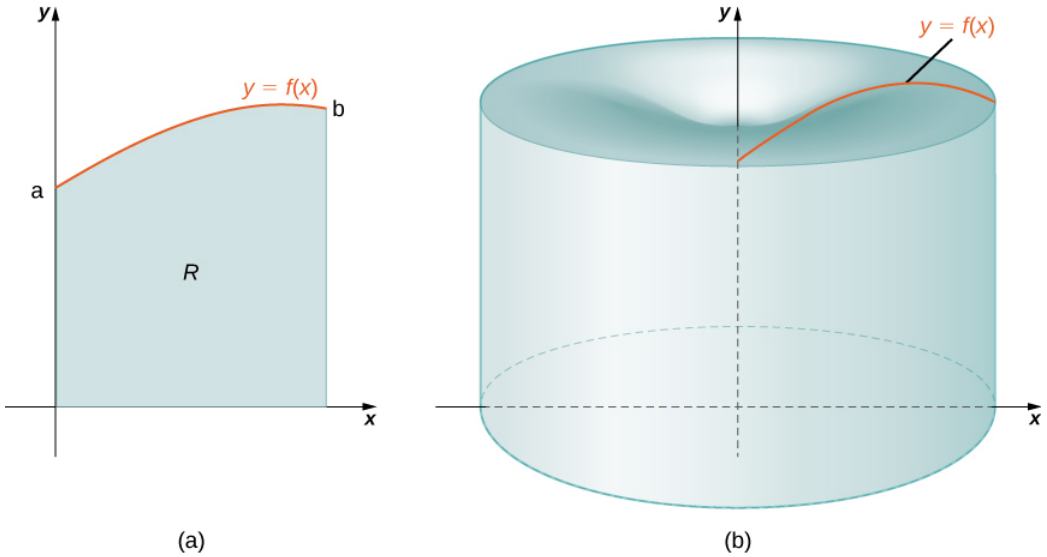


Figure 6.3.1: (a) A region bounded by the graph of a function of x. (b) The solid of revolution formed when the region is revolved around the y**-axis.**

As we have done many times before, partition the interval $[a, b]$ using a regular partition, $P = x_0, x_1, \ldots, x_n$ and, for $i = 1, 2, \ldots, n$, choose a point $x_i^* \in [x_{i-1}, x_i]$. Then, construct a rectangle over the interval $[x_{i-1}, x_i]$ of height $f(x_i^*)$ and width Δx. A representative rectangle is shown in Figure 6.3.2a. When that rectangle is revolved around the y-axis, instead of a disk or a washer, we get a cylindrical shell, as shown in Figure 6.3.2.

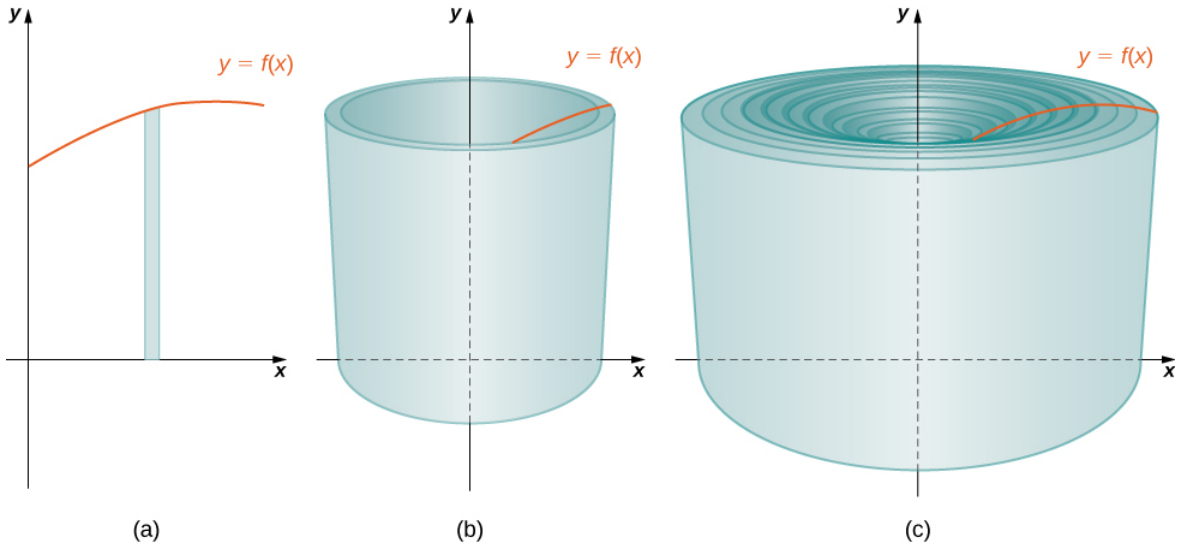


Figure 6.3.2: (a) A representative rectangle. (b) When this rectangle is revolved around the y-**axis**, the result is a cylindrical shell. (c) When we put all the shells together, we get an approximation of the original solid.

To calculate the volume of this shell, consider Figure 6.3.3.

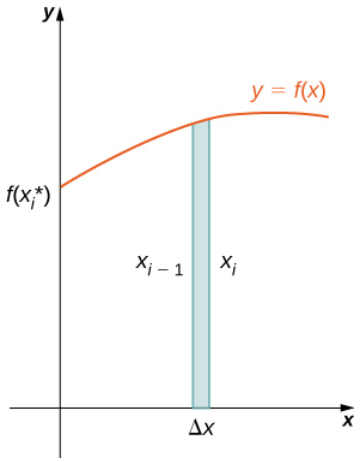


Figure 6.3.3: Calculating the volume of the shell.

The shell is a cylinder, so its volume is the cross-sectional area multiplied by the height of the cylinder. The cross-sections are annuli (ring-shaped regions—essentially, circles with a hole in the center), with outer radius x_i and inner radius x_{i-1}. Thus, the cross-sectional area is $\pi x_i^2 - \pi x_{i-1}^2$. The height of the cylinder is $f(x_i^*)$. Then the volume of the shell is

$$\begin{aligned} V_{shell} &= f(x_i^*)(\pi\, x_i^2 - \pi\, x_{i-1}^2) \\ &= \pi\, f(x_i^*)(x_i^2 - x_{i-1}^2) \\ &= \pi\, f(x_i^*)(x_i + x_{i-1})(x_i - x_{i-1}) \\ &= 2\pi\, f(x_i^*)\left(\frac{x_i + x_{i-1}}{2}\right)(x_i - x_{i-1}). \end{aligned}$$

Note that $x_i - x_{i-1} = \Delta x$, so we have

$$V_{shell} = 2\pi\, f(x_i^*)\left(\frac{x_i + x_{i-1}}{2}\right)\Delta x.$$

Furthermore, $\dfrac{x_i + x_{i-1}}{2}$ is both the midpoint of the interval $[x_{i-1}, x_i]$ and the average radius of the shell, and we can approximate this by x_i^*. We then have

$$V_{shell} \approx 2\pi\, f(x_i^*) x_i^*\, \Delta x.$$

Another way to think of this is to think of making a vertical cut in the shell and then opening it up to form a flat plate (Figure 6.3.4).

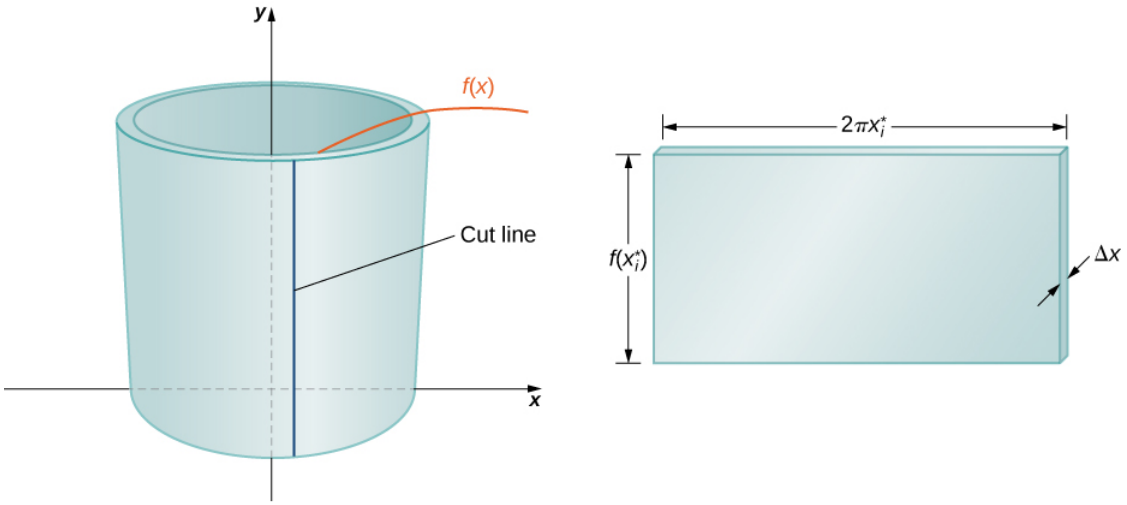


Figure 6.3.4: (a) Make a vertical cut in a representative shell. (b) Open the shell up to form a flat plate.

In reality, the outer radius of the shell is greater than the inner radius, and hence the back edge of the plate would be slightly longer than the front edge of the plate. However, we can approximate the flattened shell by a flat plate of height $f(x_i^*)$, width $2\pi x_i^*$, and thickness Δx (Figure). The volume of the shell, then, is approximately the volume of the flat plate. Multiplying the height, width, and depth of the plate, we get

$$V_{shell} \approx f(x_i^*)(2\pi\, x_i^*)\,\Delta x,$$

which is the same formula we had before.

To calculate the volume of the entire solid, we then add the volumes of all the shells and obtain

$$V \approx \sum_{i=1}^{n} (2\pi\, x_i^* f(x_i^*)\,\Delta x).$$

Here we have another Riemann sum, this time for the function $2\pi\, x\, f(x)$. Taking the limit as $n \to \infty$ gives us

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} (2\pi\, x_i^* f(x_i^*)\,\Delta x) = \int_a^b (2\pi\, x\, f(x))\, dx.$$

This leads to the following rule for the *method of cylindrical shells*.

Rule: The Method of Cylindrical Shells

Let $f(x)$ be continuous and nonnegative. Define R as the region bounded above by the graph of $f(x)$, below by the x**-axis,** on the left by the line $x = a$, and on the right by the line $x = b$. Then the volume of the solid of revolution formed by revolving R around the y-axis is given by

$$V = \int_a^b (2\pi\, x\, f(x))\, dx.$$

Now let's consider an example.

Example 6.3.1: The Method of Cylindrical Shells I

Define R as the region bounded above by the graph of $f(x) = 1/x$ and below by the x**-axis** over the interval $[1, 3]$. Find the volume of the solid of revolution formed by revolving R around the y-axis.

Solution

First we must graph the region R and the associated solid of revolution, as shown in Figure 6.3.5.

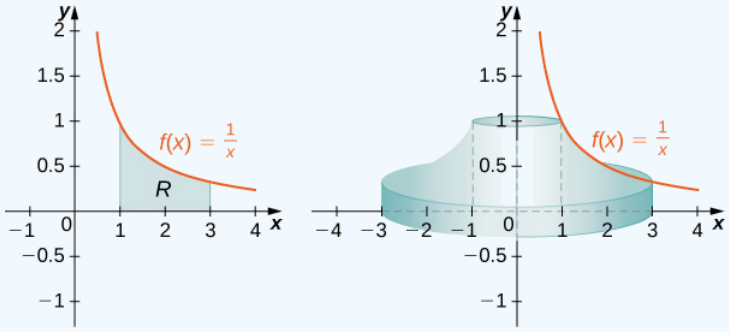


Figure 6.3.5: (a) The region R under the graph of $f(x) = 1/x$ over the interval $[1, 3]$. (b) The solid of revolution generated by revolving R about the y-axis.

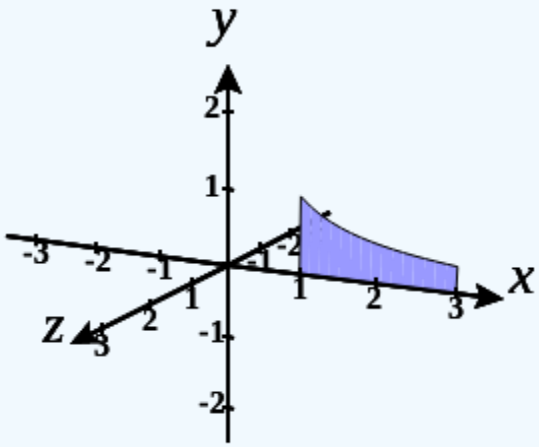


Figure 6.3.5 (c) Visualizing the solid of revolution with CalcPlot3D.

Then the volume of the solid is given by

https://math.libretexts.org/@go/page/2521

$$V = \int_a^b (2\pi\, x\, f(x))\, dx$$
$$= \int_1^3 \left(2\pi\, x \left(\frac{1}{x}\right)\right) dx$$
$$= \int_1^3 2\pi\, dx$$
$$= 2\pi\, x\Big|_1^3 = 4\pi \text{ units}^3.$$

Exercise 6.3.1

Define R as the region bounded above by the graph of $f(x) = x^2$ and below by the x-axis over the interval $[1, 2]$. Find the volume of the solid of revolution formed by revolving R around the y-axis.

Hint

Use the procedure from Example 6.3.1.

Answer

$\frac{15\pi}{2}$ units^3

Example 6.3.2: The Method of Cylindrical Shells II

Define R as the region bounded above by the graph of $f(x) = 2x - x^2$ and below by the x-axis over the interval $[0, 2]$. Find the volume of the solid of revolution formed by revolving R around the y-**axis.**

Solution

First graph the region R and the associated solid of revolution, as shown in Figure 6.3.6.

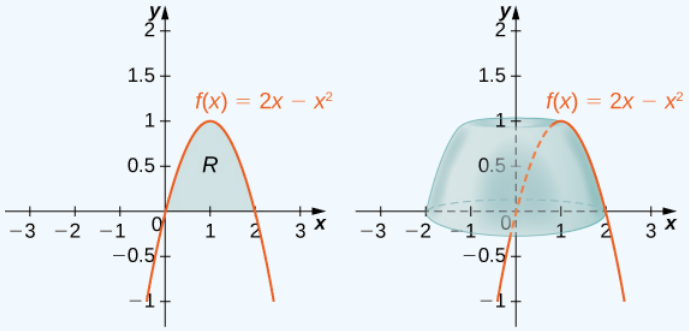


Figure 6.3.6: (a) The region R *under the graph of* $f(x) = 2x - x^2$ *over the interval* $[0, 2]$. *(b) The volume of revolution obtained by revolving* R *about the* y*-axis.*

Then the volume of the solid is given by

$$V = \int_a^b (2\pi\, x\, f(x))\, dx$$
$$= \int_0^2 (2\pi\, x(2x - x^2))\, dx$$
$$= 2\pi \int_0^2 (2x^2 - x^3)\, dx$$
$$= 2\pi \left[\frac{2x^3}{3} - \frac{x^4}{4}\right]\Big|_0^2$$
$$= \frac{8\pi}{3} \text{ units}^3$$

Exercise 6.3.2

Define R as the region bounded above by the graph of $f(x) = 3x - x^2$ and below by the x-axis over the interval $[0, 2]$. Find the volume of the solid of revolution formed by revolving R around the y-axis.

Hint

Use the process from Example 6.3.2.

Answer

8π units^3

As with the disk method and the washer method, we can use the method of cylindrical shells with solids of revolution, revolved around the x-axis, when we want to integrate with respect to y. The analogous rule for this type of solid is given here.

Rule: The Method of Cylindrical Shells for Solids of Revolution around the x-axis

Let $g(y)$ be continuous and nonnegative. Define Q as the region bounded on the right by the graph of $g(y)$, on the left by the y-axis, below by the line $y = c$, and above by the line $y = d$. Then, the volume of the solid of revolution formed by revolving Q around the x-axis is given by

$$V = \int_c^d (2\pi\, y\, g(y))\, dy.$$

Example 6.3.3: The Method of Cylindrical Shells for a Solid Revolved around the x-axis

Define Q as the region bounded on the right by the graph of $g(y) = 2\sqrt{y}$ and on the left by the y-axis for $y \in [0, 4]$. Find the volume of the solid of revolution formed by revolving Q around the x-axis.

Solution

First, we need to graph the region Q and the associated solid of revolution, as shown in Figure 6.3.7.

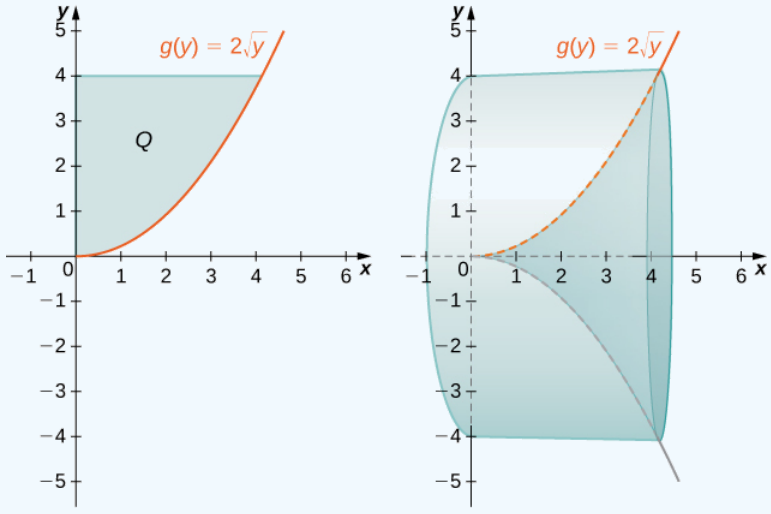


Figure 6.3.7: (a) The region Q to the left of the function $g(y)$ over the interval $[0, 4]$. (b) The solid of revolution generated by revolving Q around the x-axis.

Label the shaded region Q. Then the volume of the solid is given by

$$\begin{aligned} V &= \int_c^d (2\pi\, y\, g(y))\, dy \\ &= \int_0^4 (2\pi\, y(2\sqrt{y}))\, dy \\ &= 4\pi \int_0^4 y^{3/2}\, dy \\ &= 4\pi \left[\frac{2y^{5/2}}{5} \right] |_0^4 \\ &= \frac{256\pi}{5} \text{ units}^3 \end{aligned}$$

Exercise 6.3.3

Define Q as the region bounded on the right by the graph of $g(y) = 3/y$ and on the left by the y-axis for $y \in [1, 3]$. Find the volume of the solid of revolution formed by revolving Q around the x-axis.

Hint

Use the process from Example 6.3.3.

Answer

12π units^3

For the next example, we look at a solid of revolution for which the graph of a function is revolved around a line other than one of the two coordinate axes. To set this up, we need to revisit the development of the method of cylindrical shells. Recall that we found the volume of one of the shells to be given by

$$\begin{aligned} V_{shell} &= f(x_i^*)(\pi\, x_i^2 - \pi\, x_{i-1}^2) \\ &= \pi\, f(x_i^*)(x_i^2 - x_{i-1}^2) \\ &= \pi\, f(x_i^*)(x_i + x_{i-1})(x_i - x_{i-1}) \\ &= 2\pi\, f(x_i^*) \left(\frac{x_i + x_{i-1}}{2} \right) (x_i - x_{i-1}). \end{aligned}$$

This was based on a shell with an outer radius of x_i and an inner radius of x_{i-1}. If, however, we rotate the region around a line other than the y-axis, we have a different outer and inner radius. Suppose, for example, that we rotate the region around the line $x = -k$, where k is some positive constant. Then, the outer radius of the shell is $x_i + k$ and the inner radius of the shell is $x_{i-1} + k$. Substituting these terms into the expression for volume, we see that when a plane region is rotated around the line $x = -k$, the volume of a shell is given by

$$\begin{aligned} V_{shell} &= 2\pi\, f(x_i^*)(\frac{(x_i + k) + (x_{i-1} + k)}{2})((x_i + k) - (x_{i-1} + k)) \\ &= 2\pi\, f(x_i^*) \left(\left(\frac{x_i + x_{i-2}}{2} \right) + k \right) \Delta x. \end{aligned}$$

As before, we notice that $\frac{x_i + x_{i-1}}{2}$ is the midpoint of the interval $[x_{i-1}, x_i]$ and can be approximated by x_i^*. Then, the approximate volume of the shell is

$$V_{shell} \approx 2\pi(x_i^* + k) f(x_i^*) \Delta x.$$

The remainder of the development proceeds as before, and we see that

$$V = \int_a^b (2\pi(x + k) f(x)) dx.$$

We could also rotate the region around other horizontal or vertical lines, such as a vertical line in the right half plane. In each case, the volume formula must be adjusted accordingly. Specifically, the x-term in the integral must be replaced with an expression representing the radius of a shell. To see how this works, consider the following example.

Example 6.3.4: A Region of Revolution Revolved around a Line

Define R as the region bounded above by the graph of $f(x) = x$ and below by the x-axis over the interval $[1, 2]$. Find the volume of the solid of revolution formed by revolving R around the line $x = -1$.

Solution

First, graph the region R and the associated solid of revolution, as shown in Figure 6.3.8.

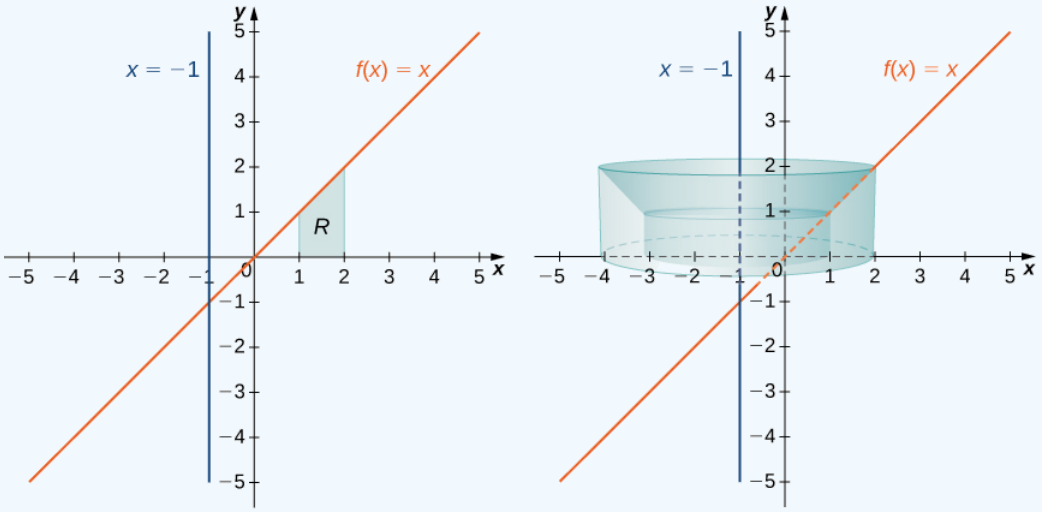


Figure 6.3.8: (a) The region R between the graph of $f(x)$ and the x-axis over the interval $[1, 2]$. (b) The solid of revolution generated by revolving R around the line $x = -1$.

Note that the radius of a shell is given by $x + 1$. Then the volume of the solid is given by

$$\begin{aligned} V &= \int_1^2 2\pi(x+1)f(x)\,dx \\ &= \int_1^2 2\pi(x+1)x\,dx = 2\pi\int_1^2 x^2 + x\,dx \\ &= 2\pi\left[\frac{x^3}{3} + \frac{x^2}{2}\right]\Big|_1^2 \\ &= \frac{23\pi}{3}\ \text{units}^3 \end{aligned}$$

Exercise 6.3.4

Define R as the region bounded above by the graph of $f(x) = x^2$ and below by the x-axis over the interval $[0, 1]$. Find the volume of the solid of revolution formed by revolving R around the line $x = -2$.

Hint

Use the process from Example 6.3.4.

Answer

$\frac{11\pi}{6}\ \text{units}^3$

For our final example in this section, let's look at the volume of a solid of revolution for which the region of revolution is bounded by the graphs of two functions.

Example 6.3.5: A Region of Revolution Bounded by the Graphs of Two Functions

Define R as the region bounded above by the graph of the function $f(x) = \sqrt{x}$ and below by the graph of the function $g(x) = 1/x$ over the interval $[1, 4]$. Find the volume of the solid of revolution generated by revolving R around the y-axis.

Solution

First, graph the region R and the associated solid of revolution, as shown in Figure 6.3.9.

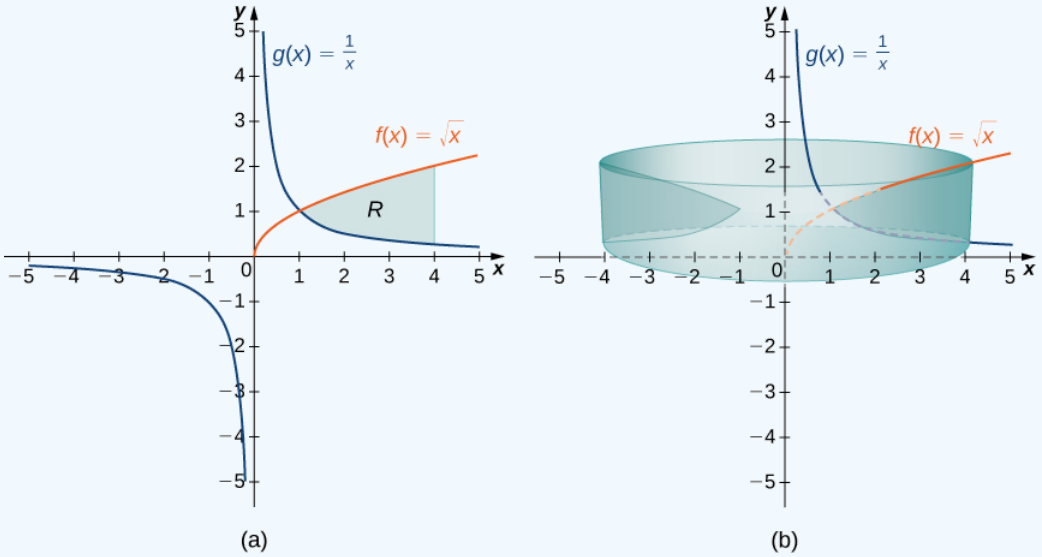


Figure 6.3.9: (a) The region R between the graph of $f(x)$ and the graph of $g(x)$ over the interval $[1, 4]$. (b) The solid of revolution generated by revolving R around the y-axis.

Note that the axis of revolution is the y-axis, so the radius of a shell is given simply by x. We don't need to make any adjustments to the x-term of our integrand. The height of a shell, though, is given by $f(x) - g(x)$, so in this case we need to adjust the $f(x)$ term of the integrand. Then the volume of the solid is given by

$$V = \int_1^4 (2\pi x(f(x) - g(x)))\, dx$$

$$= \int_1^4 (2\pi x(\sqrt{x} - \frac{1}{x}))\, dx = 2\pi \int_1^4 (x^{3/2} - 1) dx$$

$$= 2\pi \left[\frac{2x^{5/2}}{5} - x\right]\Big|_1^4 = \frac{94\pi}{5} \text{ units}^3.$$

Exercise 6.3.5

Define R as the region bounded above by the graph of $f(x) = x$ and below by the graph of $g(x) = x^2$ over the interval $[0, 1]$. Find the volume of the solid of revolution formed by revolving R around the y-axis.

Hint

Hint: Use the process from Example 6.3.5.

Answer

$\frac{\pi}{6}$ units^3

Which Method Should We Use?

We have studied several methods for finding the volume of a solid of revolution, but how do we know which method to use? It often comes down to a choice of which integral is easiest to evaluate. Figure 6.3.10 describes the different approaches for solids of revolution around the x-axis. It's up to you to develop the analogous table for solids of revolution around the y-axis.

Comparing the Methods for Finding the Volume of a Solid Revolution around the x-axis

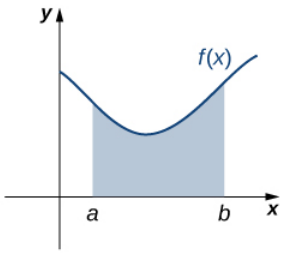
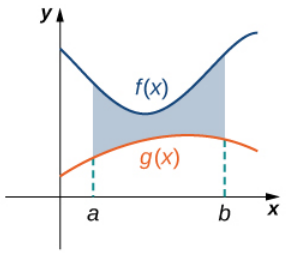
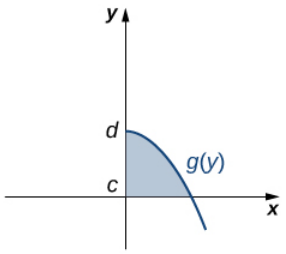
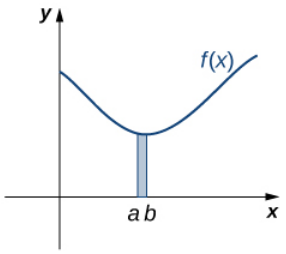
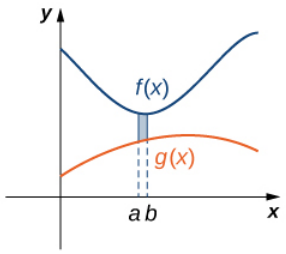
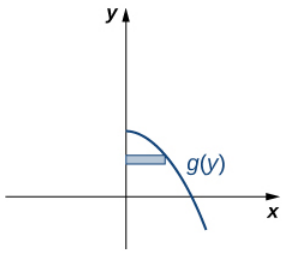
Compare	Disk Method	Washer Method	Shell Method
Volume formula	$V = \int_a^b \pi[f(x)]^2\,dx$	$V = \int_a^b \pi\left[(f(x))^2 - (g(x))^2\right]dx$	$V = \int_c^d 2\pi y\, g(y)\,dy$
Solid	No cavity in the center	Cavity in the center	With or without a cavity in the center
Interval to partition	$[a, b]$ on x-axis	$[a, b]$ on x-axis	$[c, d]$ on y-axis
Rectangle	Vertical	Vertical	Horizontal
Typical region			
Typical element			

Figure 6.3.10

Let's take a look at a couple of additional problems and decide on the best approach to take for solving them.

Example 6.3.6: Selecting the Best Method

For each of the following problems, select the best method to find the volume of a solid of revolution generated by revolving the given region around the x-axis, and set up the integral to find the volume (do not evaluate the integral).

a. The region bounded by the graphs of $y = x, y = 2 - x$, and the x-axis.
b. The region bounded by the graphs of $y = 4x - x^2$ and the x-axis.

Solution

a.

First, sketch the region and the solid of revolution as shown.

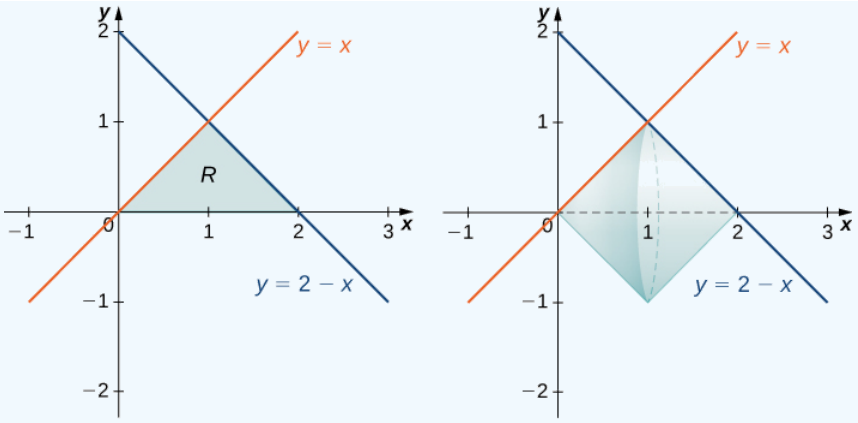


Figure 6.3.11: (a) The region R bounded by two lines and the x-axis. (b) The solid of revolution generated by revolving R about the x-axis.

Looking at the region, if we want to integrate with respect to x, we would have to break the integral into two pieces, because we have different functions bounding the region over $[0, 1]$ and $[1, 2]$. In this case, using the disk method, we would have

$$V = \int_0^1 \pi x^2 \, dx + \int_1^2 \pi (2 - x)^2 \, dx.$$

If we used the shell method instead, we would use functions of y to represent the curves, producing

$$V = \int_0^1 2\pi y[(2 - y) - y] \, dy = \int_0^1 2\pi y[2 - 2y] \, dy.$$

Neither of these integrals is particularly onerous, but since the shell method requires only one integral, and the integrand requires less simplification, we should probably go with the shell method in this case.

b.

First, sketch the region and the solid of revolution as shown.

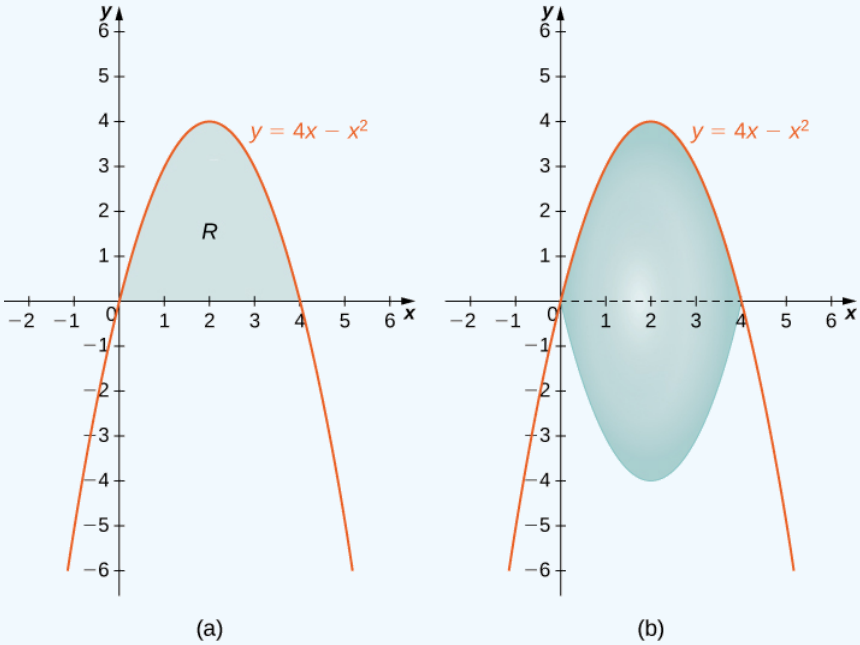


Figure 6.3.12: (a) The region R between the curve and the x-axis. (b) The solid of revolution generated by revolving R about the x-axis.

Looking at the region, it would be problematic to define a horizontal rectangle; the region is bounded on the left and right by the same function. Therefore, we can dismiss the method of shells. The solid has no cavity in the middle, so we can use the method of disks. Then

$$V = \int_0^4 \pi \left(4x - x^2\right)^2 \, dx$$

? Exercise 6.3.6

Select the best method to find the volume of a solid of revolution generated by revolving the given region around the x-axis, and set up the integral to find the volume (do not evaluate the integral): the region bounded by the graphs of $y = 2 - x^2$ and $y = x^2$.

Hint

Sketch the region and use Figure 6.3.12 to decide which integral is easiest to evaluate.

Answer

Use the method of washers;

$$V = \int_{-1}^1 \pi \left[\left(2 - x^2\right)^2 - \left(x^2\right)^2\right] \, dx$$

Key Concepts

- The method of cylindrical shells is another method for using a definite integral to calculate the volume of a solid of revolution. This method is sometimes preferable to either the method of disks or the method of washers because we integrate with respect to the other variable. In some cases, one integral is substantially more complicated than the other.
- The geometry of the functions and the difficulty of the integration are the main factors in deciding which integration method to use.

Key Equations

- **Method of Cylindrical Shells**

$$V = \int_a^b (2\pi \, x \, f(x)) \, dx$$

Glossary

method of cylindrical shells

a method of calculating the volume of a solid of revolution by dividing the solid into nested cylindrical shells; this method is different from the methods of disks or washers in that we integrate with respect to the opposite variable

This page titled 6.3: Volumes of Revolution - Cylindrical Shells is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

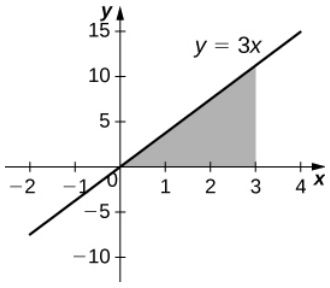
6.3E: Exercises for Section 6.3

For exercises 1 - 6, find the volume generated when the region between the two curves is rotated around the given axis. Use both the shell method and the washer method. Use technology to graph the functions and draw a typical slice by hand.

1) [T] Over the curve of $y = 3x$, $x = 0$, and $y = 3$ rotated around the y-axis.

2) [T] Under the curve of $y = 3x$, $x = 0$, and $x = 3$ rotated around the y-axis.

Answer

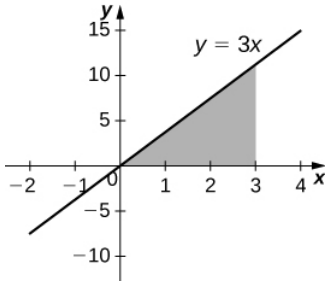


$V = 54\pi$ units^3

3) [T] Over the curve of $y = 3x$, $x = 0$, and $y = 3$ rotated around the x-axis.

4) [T] Under the curve of $y = 3x$, $x = 0$, and $x = 3$ rotated around the x-axis.

Answer

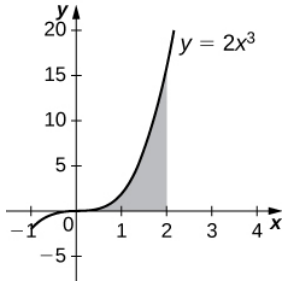


$V = 81\pi$ units^3

5) [T] Under the curve of $y = 2x^3,\ x = 0$, and $x = 2$ rotated around the y-axis.

6) [T] Under the curve of $y = 2x^3,\ x = 0$, and $x = 2$ rotated around the x-axis.

Answer



$V = \frac{512\pi}{7}$ units^3

For exercises 7 - 16, use shells to find the volumes of the given solids. Note that the rotated regions lie between the curve and the x-axis and are rotated around the y-axis.

7) $y = 1 - x^2$, $x = 0$, and $x = 1$

8) $y = 5x^3$, $x = 0$, and $x = 1$

Answer

$V = 2\pi$ units^3

9) $y = \dfrac{1}{x}$, $x = 1$, and $x = 100$

10) $y = \sqrt{1 - x^2}$, $x = 0$, and $x = 1$

Answer

$V = \frac{2\pi}{3}$ units^3

11) $y = \dfrac{1}{1 + x^2}$, $x = 0$,and $x = 3$

12) $y = \sin x^2, x = 0$, and $x = \sqrt{\pi}$

Answer

$V = 2\pi$ units^3

13) $y = \dfrac{1}{\sqrt{1 - x^2}}$, $x = 0$, and $x = \frac{1}{2}$

14) $y = \sqrt{x}$, $x = 0$, and $x = 1$

Answer

$V = \frac{4\pi}{5}$ units^3

15) $y = (1 + x^2)^3$, $x = 0$, and $x = 1$

16) $y = 5x^3 - 2x^4$, $x = 0$, and $x = 2$

Answer

$V = \frac{64\pi}{3}$ units^3

For exercises 17 - 26, use shells to find the volume generated by rotating the regions between the given curve and $y = 0$ around the x-axis.

17) $y = \sqrt{1 - x^2}$, $x = 0$, and $x = 1$

18) $y = x^2$, $x = 0$, and $x = 2$

Answer

$V = \frac{32\pi}{5}$ units^3

19) $y = e^x$, $x = 0$, and $x = 1$

20) $y = \ln(x)$, $x = 1$, and $x = e$

Answer

$V = \pi(e - 2)$ units^3

21) $x = \dfrac{1}{1 + y^2}$, $y = 1$, and $y = 4$

22) $x = \dfrac{1 + y^2}{y}$, $y = 0$, and $y = 2$

Answer

$V = \frac{28\pi}{3}$ units^3

23) $x = \cos y$, $y = 0$, and $y = \pi$

24) $x = y^3 - 4y^2$, $x = -1$, and $x = 2$

Answer

$V = \frac{84\pi}{5}$ units^3

25) $x = ye^y$, $x = -1$, and $x = 2$

26) $x = e^y \cos y$, $x = 0$, and $x = \pi$

Answer

$V = e^{\pi}\pi^2$ units^3

For exercises 27 - 36, find the volume generated when the region between the curves is rotated around the given axis.

27) $y = 3 - x$, $y = 0$, $x = 0$, and $x = 2$ rotated around the y-axis.

28) $y = x^3$, $y = 0$, $x = 0$, and $y = 8$ rotated around the y-axis.

Answer

$V = \frac{64\pi}{5}$ units^3

29) $y = x^2$, $y = x$, rotated around the y-axis.

30) $y = \sqrt{x}$, $x = 0$, and $x = 1$ rotated around the line $x = 2$.

Answer

$V = \frac{28\pi}{15}$ units^3

31) $y = \dfrac{1}{4 - x}$, $x = 1$, and $x = 2$ rotated around the line $x = 4$.

32) $y = \sqrt{x}$ and $y = x^2$ rotated around the y-axis.

Answer

$V = \frac{3\pi}{10}$ units^3

33) $y = \sqrt{x}$ and $y = x^2$ rotated around the line $x = 2$.

34) $x = y^3$, $y = \dfrac{1}{x}$, $x = 1$, and $y = 2$ rotated around the x-axis.

Answer

$\frac{52\pi}{5}$ units^3

35) $x = y^2$ and $y = x$ rotated around the line $y = 2$.

36) [T] Left of $x = \sin(\pi y)$, right of $y = x$, around the y-axis.

Answer

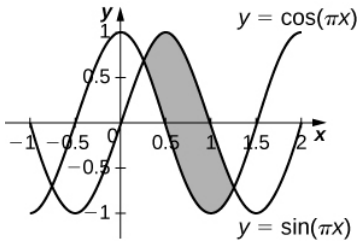
$V \approx 0.9876$ units^3

For exercises 37 - 44, use technology to graph the region. Determine which method you think would be easiest to use to calculate the volume generated when the function is rotated around the specified axis. Then, use your chosen method to find the volume.

37) [T] $y = x^2$ and $y = 4x$ rotated around the y-axis.

38) [T] $y = \cos(\pi x), y = \sin(\pi x), x = \frac{1}{4}$, and $x = \frac{5}{4}$ rotated around the y-axis.

Answer

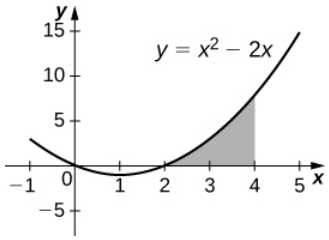


$V = 3\sqrt{2}$ units^3

39) [T] $y = x^2 - 2x,\ x = 2$, and $x = 4$ rotated around the y-axis.

40) [T] $y = x^2 - 2x,\ x = 2$, and $x = 4$ rotated around the x-axis.

Answer

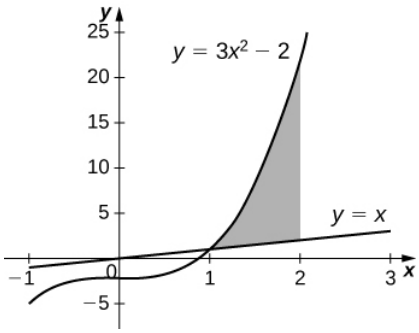


$V = \frac{496\pi}{15}$ units^3

41) [T] $y = 3x^3 - 2,\ y = x$, and $x = 2$ rotated around the x-axis.

42) [T] $y = 3x^3 - 2,\ y = x$, and $x = 2$ rotated around the y-axis.

Answer

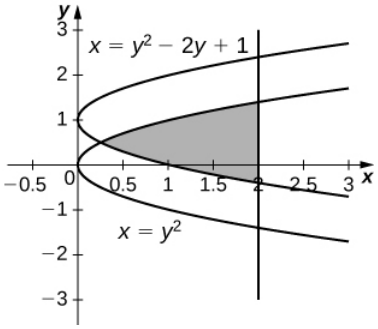


$V = \frac{398\pi}{15}$ units^3

43) [T] $x = \sin(\pi y^2)$ and $x = \sqrt{2}y$ rotated around the x-axis.

44) [T] $x = y^2,\ x = y^2 - 2y + 1$, and $x = 2$ rotated around the y-axis.

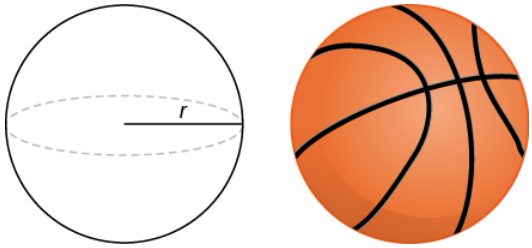
Answer



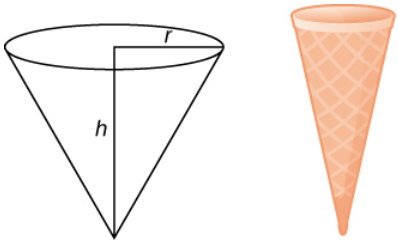
$V = 15.9074$ units^3

For exercises 45 - 51, use the method of shells to approximate the volumes of some common objects, which are pictured in accompanying figures.

45) Use the method of shells to find the volume of a sphere of radius r.



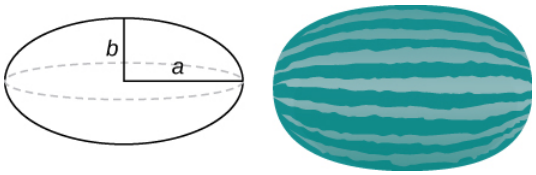
46) Use the method of shells to find the volume of a cone with radius r and height h.



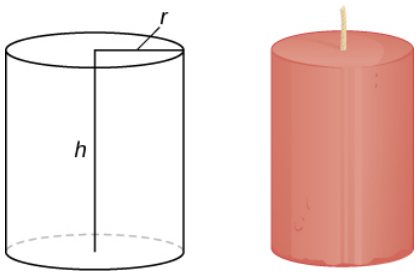
Answer

$V = \frac{1}{3}\pi r^2 h$ units^3

47) Use the method of shells to find the volume of an ellipse $(x^2/a^2) + (y^2/b^2) = 1$ rotated around the x-axis.



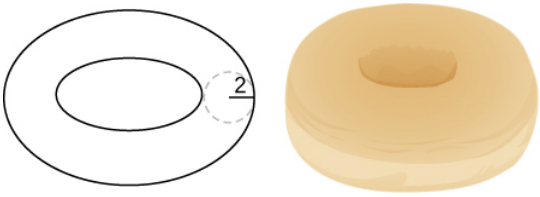
48) Use the method of shells to find the volume of a cylinder with radius r and height h.



Answer

$V = \pi r^2 h$ units^3

49) Use the method of shells to find the volume of the donut created when the circle $x^2 + y^2 = 4$ is rotated around the line $x = 4$.



50) Consider the region enclosed by the graphs of $y = f(x),\ y = 1 + f(x),\ x = 0,\ y = 0$, and $x = a > 0$. What is the volume of the solid generated when this region is rotated around the y-axis? Assume that the function is defined over the interval $[0, a]$.

Answer

$V = \pi a^2$ units^3

51) Consider the function $y = f(x)$, which decreases from $f(0) = b$ to $f(1) = 0$. Set up the integrals for determining the volume, using both the shell method and the disk method, of the solid generated when this region, with $x = 0$ and $y = 0$, is rotated around the y-axis. Prove that both methods approximate the same volume. Which method is easier to apply? (Hint: Since $f(x)$ is one-to-one, there exists an inverse $f^{-1}(y)$.)

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 6.3E: Exercises for Section 6.3 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.4: Arc Length of a Curve and Surface Area

Learning Objectives

- Determine the length of a curve, $y = f(x)$, between two points.
- Determine the length of a curve, $x = g(y)$, between two points.
- Find the surface area of a solid of revolution.

In this section, we use definite integrals to find the arc length of a curve. We can think of **arc length** as the distance you would travel if you were walking along the path of the curve. Many real-world applications involve arc length. If a rocket is launched along a parabolic path, we might want to know how far the rocket travels. Or, if a curve on a map represents a road, we might want to know how far we have to drive to reach our destination.

We begin by calculating the arc length of curves defined as functions of x, then we examine the same process for curves defined as functions of y. (The process is identical, with the roles of x and y reversed.) The techniques we use to find arc length can be extended to find the surface area of a surface of revolution, and we close the section with an examination of this concept.

Arc Length of the Curve y = f(x)

In previous applications of integration, we required the function $f(x)$ to be integrable, or at most continuous. However, for calculating arc length we have a more stringent requirement for $f(x)$. Here, we require $f(x)$ to be differentiable, and furthermore we require its derivative, $f'(x)$, to be continuous. Functions like this, which have continuous derivatives, are called **smooth.** (This property comes up again in later chapters.)

Let $f(x)$ be a smooth function defined over $[a, b]$. We want to calculate the length of the curve from the point $(a, f(a))$ to the point $(b, f(b))$. We start by using line segments to approximate the length of the curve. For $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of $[a, b]$. Then, for $i = 1, 2, \ldots, n$, construct a line segment from the point $(x_{i-1}, f(x_{i-1}))$ to the point $(x_i, f(x_i))$. Although it might seem logical to use either horizontal or vertical line segments, we want our line segments to approximate the curve as closely as possible. Figure 6.4.1 depicts this construct for $n = 5$.

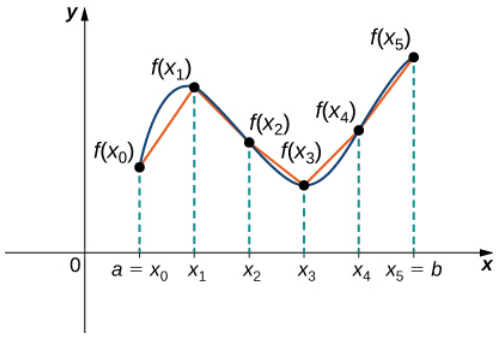


Figure 6.4.1: We can approximate the length of a curve by adding line segments.

To help us find the length of each line segment, we look at the change in vertical distance as well as the change in horizontal distance over each interval. Because we have used a regular partition, the change in horizontal distance over each interval is given by Δx. The change in vertical distance varies from interval to interval, though, so we use $\Delta y_i = f(x_i) - f(x_{i-1})$ to represent the change in vertical distance over the interval $[x_{i-1}, x_i]$, as shown in Figure 6.4.2. Note that some (or all) Δy_i may be negative.

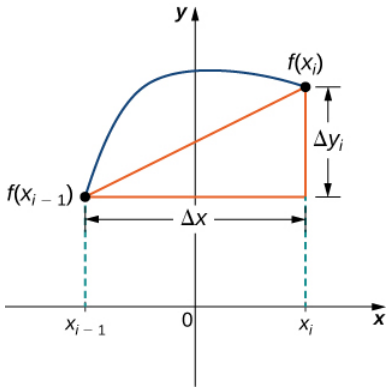


Figure 6.4.2: A representative line segment approximates the curve over the interval $[x_{i-1}, x_i]$.

By the Pythagorean theorem, the length of the line segment is

$$\sqrt{(\Delta x)^2 + (\Delta y_i)^2}.$$

We can also write this as

$$\Delta x\sqrt{1 + ((\Delta y_i)/(\Delta x))^2}.$$

Now, by the Mean Value Theorem, there is a point $x_i^* \in [x_{i-1}, x_i]$ such that $f'(x_i^*) = (\Delta y_i)/(\Delta x)$. Then the length of the line segment is given by

$$\Delta x\sqrt{1 + [f'(x_i^*)]^2}.$$

Adding up the lengths of all the line segments, we get

$$\text{Arc Length} \approx \sum_{i=1}^{n} \sqrt{1 + [f'(x_i^*)]^2}\Delta x.$$

This is a Riemann sum. Taking the limit as $n \to \infty$, we have

$$\begin{aligned}\text{Arc Length} &= \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{1 + [f'(x_i^*)]^2}\Delta x \\ &= \int_a^b \sqrt{1 + [f'(x)]^2}dx.\end{aligned}$$

We summarize these findings in the following theorem.

Arc Length for $y = f(x)$

Let $f(x)$ be a smooth function over the interval $[a, b]$. Then the arc length of the portion of the graph of $f(x)$ from the point $(a, f(a))$ to the point $(b, f(b))$ is given by

$$\text{Arc Length} = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx.$$

Note that we are integrating an expression involving $f'(x)$, so we need to be sure $f'(x)$ is integrable. This is why we require $f(x)$ to be smooth. The following example shows how to apply the theorem.

Example 6.4.1: Calculating the Arc Length of a Function of x

Let $f(x) = 2x^{3/2}$. Calculate the arc length of the graph of $f(x)$ over the interval $[0, 1]$. Round the answer to three decimal places.

Solution

We have $f'(x) = 3x^{1/2}$, so $[f'(x)]^2 = 9x$. Then, the arc length is

$$\text{Arc Length} = \int_a^b \sqrt{1+[f'(x)]^2}dx$$
$$= \int_0^1 \sqrt{1+9x}\,dx.$$

Substitute $u = 1 + 9x$. Then, $du = 9\,dx$. When $x = 0$, then $u = 1$, and when $x = 1$, then $u = 10$. Thus,

$$\text{Arc Length} = \int_0^1 \sqrt{1+9x}\,dx$$
$$= \frac{1}{9}\int_0^1 \sqrt{1+9x}\cdot 9\,dx$$
$$= \frac{1}{9}\int_1^{10} \sqrt{u}\,du$$
$$= \frac{1}{9}\cdot\frac{2}{3}u^{3/2}\Big|_1^{10} = \frac{2}{27}[10\sqrt{10}-1]$$
$$\approx 2.268 \text{ units}.$$

? Exercise 6.4.1

Let $f(x) = (4/3)x^{3/2}$. Calculate the arc length of the graph of $f(x)$ over the interval $[0, 1]$. Round the answer to three decimal places.

Hint

Use the process from the previous example. Don't forget to change the limits of integration.

Answer

$$\frac{1}{6}(5\sqrt{5}-1) \approx 1.697 \text{ units}$$

Although it is nice to have a formula for calculating arc length, this particular theorem can generate expressions that are difficult to integrate. We study some techniques for integration in Introduction to Techniques of Integration. In some cases, we may have to use a computer or calculator to approximate the value of the integral.

✓ Example 6.4.2: Using a Computer or Calculator to Determine the Arc Length of a Function of x

Let $f(x) = x^2$. Calculate the arc length of the graph of $f(x)$ over the interval $[1, 3]$.

Solution

We have $f'(x) = 2x$, so $[f'(x)]^2 = 4x^2$. Then the arc length is given by

$$\text{Arc Length} = \int_a^b \sqrt{1+[f'(x)]^2}\,dx$$
$$= \int_1^3 \sqrt{1+4x^2}\,dx.$$

Using a computer to approximate the value of this integral, we get

$$\int_1^3 \sqrt{1+4x^2}\,dx \approx 8.26815 \text{ units}.$$

? Exercise 6.4.2

Let $f(x) = \sin x$. Calculate the arc length of the graph of $f(x)$ over the interval $[0, \pi]$. Use a computer or calculator to approximate the value of the integral.

Hint

Use the process from the previous example.

Answer

$$\text{Arc Length} \approx 3.8202 \text{ units}$$

Arc Length of the Curve x = g(y)

We have just seen how to approximate the length of a curve with line segments. If we want to find the arc length of the graph of a function of y, we can repeat the same process, except we partition the y-axis instead of the x-axis. Figure 6.4.3 shows a representative line segment.

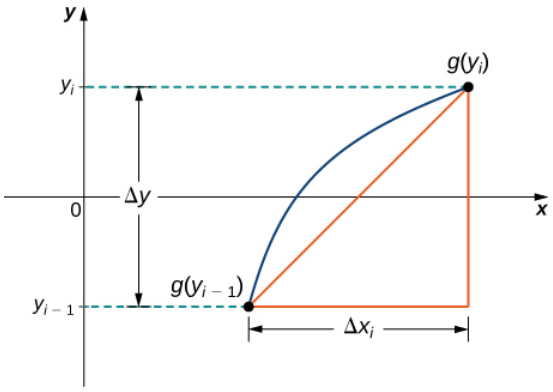


Figure 6.4.3: A representative line segment over the interval $[y_{i-1}, y_i]$.

Then the length of the line segment is

$$\sqrt{(\Delta y)^2 + (\Delta x_i)^2},$$

which can also be written as

$$\Delta y \sqrt{1 + \left(\frac{\Delta x_i}{\Delta y}\right)^2}.$$

If we now follow the same development we did earlier, we get a formula for arc length of a function $x = g(y)$.

Arc Length for $x = g(y)$

Let $g(y)$ be a smooth function over an interval $[c, d]$. Then, the arc length of the graph of $g(y)$ from the point $(c, g(c))$ to the point $(d, g(d))$ is given by

$$\text{Arc Length} = \int_c^d \sqrt{1 + [g'(y)]^2}\, dy.$$

Example 6.4.3: Calculating the Arc Length of a Function of y

Let $g(y) = 3y^3$. Calculate the arc length of the graph of $g(y)$ over the interval $[1, 2]$.

Solution

We have $g'(y) = 9y^2$, so $[g'(y)]^2 = 81y^4$. Then the arc length is

$$\begin{aligned} \text{Arc Length} &= \int_c^d \sqrt{1 + [g'(y)]^2}\, dy \\ &= \int_1^2 \sqrt{1 + 81y^4}\, dy. \end{aligned}$$

Using a computer to approximate the value of this integral, we obtain

$$\int_1^2 \sqrt{1+81y^4}dy \approx 21.0277 \text{ units}.$$

? Exercise 6.4.3

Let $g(y) = 1/y$. Calculate the arc length of the graph of $g(y)$ over the interval $[1, 4]$. Use a computer or calculator to approximate the value of the integral.

Hint

Use the process from the previous example.

Answer

$$\text{Arc Length} = 3.15018 \text{ units}$$

Area of a Surface of Revolution

The concepts we used to find the arc length of a curve can be extended to find the surface area of a surface of revolution. **Surface area** is the total area of the outer layer of an object. For objects such as cubes or bricks, the surface area of the object is the sum of the areas of all of its faces. For curved surfaces, the situation is a little more complex. Let $f(x)$ be a nonnegative smooth function over the interval $[a, b]$. We wish to find the surface area of the surface of revolution created by revolving the graph of $y = f(x)$ around the x-axis as shown in the following figure.

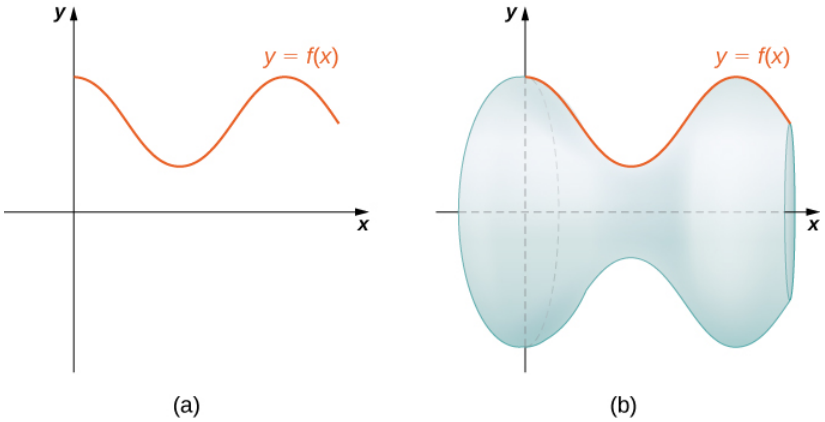


Figure 6.4.4: (a) A curve representing the function $f(x)$. (b) The surface of revolution formed by revolving the graph of $f(x)$ around the $x-axis$.

As we have done many times before, we are going to partition the interval $[a, b]$ and approximate the surface area by calculating the surface area of simpler shapes. We start by using line segments to approximate the curve, as we did earlier in this section. For $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of $[a, b]$. Then, for $i = 1, 2, \ldots, n$, construct a line segment from the point $(x_{i-1}, f(x_{i-1}))$ to the point $(x_i, f(x_i))$. Now, revolve these line segments around the x-axis to generate an approximation of the surface of revolution as shown in the following figure.

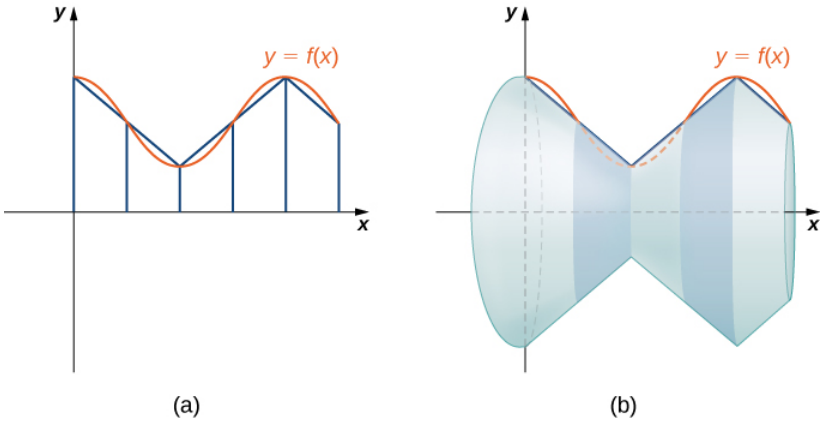


Figure 6.4.5: (a) Approximating $f(x)$ with line segments. (b) The surface of revolution formed by revolving the line segments around the x-axis.

Notice that when each line segment is revolved around the axis, it produces a band. These bands are actually pieces of cones (think of an ice cream cone with the pointy end cut off). A piece of a cone like this is called a **frustum** of a cone.

To find the surface area of the band, we need to find the lateral surface area, S, of the frustum (the area of just the slanted outside surface of the frustum, not including the areas of the top or bottom faces). Let r_1 and r_2 be the radii of the wide end and the narrow end of the frustum, respectively, and let l be the slant height of the frustum as shown in the following figure.

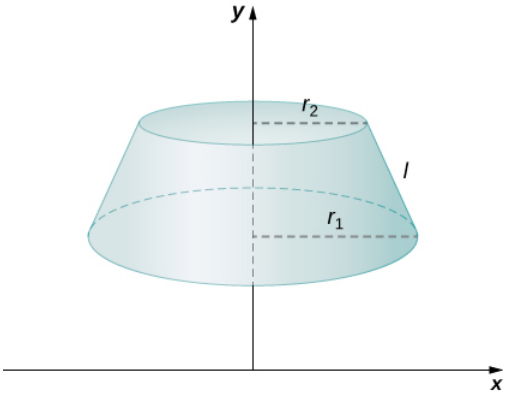


Figure 6.4.6: A frustum of a cone can approximate a small part of surface area.

We know the lateral surface area of a cone is given by

$$\text{Lateral Surface Area } = \pi rs,$$

where r is the radius of the base of the cone and s is the **slant height** (Figure 6.4.7).

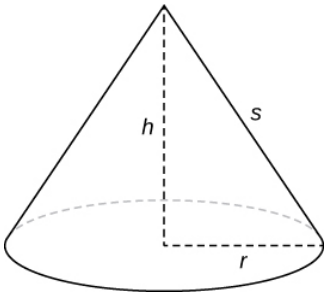


Figure 6.4.7: The lateral surface area of the cone is given by πrs.

Since a frustum can be thought of as a piece of a cone, the lateral surface area of the frustum is given by the lateral surface area of the whole cone less the lateral surface area of the smaller cone (the pointy tip) that was cut off (Figure 6.4.8).

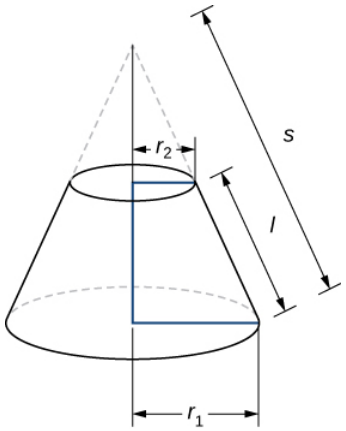


Figure 6.4.8: Calculating the lateral surface area of a frustum of a cone.

The cross-sections of the small cone and the large cone are similar triangles, so we see that

$$\frac{r_2}{r_1} = \frac{s-l}{s}$$

Solving for s, we get

$$\begin{aligned} \frac{r_2}{r_1} &= \frac{s-l}{s} \\ r_2 s &= r_1(s-l) \\ r_2 s &= r_1 s - r_1 l \\ r_1 l &= r_1 s - r_2 s \\ r_1 l &= (r_1 - r_2)s \\ \frac{r_1 l}{r_1 - r_2} &= s \end{aligned}$$

Then the lateral surface area (SA) of the frustum is

$$\begin{aligned} S &= (\text{Lateral SA of large cone}) - (\text{Lateral SA of small cone}) \\ &= \pi r_1 s - \pi r_2 (s-l) \\ &= \pi r_1 \left(\frac{r_1 l}{r_1 - r_2}\right) - \pi r_2 \left(\frac{r_1 l}{r_1 - r_2} - l\right) \\ &= \frac{\pi r_1^2 l}{r^1 - r^2} - \frac{\pi r_1 r_2 l}{r_1 - r_2} + \pi r_2 l \\ &= \frac{\pi r_1^2 l}{r_1 - r_2} - \frac{\pi r_1 r 2_l}{r_1 - r_2} + \frac{\pi r_2 l (r_1 - r_2)}{r_1 - r_2} \\ &= \frac{\pi r_1^2}{l r_1 - r_2} - \frac{\pi r_1 r_2 l}{r_1 - r_2} + \frac{\pi r_1 r_2 l}{r_1 - r_2} - \frac{\pi r_2^2 l}{r_1 - r_3} \\ &= \frac{\pi(r_1^2 - r_2^2)l}{r_1 - r_2} = \frac{\pi(r_1 - r + 2)(r1 + r2)l}{r_1 - r_2} \\ &= \pi(r_1 + r_2)l. \end{aligned}$$

Let's now use this formula to calculate the surface area of each of the bands formed by revolving the line segments around the x-axis. A representative band is shown in the following figure.

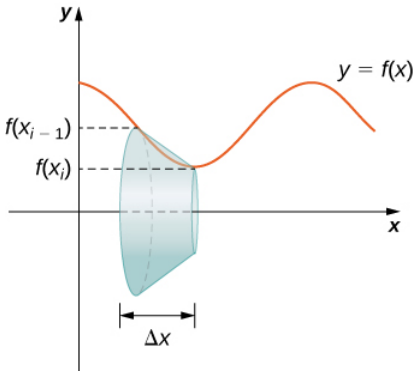


Figure 6.4.9: A representative band used for determining surface area.

Note that the slant height of this frustum is just the length of the line segment used to generate it. So, applying the surface area formula, we have

$$\begin{aligned} S &= \pi(r_1 + r_2)l \\ &= \pi\big(f(x_{i-1}) + f(x_i)\big)\sqrt{\Delta x^2 + (\Delta y_i)^2} \\ &= \pi\big(f(x_{i-1}) + f(x_i)\big)\Delta x\sqrt{1 + \left(\frac{\Delta y_i}{\Delta x}\right)^2} \end{aligned}$$

Now, as we did in the development of the arc length formula, we apply the Mean Value Theorem to select $x_i^* \in [x_{i-1}, x_i]$ such that $f'(x_i^*) = (\Delta y_i)/\Delta x$. This gives us

$$S = \pi\big(f(x_{i-1}) + f(x_i)\big)\Delta x\sqrt{1 + \big(f'(x_i^*)\big)^2}$$

Furthermore, since $f(x)$ is continuous, by the *Intermediate Value Theorem*, there is a point $x_i^{**} \in [x_{i-1}, x_i]$ such that $f(x_i^{**}) = \frac{1}{2}\left[f(x_{i-1}) + f(x_i)\right]$,

so we get

$$S = 2\pi f(x_i^{**})\Delta x\sqrt{1 + \big(f'(x_i^*)\big)^2}.$$

Then the approximate surface area of the whole surface of revolution is given by

$$\text{Surface Area} \approx \sum_{i=1}^{n} 2\pi f(x_i^{**})\Delta x\sqrt{1 + \big(f'(x_i^*)\big)^2}.$$

This almost looks like a Riemann sum, except we have functions evaluated at two different points, x_i^* and x_i^{**}, over the interval $[x_{i-1}, x_i]$. Although we do not examine the details here, it turns out that because $f(x)$ is smooth, if we let $n \to \infty$, the limit works the same as a Riemann sum even with the two different evaluation points. This makes sense intuitively. Both x_i^* and x_i^{**} are in the interval $[x_{i-1}, x_i]$, so it makes sense that as $n \to \infty$, both x_i^* and x_i^{**} approach x Those of you who are interested in the details should consult an advanced calculus text.

Taking the limit as $n \to \infty$, we get

$$\begin{aligned} \text{Surface Area} &= \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi f(x_i^{**})\Delta x\sqrt{1 + \big(f'(x_i^*)\big)^2} \\ &= \int_a^b \left(2\pi f(x)\sqrt{1 + \big(f'(x)\big)^2}\right) dx \end{aligned}$$

As with arc length, we can conduct a similar development for functions of y to get a formula for the surface area of surfaces of revolution about the y-axis. These findings are summarized in the following theorem.

Surface Area of a Surface of Revolution

Let $f(x)$ be a nonnegative smooth function over the interval $[a, b]$. Then, the surface area of the surface of revolution formed by revolving the graph of $f(x)$ around the x-axis is given by

$$\text{Surface Area} = \int_a^b \left(2\pi f(x)\sqrt{1+\left(f'(x)\right)^2}\right) dx$$

Similarly, let $g(y)$ be a nonnegative smooth function over the interval $[c, d]$. Then, the surface area of the surface of revolution formed by revolving the graph of $g(y)$ around the y-axis is given by

$$\text{Surface Area} = \int_c^d \left(2\pi g(y)\sqrt{1+\left(g'(y)\right)^2}\right) dy$$

Example 6.4.4: Calculating the Surface Area of a Surface of Revolution 1.

Let $f(x) = \sqrt{x}$ over the interval $[1, 4]$. Find the surface area of the surface generated by revolving the graph of $f(x)$ around the x-axis. Round the answer to three decimal places.

Solution

The graph of $f(x)$ and the surface of rotation are shown in Figure 6.4.10.

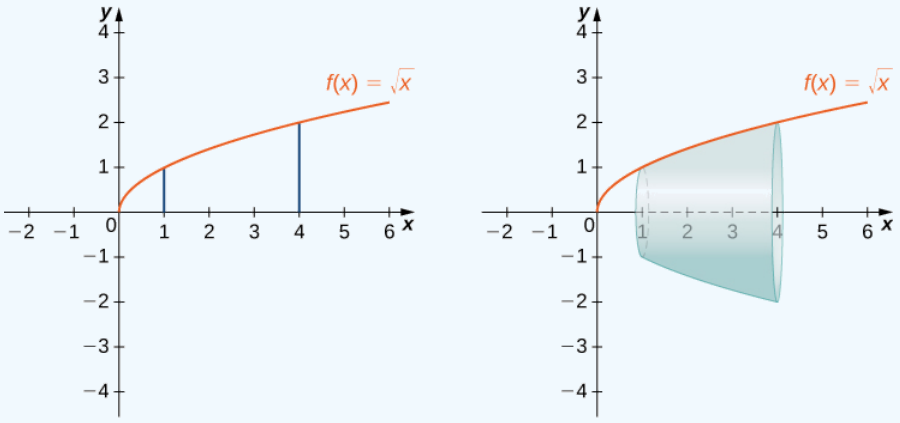


Figure 6.4.10: (a) The graph of $f(x)$. (b) The surface of revolution.

We have $f(x) = \sqrt{x}$. Then, $f'(x) = \frac{1}{2\sqrt{x}}$ and $\left(f'(x)\right)^2 = \frac{1}{4x}$. Then,

$$\begin{aligned}\text{Surface Area} &= \int_a^b \left(2\pi f(x)\sqrt{1+\left(f'(x)\right)^2}\right) dx \\ &= \int_1^4 \left(2\pi\sqrt{x}\sqrt{1+\frac{1}{4x}}\right) dx \\ &= \int_1^4 \left(2\pi\sqrt{x+\frac{1}{4}}\right) dx.\end{aligned}$$

Let $u = x + 1/4$. Then, $du = dx$. When $x = 1$, $u = 5/4$, and when $x = 4$, $u = 17/4$. This gives us

$$\begin{aligned}\int_0^1 \left(2\pi\sqrt{x+\frac{1}{4}}\right) dx &= \int_{5/4}^{17/4} 2\pi\sqrt{u}\, du \\ &= 2\pi\left[\frac{2}{3}u^{3/2}\right]\Big|_{5/4}^{17/4} \\ &= \frac{\pi}{6}[17\sqrt{17} - 5\sqrt{5}] \approx 30.846 \text{ units}^2\end{aligned}$$

Exercise 6.4.4

Let $f(x) = \sqrt{1-x}$ over the interval $[0, 1/2]$. Find the surface area of the surface generated by revolving the graph of $f(x)$ around the x-axis. Round the answer to three decimal places.

Hint

Use the process from the previous example.

Answer

$$\frac{\pi}{6}\left(5\sqrt{5} - 3\sqrt{3}\right) \approx 3.133 \text{ units}^2$$

Example 6.4.5: Calculating the Surface Area of a Surface of Revolution 2

Let $f(x) = y = \sqrt[3]{3x}$. Consider the portion of the curve where $0 \le y \le 2$. Find the surface area of the surface generated by revolving the graph of $f(x)$ around the y-axis.

Solution

Notice that we are revolving the curve around the y-axis, and the interval is in terms of y, so we want to rewrite the function as a function of y. We get $x = g(y) = (1/3)y^3$. The graph of $g(y)$ and the surface of rotation are shown in the following figure.

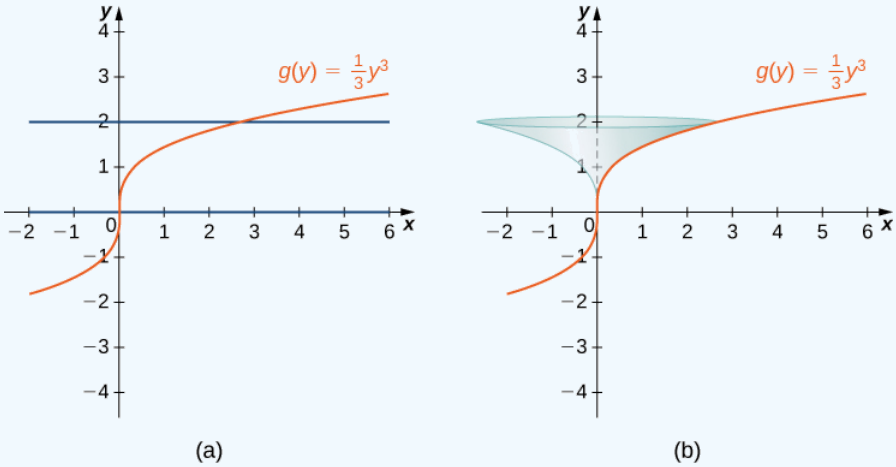


Figure 6.4.11: (a) The graph of $g(y)$. (b) The surface of revolution.

We have $g(y) = (1/3)y^3$, so $g'(y) = y^2$ and $\left(g'(y)\right)^2 = y^4$. Then

$$\begin{aligned}\text{Surface Area} &= \int_c^d \left(2\pi g(y)\sqrt{1+\left(g'(y)\right)^2}\right) dy \\ &= \int_0^2 \left(2\pi\left(\frac{1}{3}y^3\right)\sqrt{1+y^4}\right) dy \\ &= \frac{2\pi}{3}\int_0^2 \left(y^3\sqrt{1+y^4}\right) dy.\end{aligned}$$

Let $u = y^4 + 1$. Then $du = 4y^3\,dy$. When $y = 0$, $u = 1$, and when $y = 2$, $u = 17$. Then

$$\begin{aligned}\frac{2\pi}{3}\int_0^2 \left(y^3\sqrt{1+y^4}\right) dy &= \frac{2\pi}{3}\int_1^{17} \frac{1}{4}\sqrt{u}\,du \\ &= \frac{\pi}{6}\left[\frac{2}{3}u^{3/2}\right]\Big|_1^{17} = \frac{\pi}{9}\left[(17)^{3/2} - 1\right] \approx 24.118 \text{ units}^2.\end{aligned}$$

? Exercise 6.4.5

Let $g(y) = \sqrt{9 - y^2}$ over the interval $y \in [0, 2]$. Find the surface area of the surface generated by revolving the graph of $g(y)$ around the y-axis.

Hint

Use the process from the previous example.

Answer

$12\pi \text{ units}^2$

Key Concepts

- The arc length of a curve can be calculated using a definite integral.
- The arc length is first approximated using line segments, which generates a Riemann sum. Taking a limit then gives us the definite integral formula. The same process can be applied to functions of y.
- The concepts used to calculate the arc length can be generalized to find the surface area of a surface of revolution.
- The integrals generated by both the arc length and surface area formulas are often difficult to evaluate. It may be necessary to use a computer or calculator to approximate the values of the integrals.

Key Equations

- **Arc Length of a Function of x**

$$\textbf{Arc Length} = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$$

- **Arc Length of a Function of y**

$$\textbf{Arc Length} = \int_c^d \sqrt{1 + [g'(y)]^2}\, dy$$

- **Surface Area of a Function of x**

$$\textbf{Surface Area} = \int_a^b \left(2\pi f(x)\sqrt{1 + \left(f'(x)\right)^2}\right)\, dx$$

Glossary

arc length

the arc length of a curve can be thought of as the distance a person would travel along the path of the curve

frustum

a portion of a cone; a frustum is constructed by cutting the cone with a plane parallel to the base

surface area

the surface area of a solid is the total area of the outer layer of the object; for objects such as cubes or bricks, the surface area of the object is the sum of the areas of all of its faces

This page titled 6.4: Arc Length of a Curve and Surface Area is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.4E: Exercises for Section 6.4

For exercises 1 - 3, find the length of the functions over the given interval.

1) $y = 5x$ from $x = 0$ to $x = 2$

Answer

$s = 2\sqrt{26}$ units

2) $y = -\frac{1}{2}x + 25$ from $x = 1$ to $x = 4$

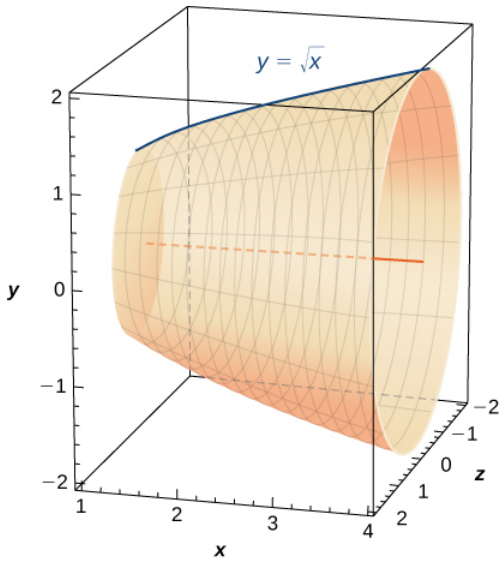
3) $x = 4y$ from $y = -1$ to $y = 1$

Answer

$s = 2\sqrt{17}$ units

4) Pick an arbitrary linear function $x = g(y)$ over any interval of your choice (y_1, y_2). Determine the length of the function and then prove the length is correct by using geometry.

5) Find the surface area of the volume generated when the curve $y = \sqrt{x}$ revolves around the x-axis from $(1, 1)$ to $(4, 2)$, as seen here.



Answer

$A = \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$ units^2

6) Find the surface area of the volume generated when the curve $y = x^2$ revolves around the y-axis from $(1, 1)$ to $(3, 9)$.

For exercises 7 - 16, find the lengths of the functions of x over the given interval. If you cannot evaluate the integral exactly, use technology to approximate it.

7) $y = x^{3/2}$ from $(0, 0)$ to $(1, 1)$

Answer

$s = \frac{13\sqrt{13}-8}{27}$ units

8) $y = x^{2/3}$ from $(1, 1)$ to $(8, 4)$

9) $y = \frac{1}{3}(x^2 + 2)^{3/2}$ from $x = 0$ to $x = 1$

Answer

$s = \frac{4}{3}$ units

10) $y = \frac{1}{3}(x^2 - 2)^{3/2}$ from $x = 2$ to $x = 4$

11) [T] $y = e^x$ on $x = 0$ to $x = 1$

Answer

$s \approx 2.0035$ units

12) $y = \dfrac{x^3}{3} + \dfrac{1}{4x}$ from $x = 1$ to $x = 3$

13) $y = \dfrac{x^4}{4} + \dfrac{1}{8x^2}$ from $x = 1$ to $x = 2$

Answer

$s = \frac{123}{32}$ units

14) $y = \dfrac{2x^{3/2}}{3} - \dfrac{x^{1/2}}{2}$ from $x = 1$ to $x = 4$

15) $y = \frac{1}{27}(9x^2 + 6)^{3/2}$ from $x = 0$ to $x = 2$

Answer

$s = 10$ units

16) [T] $y = \sin x$ on $x = 0$ to $x = \pi$

For exercises 17 - 26, find the lengths of the functions of y over the given interval. If you cannot evaluate the integral exactly, use technology to approximate it.

17) $y = \dfrac{5 - 3x}{4}$ from $y = 0$ to $y = 4$

Answer

$s = \frac{20}{3}$ units

18) $x = \frac{1}{2}(e^y + e^{-y})$ from $y = -1$ to $y = 1$

19) $x = 5y^{3/2}$ from $y = 0$ to $y = 1$

Answer

$s = \frac{1}{675}(229\sqrt{229} - 8)$ units

20) [T] $x = y^2$ from $y = 0$ to $y = 1$

21) $x = \sqrt{y}$ from $y = 0$ to $y = 1$

Answer

$s = \frac{1}{8}(4\sqrt{5} + \ln(9 + 4\sqrt{5}))$ units

22) $x = \frac{2}{3}(y^2 + 1)^{3/2}$ from $y = 1$ to $y = 3$

23) [T] $x = \tan y$ from $y = 0$ to $y = \frac{3}{4}$

Answer

$s \approx 1.201$ units

24) [T] $x = \cos^2 y$ from $y = -\frac{\pi}{2}$ to $y = \frac{\pi}{2}$

25) [T] $x = 4^y$ from $y = 0$ to $y = 2$

Answer

$s \approx 15.2341$ units

26) [T] $x = \ln(y)$ on $y = \dfrac{1}{e}$ to $y = e$

For exercises 27 - 34, find the surface area of the volume generated when the following curves revolve around the x-axis. If you cannot evaluate the integral exactly, use your calculator to approximate it.

27) $y = \sqrt{x}$ from $x = 2$ to $x = 6$

Answer

$A = \frac{49\pi}{3}$ units^2

28) $y = x^3$ from $x = 0$ to $x = 1$

29) $y = 7x$ from $x = -1$ to $x = 1$

Answer

$A = 70\pi\sqrt{2}$ units^2

30) [T] $y = \frac{1}{x^2}$ from $x = 1$ to $x = 3$

31) $y = \sqrt{4 - x^2}$ from $x = 0$ to $x = 2$

Answer

$A = 8\pi$ units^2

32) $y = \sqrt{4 - x^2}$ from $x = -1$ to $x = 1$

33) $y = 5x$ from $x = 1$ to $x = 5$

Answer

$A = 120\pi\sqrt{26}$ units^2

34) [T] $y = \tan x$ from $x = -\frac{\pi}{4}$ to $x = \frac{\pi}{4}$

For exercises 35 - 42, find the surface area of the volume generated when the following curves revolve around the y-axis. If you cannot evaluate the integral exactly, use your calculator to approximate it.

35) $y = x^2$ from $x = 0$ to $x = 2$

Answer

$A = \frac{\pi}{6}(17\sqrt{17} - 1)$ units^2

36) $y = \frac{1}{2}x^2 + \frac{1}{2}$ from $x = 0$ to $x = 1$

37) $y = x + 1$ from $x = 0$ to $x = 3$

Answer

$A = 9\sqrt{2}\pi$ units^2

38) [T] $y = \dfrac{1}{x}$ from $x = \dfrac{1}{2}$ to $x = 1$

39) $y = \sqrt[3]{x}$ from $x = 1$ to $x = 27$

Answer

$A = \frac{10\sqrt{10}\pi}{27}(73\sqrt{73} - 1)$ units^2

40) [T] $y = 3x^4$ from $x = 0$ to $x = 1$

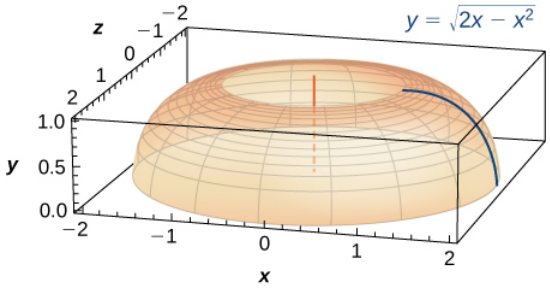
41) [T] $y = \dfrac{1}{\sqrt{x}}$ from $x = 1$ to $x = 3$

Answer

$A \approx 25.645$ units^2

42) [T] $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{2}$

43) The base of a lamp is constructed by revolving a quarter circle $y = \sqrt{2x - x^2}$ around the y-axis from $x = 1$ to $x = 2$, as seen here. Create an integral for the surface area of this curve and compute it.



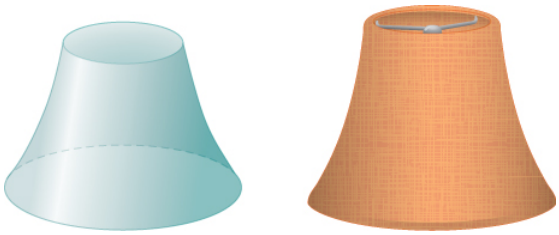
Answer

$A = 2\pi$ units^2

44) A light bulb is a sphere with radius $1/2$ in. with the bottom sliced off to fit exactly onto a cylinder of radius $1/4$ in. and length $1/3$ in., as seen here. The sphere is cut off at the bottom to fit exactly onto the cylinder, so the radius of the cut is $1/4$ in. Find the surface area (not including the top or bottom of the cylinder).



45) [T] A lampshade is constructed by rotating $y = 1/x$ around the x-axis from $y = 1$ to $y = 2$, as seen here. Determine how much material you would need to construct this lampshade—that is, the surface area—accurate to four decimal places.



Answer

10.5017 units^2

46) [T] An anchor drags behind a boat according to the function $y = 24e^{-x/2} - 24$, where y represents the depth beneath the boat and x is the horizontal distance of the anchor from the back of the boat. If the anchor is 23 ft below the boat, how much rope do you have to pull to reach the anchor? Round your answer to three decimal places.

47) [T] You are building a bridge that will span 10 ft. You intend to add decorative rope in the shape of $y = 5|\sin((x\pi)/5)|$, where x is the distance in feet from one end of the bridge. Find out how much rope you need to buy, rounded to the nearest foot.

Answer

23 ft

For exercise 48, find the exact arc length for the following problems over the given interval.

48) $y = \ln(\sin x)$ from $x = \frac{\pi}{4}$ to $x = \frac{3\pi}{4}$. (Hint: Recall trigonometric identities.)

49) Draw graphs of $y = x^2, y = x^6$, and $y = x^{10}$. For $y = x^n$, as n increases, formulate a prediction on the arc length from $(0, 0)$ to $(1, 1)$. Now, compute the lengths of these three functions and determine whether your prediction is correct.

Answer

2

50) Compare the lengths of the parabola $x = y^2$ and the line $x = by$ from $(0, 0)$ to (b^2, b) as b increases. What do you notice?

51) Solve for the length of $x = y^2$ from $(0, 0)$ to $(1, 1)$. Show that $x = \dfrac{y^2}{2}$ from $(0, 0)$ to $(2, 2)$ is twice as long. Graph both functions and explain why this is so.

Answer

Answers may vary

52) [T] Which is longer between $(1, 1)$ and $(2, \frac{1}{2})$: the hyperbola $y = \dfrac{1}{x}$ or the graph of $x + 2y = 3$?

53) Explain why the surface area is infinite when $y = 1/x$ is rotated around the x-axis for $1 \leq x < \infty$, but the volume is finite.

Answer

For more information, look up Gabriel's Horn.

Contributors

Gilbert Strang (MIT) and Edwin "Jed" Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 6.4E: Exercises for Section 6.4 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.5: Physical Applications of Integration

Learning Objectives

- Determine the mass of a one-dimensional object from its linear density function.
- Determine the mass of a two-dimensional circular object from its radial density function.
- Calculate the work done by a variable force acting along a line.
- Calculate the work done in pumping a liquid from one height to another.
- Find the hydrostatic force against a submerged vertical plate.

In this section, we examine some physical applications of integration. Let's begin with a look at calculating mass from a density function. We then turn our attention to work, and close the section with a study of hydrostatic force.

Mass and Density

We can use integration to develop a formula for calculating mass based on a density function. First we consider a thin rod or wire. Orient the rod so it aligns with the x-axis, with the left end of the rod at $x = a$ and the right end of the rod at $x = b$ (Figure 6.5.1). Note that although we depict the rod with some thickness in the figures, for mathematical purposes we assume the rod is thin enough to be treated as a one-dimensional object.

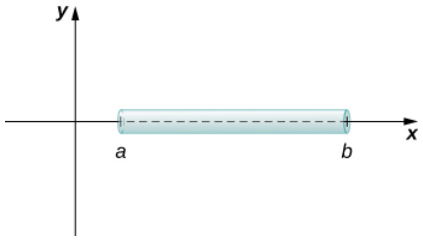


Figure 6.5.1: We can calculate the mass of a thin rod oriented along the x-axis by integrating its density function.

If the rod has constant density ρ, given in terms of mass per unit length, then the mass of the rod is just the product of the density and the length of the rod: $(b-a)\rho$. If the density of the rod is not constant, however, the problem becomes a little more challenging. When the density of the rod varies from point to point, we use a linear density function, $\rho(x)$, to denote the density of the rod at any point, x. Let $\rho(x)$ be an integrable linear density function. Now, for $i = 0, 1, 2, \ldots, n$ let $P = x_i$ be a regular partition of the interval $[a, b]$, and for $i = 1, 2, \ldots, n$ choose an arbitrary point $x_i^* \in [x_{i-1}, x_i]$. Figure 6.5.2 shows a representative segment of the rod.

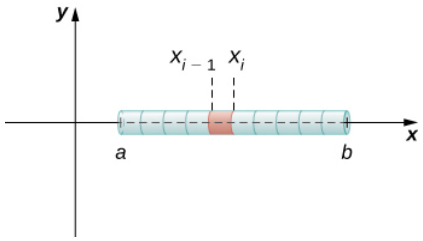


Figure 6.5.2: A representative segment of the rod.

The mass m_i of the segment of the rod from x_{i-1} to x_i is approximated by

$$\begin{aligned} m_i &\approx \rho(x_i^*)(x_i - x_{i-1}) \\ &= \rho(x_i^*)\Delta x. \end{aligned}$$

Adding the masses of all the segments gives us an approximation for the mass of the entire rod:

$$\begin{aligned} m &= \sum_{i=1}^{n} m_i \\ &\approx \sum_{i=1}^{n} \rho(x_i^*)\Delta x. \end{aligned}$$

This is a Riemann sum. Taking the limit as $n \to \infty$, we get an expression for the exact mass of the rod:

$$\begin{aligned} m &= \lim_{n\to\infty} \sum_{i=1}^{n} \rho(x_i^*)\Delta x \\ &= \int_a^b \rho(x)dx. \end{aligned}$$

We state this result in the following theorem.

Mass–Density Formula of a One-Dimensional Object

Given a thin rod oriented along the x-axis over the interval $[a, b]$, let $\rho(x)$ denote a linear density function giving the density of the rod at a point x in the interval. Then the mass of the rod is given by

$$m = \int_a^b \rho(x)dx. \tag{6.5.1}$$

We apply this theorem in the next example.

Example 6.5.1: Calculating Mass from Linear Density

Consider a thin rod oriented on the x-axis over the interval $[\pi/2, \pi]$. If the density of the rod is given by $\rho(x) = \sin x$, what is the mass of the rod?

Solution

Applying Equation 6.5.1 directly, we have

$$\begin{aligned} m &= \int_a^b \rho(x)dx \\ &= \int_{\pi/2}^{\pi} \sin x\, dx \\ &= -\cos x\Big|_{\pi/2}^{\pi} \\ &= 1. \end{aligned}$$

Exercise 6.5.1

Consider a thin rod oriented on the x-axis over the interval $[1, 3]$. If the density of the rod is given by $\rho(x) = 2x^2 + 3$, what is the mass of the rod?

Hint

Use the process from the previous example.

Solution

$70/3$

We now extend this concept to find the mass of a two-dimensional disk of radius r. As with the rod we looked at in the one-dimensional case, here we assume the disk is thin enough that, for mathematical purposes, we can treat it as a two-dimensional object. We assume the density is given in terms of mass per unit area (called **area density**), and further assume the density varies only along the disk's radius (called **radial density**). We orient the disk in the $xy - plane$, with the center at the origin. Then, the density of the disk can be treated as a function of x, denoted $\rho(x)$. We assume $\rho(x)$ is integrable. Because density is a function of x, we partition the interval from $[0, r]$ along the x-axis. For $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of the interval $[0, r]$, and for $i = 1, 2, \ldots, n$, choose an arbitrary point $x_i^* \in [x_{i-1}, x_i]$. Now, use the partition to break up the disk into thin (two-dimensional) washers. A disk and a representative washer are depicted in the following figure.

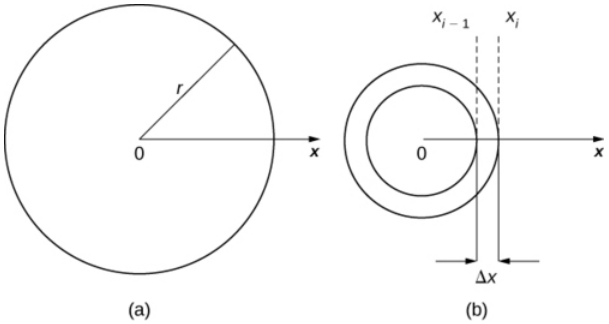


Figure 6.5.3: (a) A thin disk in the *xy*-plane. (b) A representative washer.

We now approximate the density and area of the washer to calculate an approximate mass, m_i. Note that the area of the washer is given by

$$\begin{aligned} A_i &= \pi(x_i)^2 - \pi(x_{i-1})^2 \\ &= \pi[x_i^2 - x_{i-1}^2] \\ &= \pi(x_i + x_{i-1})(x_i - x_{i-1}) \\ &= \pi(x_i + x_{i-1})\Delta x. \end{aligned}$$

You may recall that we had an expression similar to this when we were computing volumes by shells. As we did there, we use $x_i^* \approx (x_i + x_{i-1})/2$ to approximate the average radius of the washer. We obtain

$$A_i = \pi(x_i + x_{i-1})\Delta x \approx 2\pi x_i^* \Delta x.$$

Using $\rho(x_i^*)$ to approximate the density of the washer, we approximate the mass of the washer by

$$m_i \approx 2\pi x_i^* \rho(x_i^*)\Delta x.$$

Adding up the masses of the washers, we see the mass m of the entire disk is approximated by

$$m = \sum_{i=1}^{n} m_i \approx \sum_{i=1}^{n} 2\pi x_i^* \rho(x_i^*)\Delta x.$$

We again recognize this as a Riemann sum, and take the limit as $n \to \infty$. This gives us

$$\begin{aligned} m &= \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi x_i^* \rho(x_i^*)\Delta x \\ &= \int_0^r 2\pi x \rho(x) dx. \end{aligned}$$

We summarize these findings in the following theorem.

Mass–Density Formula of a Circular Object

Let $\rho(x)$ be an integrable function representing the radial density of a disk of radius r. Then the mass of the disk is given by

$$m = \int_0^r 2\pi x \rho(x) dx. \tag{6.5.2}$$

Example 6.5.2: Calculating Mass from Radial Density

Let $\rho(x) = \sqrt{x}$ represent the radial density of a disk. Calculate the mass of a disk of radius 4.

Solution

Applying Equation 6.5.2, we find

$$m = \int_0^r 2\pi x \rho(x) dx$$
$$= \int_0^4 2\pi x\sqrt{x} dx = 2\pi \int_0^4 x^{3/2} dx$$
$$= 2\pi \frac{2}{5} x^{5/2} \,|_0^4 = \frac{4\pi}{5}[32]$$
$$= \frac{128\pi}{5}.$$

? Exercise 6.5.2

Let $\rho(x) = 3x + 2$ represent the radial density of a disk. Calculate the mass of a disk of radius 2.

Hint

Use the process from the previous example.

Solution

24π

Work Done by a Force

We now consider work. In physics, work is related to force, which is often intuitively defined as a push or pull on an object. When a force moves an object, we say the force does work on the object. In other words, work can be thought of as the amount of energy it takes to move an object. According to physics, when we have a constant force, work can be expressed as the product of force and distance.

In the English system, the unit of force is the pound and the unit of distance is the foot, so work is given in foot-pounds. In the metric system, kilograms and meters are used. One newton is the force needed to accelerate 1 kilogram of mass at the rate of 1 m/sec^2. Thus, the most common unit of work is the newton-meter. This same unit is also called the joule. Both are defined as kilograms times meters squared over seconds squared $(kg \cdot m^2/s^2)$.

When we have a constant force, things are pretty easy. It is rare, however, for a force to be constant. The work done to compress (or elongate) a spring, for example, varies depending on how far the spring has already been compressed (or stretched). We look at springs in more detail later in this section.

Suppose we have a variable force $F(x)$ that moves an object in a positive direction along the x-axis from point a to point b. To calculate the work done, we partition the interval $[a, b]$ and estimate the work done over each subinterval. So, for $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of the interval $[a, b]$, and for $i = 1, 2, \ldots, n$, choose an arbitrary point $x_i^* \in [x_{i-1}, x_i]$. To calculate the work done to move an object from point x_{i-1} to point x_i, we assume the force is roughly constant over the interval, and use $F(x_i^*)$ to approximate the force. The work done over the interval $[x_{i-1}, x_i]$, then, is given by

$$W_i \approx F(x_i^*)(x_i - x_{i-1}) = F(x_i^*)\Delta x.$$

Therefore, the work done over the interval $[a, b]$ is approximately

$$W = \sum_{i=1}^{n} W_i \approx \sum_{i=1}^{n} F(x_i^*)\Delta x.$$

Taking the limit of this expression as $n \to \infty$ gives us the exact value for work:

$$W = \lim_{n\to\infty} \sum_{i=1}^{n} F(x_i^*)\Delta x$$
$$= \int_a^b F(x) dx.$$

Thus, we can define work as follows.

Definition: Work

If a variable force $F(x)$ moves an object in a positive direction along the x-axis from point a to point b, then the work done on the object is

$$W = \int_a^b F(x)dx. \tag{6.5.3}$$

Note that if F is constant, the integral evaluates to $F \cdot (b-a) = F \cdot d$, which is the formula we stated at the beginning of this section.

Now let's look at the specific example of the work done to compress or elongate a spring. Consider a block attached to a horizontal spring. The block moves back and forth as the spring stretches and compresses. Although in the real world we would have to account for the force of friction between the block and the surface on which it is resting, we ignore friction here and assume the block is resting on a frictionless surface. When the spring is at its natural length (at rest), the system is said to be at equilibrium. In this state, the spring is neither elongated nor compressed, and in this equilibrium position the block does not move until some force is introduced. We orient the system such that $x = 0$ corresponds to the equilibrium position (Figure 6.5.4).

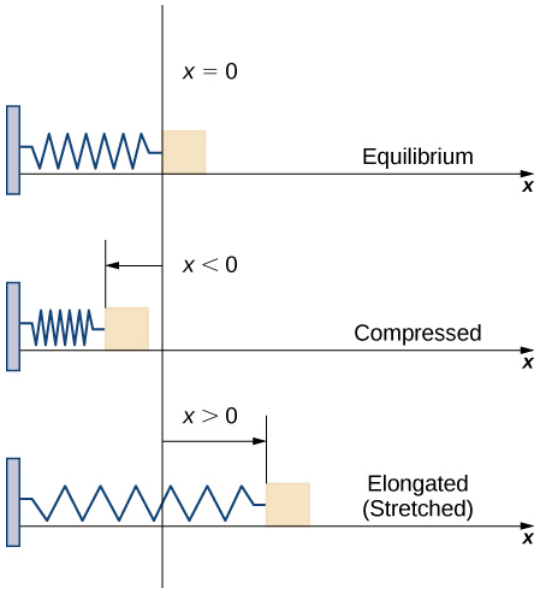


Figure 6.5.4: A block attached to a horizontal spring at equilibrium, compressed, and elongated.

According to Hooke's law, the force required to compress or stretch a spring from an equilibrium position is given by $F(x) = kx$, for some constant k. The value of k depends on the physical characteristics of the spring. The constant k is called the spring constant and is always positive. We can use this information to calculate the work done to compress or elongate a spring, as shown in the following example.

Example 6.5.3: The Work Required to Stretch or Compress a Spring

Suppose it takes a force of 10 N (in the negative direction) to compress a spring 0.2 m from the equilibrium position. How much work is done to stretch the spring 0.5 m from the equilibrium position?

Solution

First find the spring constant, k. When $x = -0.2$, we know $F(x) = -10$, so

$$\begin{aligned} F(x) &= kx \\ -10 &= k(-0.2) \\ k &= 50 \end{aligned}$$

and $F(x) = 50x$. Then, to calculate work, we integrate the force function, obtaining

$$W = \int_a^b F(x)dx$$
$$= \int_0^{0.5} 50x\,dx$$
$$= 25x^2\big|_0^{0.5}$$
$$= 6.25.$$

The work done to stretch the spring is 6.25 J.

? Exercise 6.5.3

Suppose it takes a force of 8 lb to stretch a spring 6 in. from the equilibrium position. How much work is done to stretch the spring 1 ft from the equilibrium position?

Hint

Use the process from the previous example. Be careful with units.

Solution

8 ft-lb

Work Done in Pumping

Consider the work done to pump water (or some other liquid) out of a tank. Pumping problems are a little more complicated than spring problems because many of the calculations depend on the shape and size of the tank. In addition, instead of being concerned about the work done to move a single mass, we are looking at the work done to move a volume of water, and it takes more work to move the water from the bottom of the tank than it does to move the water from the top of the tank.

We examine the process in the context of a cylindrical tank, then look at a couple of examples using tanks of different shapes. Assume a cylindrical tank of radius 4 m and height 10 m is filled to a depth of 8 m. How much work does it take to pump all the water over the top edge of the tank?

The first thing we need to do is define a frame of reference. We let x represent the vertical distance below the top of the tank. That is, we orient the x-axis vertically, with the origin at the top of the tank and the downward direction being positive (Figure $6.5.5$).

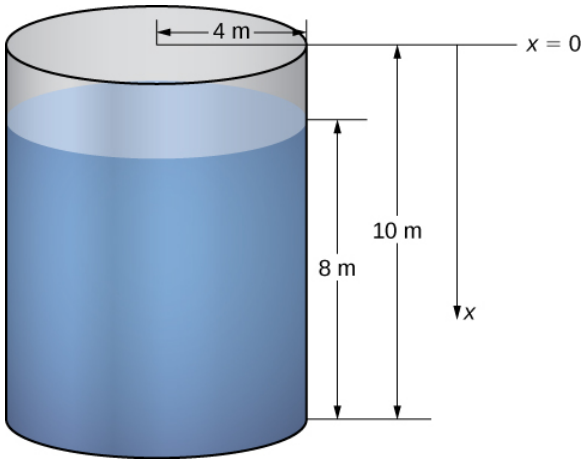


Figure $6.5.5$: How much work is needed to empty a tank partially filled with water?

Using this coordinate system, the water extends from $x = 2$ to $x = 10$. Therefore, we partition the interval $[2, 10]$ and look at the work required to lift each individual "layer" of water. So, for $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of the interval $[2, 10]$, and for $i = 1, 2, \ldots, n$, choose an arbitrary point $x_i^* \in [x_{i-1}, x_i]$. Figure $6.5.6$ shows a representative layer.

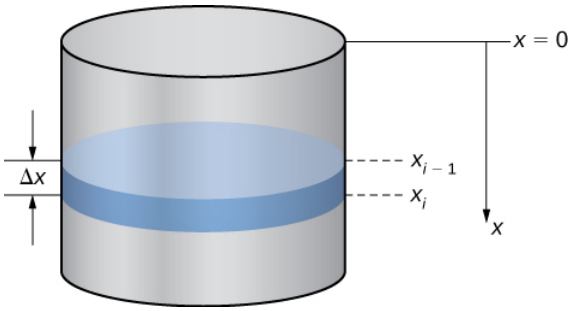


Figure 6.5.6: A representative layer of water.

In pumping problems, the force required to lift the water to the top of the tank is the force required to overcome gravity, so it is equal to the weight of the water. Given that the weight-density of water is $9800\ \text{N/m}^3$, or $62.4\ \text{lb/ft}^3$, calculating the volume of each layer gives us the weight. In this case, we have

$$V = \pi(4)^2\Delta x = 16\pi\Delta x.$$

Then, the force needed to lift each layer is

$$F = 9800 \cdot 16\pi\Delta x = 156,800\pi\Delta x.$$

Note that this step becomes a little more difficult if we have a noncylindrical tank. We look at a noncylindrical tank in the next example.

We also need to know the distance the water must be lifted. Based on our choice of coordinate systems, we can use x_i^* as an approximation of the distance the layer must be lifted. Then the work to lift the ith layer of water W_i is approximately

$$W_i \approx 156,800\pi x_i^*\Delta x.$$

Adding the work for each layer, we see the approximate work to empty the tank is given by

$$\begin{aligned} W &= \sum_{i=1}^{n} W_i \\ &\approx \sum_{i=1}^{n} 156,800\pi x_i^*\Delta x. \end{aligned}$$

This is a Riemann sum, so taking the limit as $n \to \infty$, we get

$$\begin{aligned} W &= \lim_{n\to\infty} \sum_{i=1}^{n} 156,800\pi x_i^*\Delta x \\ &= 156,800\pi \int_2^{10} x dx \\ &= 156,800\pi \left(\frac{x^2}{2}\right)\Big|_2^{10} = 7,526,400\pi \approx 23,644,883. \end{aligned}$$

The work required to empty the tank is approximately 23,650,000 J.

For pumping problems, the calculations vary depending on the shape of the tank or container. The following problem-solving strategy lays out a step-by-step process for solving pumping problems.

Problem-Solving Strategy: Solving Pumping Problems

1. Sketch a picture of the tank and select an appropriate frame of reference.
2. Calculate the volume of a representative layer of water.
3. Multiply the volume by the weight-density of water to get the force.
4. Calculate the distance the layer of water must be lifted.
5. Multiply the force and distance to get an estimate of the work needed to lift the layer of water.

6. Sum the work required to lift all the layers. This expression is an estimate of the work required to pump out the desired amount of water, and it is in the form of a Riemann sum.
7. Take the limit as $n \to \infty$ and evaluate the resulting integral to get the exact work required to pump out the desired amount of water.

We now apply this problem-solving strategy in an example with a noncylindrical tank.

Example 6.5.4: A Pumping Problem with a Noncylindrical Tank

Assume a tank in the shape of an inverted cone, with height 12 ft and base radius 4 ft. The tank is full to start with, and water is pumped over the upper edge of the tank until the height of the water remaining in the tank is 4 ft. How much work is required to pump out that amount of water?

Solution

The tank is depicted in Figure 6.5.7. As we did in the example with the cylindrical tank, we orient the x-axis vertically, with the origin at the top of the tank and the downward direction being positive (step 1).

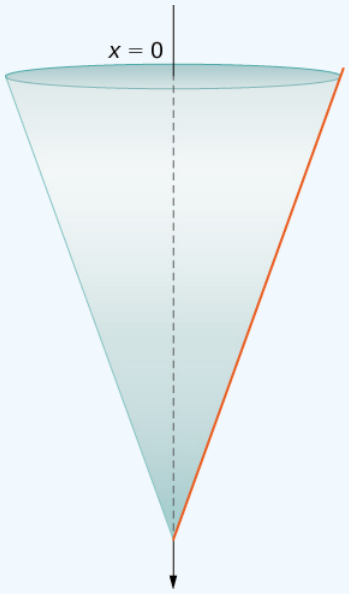


Figure 6.5.7: A water tank in the shape of an inverted cone.

The tank starts out full and ends with 4 ft of water left, so, based on our chosen frame of reference, we need to partition the interval $[0, 8]$. Then, for $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of the interval $[0, 8]$, and for $i = 1, 2, \ldots, n$, choose an arbitrary point $x_i^* \in [x_{i-1}, x_i]$. We can approximate the volume of a layer by using a disk, then use similar triangles to find the radius of the disk (Figure 6.5.8).

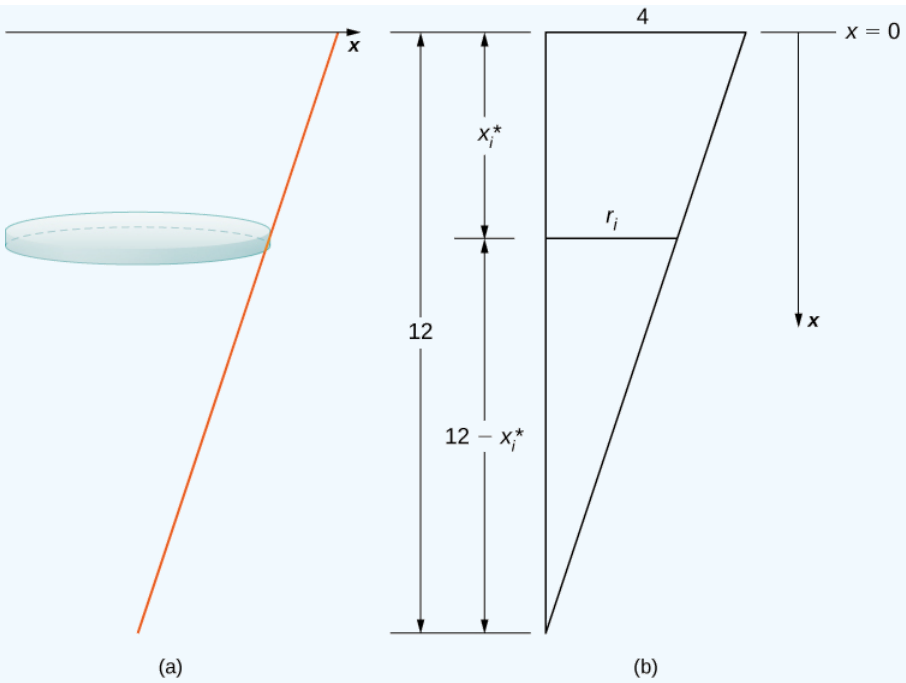


Figure 6.5.8: Using similar triangles to express the radius of a disk of water.

From properties of similar triangles, we have

$$\begin{aligned} \frac{r_i}{12 - x_i^*} &= \frac{4}{12} = \frac{1}{3} \qquad \text{(step 1)} \\ 3r_i &= 12 - x_i^* \\ r_i &= \frac{12 - x_i^*}{3} \\ &= 4 - \frac{x_i^*}{3}. \end{aligned}$$

Then the volume of the disk is

$$V_i = \pi \left(4 - \frac{x_i^*}{3}\right)^2 \Delta x. \qquad \text{(step 2)}$$

The weight-density of water is 62.4lb/ft^3, so the force needed to lift each layer is approximately

$$F_i \approx 62.4\pi \left(4 - \frac{x_i^*}{3}\right)^2 \Delta x \qquad \text{(step 3)}$$

Based on the diagram, the distance the water must be lifted is approximately x_i^* feet (step 4), so the approximate work needed to lift the layer is

$$W_i \approx 62.4\pi x_i^* \left(4 - \frac{x_i^*}{3}\right)^2 \Delta x. \qquad \text{(step 5)}$$

Summing the work required to lift all the layers, we get an approximate value of the total work:

$$W = \sum_{i=1}^{n} W_i \approx \sum_{i=1}^{n} 62.4\pi x_i^* \left(4 - \frac{x_i^*}{3}\right)^2 \Delta x. \qquad \text{(step 6)}$$

Taking the limit as $n \to \infty$, we obtain

$$W = \lim_{n\to\infty} \sum_{i=1}^{n} 62.4\pi x_i^*(4 - \frac{x_i^*}{3})^2 \Delta x$$

$$= \int_0^8 62.4\pi x \left(4 - \frac{x}{3}\right)^2 dx$$

$$= 62.4\pi \int_0^8 x\left(16 - \frac{8x}{3} + \frac{x^2}{9}\right) dx = 62.4\pi \int_0^8 \left(16x - \frac{8x^2}{3} + \frac{x^3}{9}\right) dx$$

$$= 62.4\pi \left[8x^2 - \frac{8x^3}{9} + \frac{x^4}{36}\right]\Big|_0^8 = 10,649.6\pi \approx 33,456.7.$$

It takes approximately $33,450$ ft-lb of work to empty the tank to the desired level.

? Exercise 6.5.4

A tank is in the shape of an inverted cone, with height 10 ft and base radius 6 ft. The tank is filled to a depth of 8 ft to start with, and water is pumped over the upper edge of the tank until 3 ft of water remain in the tank. How much work is required to pump out that amount of water?

Hint

Use the process from the previous example.

Solution

Approximately $43,255.2$ ft-lb

Hydrostatic Force and Pressure

In this last section, we look at the force and pressure exerted on an object submerged in a liquid. In the English system, force is measured in pounds. In the metric system, it is measured in newtons. Pressure is force per unit area, so in the English system we have pounds per square foot (or, perhaps more commonly, pounds per square inch, denoted psi). In the metric system we have newtons per square meter, also called **pascals.**

Let's begin with the simple case of a plate of area A submerged horizontally in water at a depth s (Figure $6.5.9$). Then, the force exerted on the plate is simply the weight of the water above it, which is given by $F = \rho As$, where ρ is the weight density of water (weight per unit volume). To find the **hydrostatic pressure**—that is, the pressure exerted by water on a submerged object—we divide the force by the area. So the pressure is $p = F/A = \rho s$.

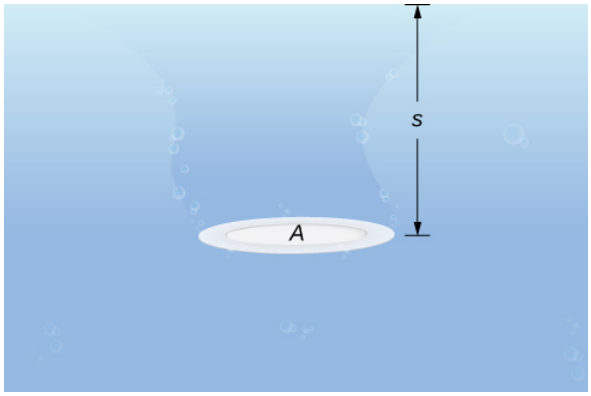


Figure $6.5.9$: A plate submerged horizontally in water.

By Pascal's principle, the pressure at a given depth is the same in all directions, so it does not matter if the plate is submerged horizontally or vertically. So, as long as we know the depth, we know the pressure. We can apply Pascal's principle to find the force exerted on surfaces, such as dams, that are oriented vertically. We cannot apply the formula $F = \rho As$ directly, because the depth varies from point to point on a vertically oriented surface. So, as we have done many times before, we form a partition, a Riemann sum, and, ultimately, a definite integral to calculate the force.

Suppose a thin plate is submerged in water. We choose our frame of reference such that the x-axis is oriented vertically, with the downward direction being positive, and point $x = 0$ corresponding to a logical reference point. Let $s(x)$ denote the depth at point x. Note we often let $x = 0$ correspond to the surface of the water. In this case, depth at any point is simply given by $s(x) = x$. However, in some cases we may want to select a different reference point for $x = 0$, so we proceed with the development in the more general case. Last, let $w(x)$ denote the width of the plate at the point x.

Assume the top edge of the plate is at point $x = a$ and the bottom edge of the plate is at point $x = b$. Then, for $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of the interval $[a, b]$, and for $i = 1, 2, \ldots, n$, choose an arbitrary point $x_i^* \in [x_{i-1}, x_i]$. The partition divides the plate into several thin, rectangular strips (Figure 6.5.10).

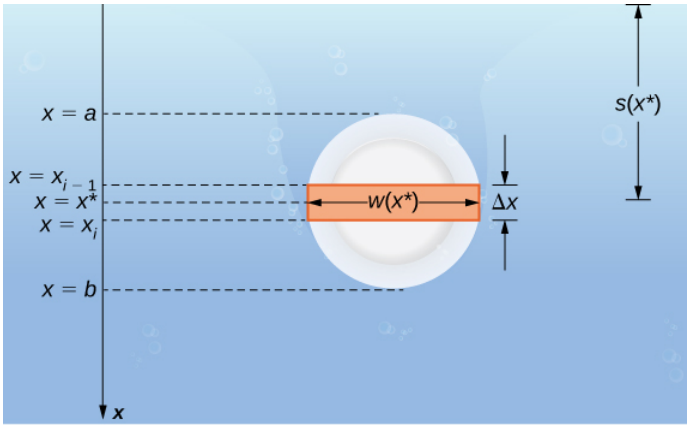


Figure 6.5.10: A thin plate submerged vertically in water.

Let's now estimate the force on a representative strip. If the strip is thin enough, we can treat it as if it is at a constant depth, $s(x_i^*)$. We then have

$$F_i = \rho A s = \rho[w(x_i^*)\Delta x]s(x_i^*).$$

Adding the forces, we get an estimate for the force on the plate:

$$F \approx \sum_{i=1}^{n} F_i = \sum_{i=1}^{n} \rho[w(x_i^*)\Delta x]s(x_i^*).$$

This is a Riemann sum, so taking the limit gives us the exact force. We obtain

$$F = \lim_{n\to\infty} \sum_{i=1}^{n} \rho[w(x_i^*)\Delta x]s(x_i^*) = \int_a^b \rho w(x)s(x)dx. \tag{6.5.4}$$

Evaluating this integral gives us the force on the plate. We summarize this in the following problem-solving strategy.

Problem-Solving Strategy: Finding Hydrostatic Force

1. Sketch a picture and select an appropriate frame of reference. (Note that if we select a frame of reference other than the one used earlier, we may have to adjust Equation 6.5.4 accordingly.)
2. Determine the depth and width functions, $s(x)$ and $w(x)$.
3. Determine the weight-density of whatever liquid with which you are working. The weight-density of water is $62.4\,\text{lb/ft}^3$, or $9800\,\text{N/m}^3$.
4. Use the equation to calculate the total force.

Example 6.5.5: Finding Hydrostatic Force

A water trough 15 ft long has ends shaped like inverted isosceles triangles, with base 8 ft and height 3 ft. Find the force on one end of the trough if the trough is full of water.

Solution

Figure 6.5.11 shows the trough and a more detailed view of one end.

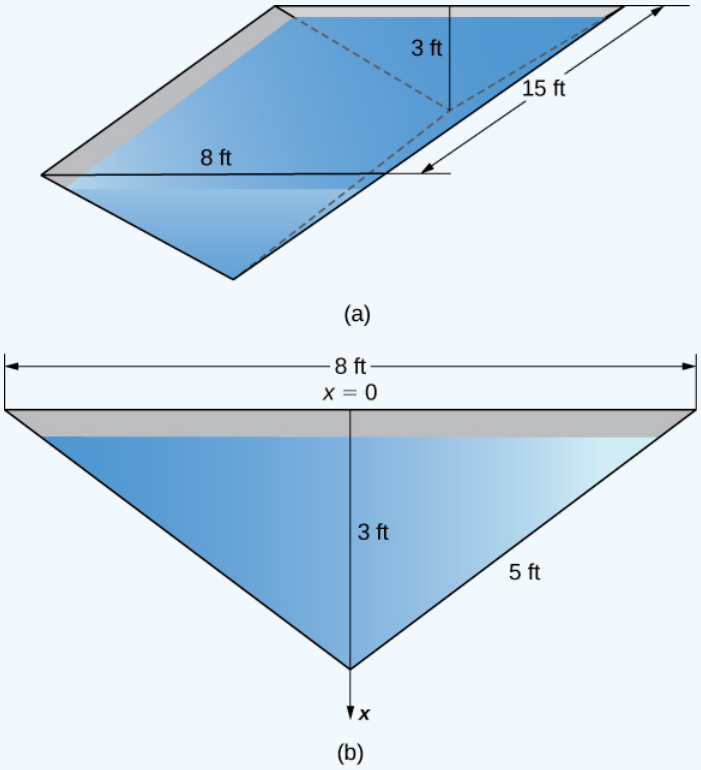


Figure 6.5.11: (a) A water trough with a triangular cross-section. (b) Dimensions of one end of the water trough.

Select a frame of reference with the x-axis oriented vertically and the downward direction being positive. Select the top of the trough as the point corresponding to $x = 0$ (step 1). The depth function, then, is $s(x) = x$. Using similar triangles, we see that $w(x) = 8 - (8/3)x$ (step 2). Now, the weight density of water is $62.4\,\text{lb/ft}^3$ (step 3), so applying Equation 6.5.4, we obtain

$$F = \int_a^b \rho w(x) s(x) dx$$

$$= \int_0^3 62.4 \left(8 - \frac{8}{3}x\right) x\, dx = 62.4 \int_0^3 \left(8x - \frac{8}{3}x^2\right) dx$$

$$= 62.4 \left[4x^2 - \frac{8}{9}x^3\right]\Big|_0^3 = 748.8.$$

The water exerts a force of 748.8 lb on the end of the trough (step 4).

? Exercise 6.5.5

A water trough 12 m long has ends shaped like inverted isosceles triangles, with base 6 m and height 4 m. Find the force on one end of the trough if the trough is full of water.

Hint

Follow the problem-solving strategy and the process from the previous example.

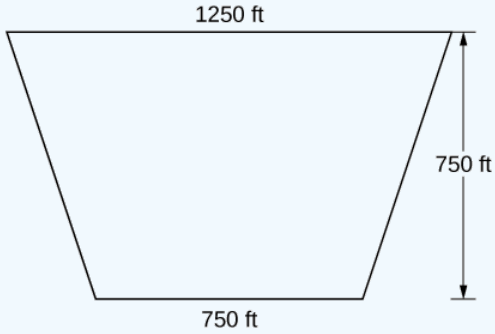
Solution

$156,800$ N

Example 6.5.6: Finding Hydrostatic Force

We now return our attention to the Hoover Dam, mentioned at the beginning of this chapter. The actual dam is arched, rather than flat, but we are going to make some simplifying assumptions to help us with the calculations. Assume the face of the Hoover Dam is shaped like an isosceles trapezoid with lower base 750 ft, upper base 1250 ft, and height 750 ft (see the following figure).





When the reservoir is full, Lake Mead's maximum depth is about 530 ft, and the surface of the lake is about 10 ft below the top of the dam (see the following figure).

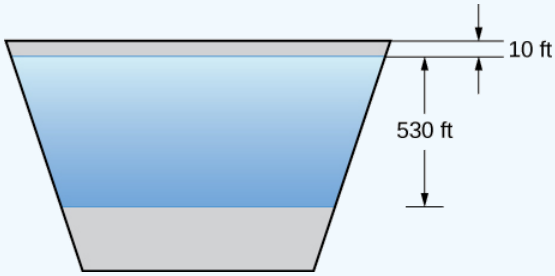


Figure 6.5.12: A simplified model of the Hoover Dam with assumed dimensions.

a. Find the force on the face of the dam when the reservoir is full.
b. The southwest United States has been experiencing a drought, and the surface of Lake Mead is about 125 ft below where it would be if the reservoir were full. What is the force on the face of the dam under these circumstances?

Solution:

a.

We begin by establishing a frame of reference. As usual, we choose to orient the x-axis vertically, with the downward direction being positive. This time, however, we are going to let $x = 0$ represent the top of the dam, rather than the surface of the water. When the reservoir is full, the surface of the water is 10 ft below the top of the dam, so $s(x) = x - 10$ (see the following figure).

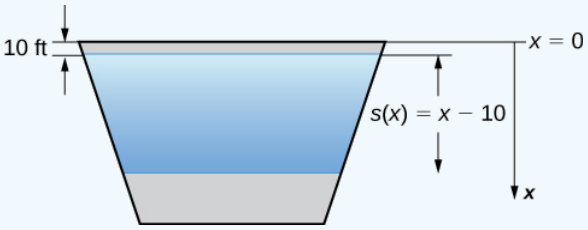


Figure 6.5.13: We first choose a frame of reference.

To find the width function, we again turn to similar triangles as shown in the figure below.

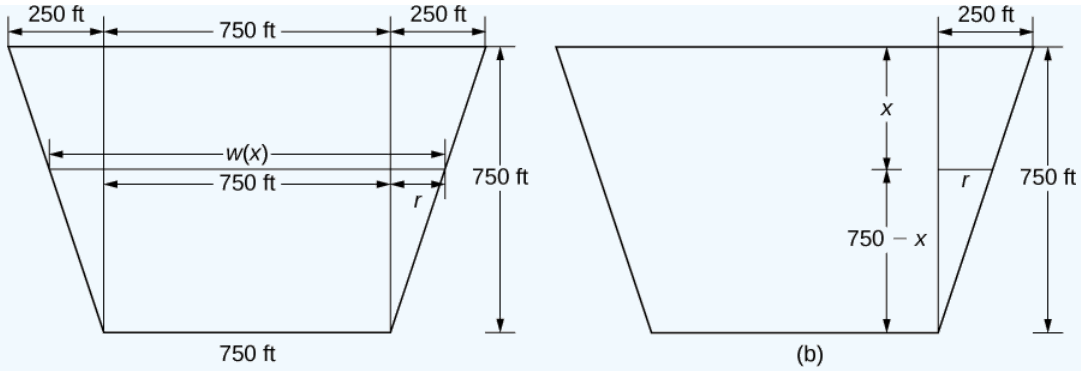


Figure 6.5.14: We use similar triangles to determine a function for the width of the dam. (a) Assumed dimensions of the dam; (b) highlighting the similar triangles.

From the figure, we see that $w(x) = 750 + 2r$. Using properties of similar triangles, we get $r = 250 - (1/3)x$. Thus,

$$w(x) = 1250 - \frac{2}{3}x \qquad \text{(step 2)}$$

Using a weight-density of 62.4lb/ft^3 (step 3) and applying Equation 6.5.4, we get

$$\begin{aligned} F &= \int_a^b \rho w(x) s(x)\, dx \\ &= \int_{10}^{540} 62.4 \left(1250 - \frac{2}{3}x\right)(x - 10)\, dx \\ &= 62.4 \int_{10}^{540} -\frac{2}{3}[x^2 - 1885x + 18750]\, dx \\ &= -62.4\left(\frac{2}{3}\right)\left[\frac{x^3}{3} - \frac{1885x^2}{2} + 18750x\right]\Big|_{10}^{540} \approx 8,832,245,000\,\text{lb} = 4,416,122.5\,\text{t}. \end{aligned}$$

Note the change from pounds to tons ($2000\text{lb} = 1$ ton) (step 4). This changes our depth function, $s(x)$, and our limits of integration. We have $s(x) = x - 135$. The lower limit of integration is 135. The upper limit remains 540. Evaluating the integral, we get

$$\begin{aligned} F &= \int_a^b \rho w(x) s(x)\, dx \\ &= \int_{135}^{540} 62.4 \left(1250 - \frac{2}{3}x\right)(x - 135)\, dx \\ &= -62.4(\tfrac{2}{3}) \int_{135}^{540} (x - 1875)(x - 135)\, dx = -62.4\left(\frac{2}{3}\right) \int_{135}^{540} (x^2 - 2010x + 253125)\, dx \\ &= -62.4\left(\frac{2}{3}\right)\left[\frac{x^3}{3} - 1005x^2 + 253125x\right]\Big|_{135}^{540} \approx 5,015,230,000\,\text{lb} = 2,507,615\,\text{t}. \end{aligned}$$

Exercise 6.5.6

When the reservoir is at its average level, the surface of the water is about 50 ft below where it would be if the reservoir were full. What is the force on the face of the dam under these circumstances?

Hint

Change the depth function, $s(x)$, and the limits of integration.

Solution

Approximately 7,164,520,000 lb or 3,582,260 t

Key Concepts

- Several physical applications of the definite integral are common in engineering and physics.
- Definite integrals can be used to determine the mass of an object if its density function is known.
- Work can also be calculated from integrating a force function, or when counteracting the force of gravity, as in a pumping problem.
- Definite integrals can also be used to calculate the force exerted on an object submerged in a liquid.

Key Equations

- **Mass of a one-dimensional object**

$$m = \int_a^b \rho(x)dx$$

- **Mass of a circular object**

$$m = \int_0^r 2\pi x\rho(x)dx$$

- **Work done on an object**

$$W = \int_a^b F(x)dx$$

- **Hydrostatic force on a plate**

$$F = \int_a^b \rho w(x)s(x)dx$$

Glossary

density function
a density function describes how mass is distributed throughout an object; it can be a linear density, expressed in terms of mass per unit length; an area density, expressed in terms of mass per unit area; or a volume density, expressed in terms of mass per unit volume; weight-density is also used to describe weight (rather than mass) per unit volume

Hooke's law
this law states that the force required to compress (or elongate) a spring is proportional to the distance the spring has been compressed (or stretched) from equilibrium; in other words, $F = kx$, where k is a constant

hydrostatic pressure
the pressure exerted by water on a submerged object

work
the amount of energy it takes to move an object; in physics, when a force is constant, work is expressed as the product of force and distance

---

This page titled 6.5: Physical Applications of Integration is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.5E: Exercises for Section 6.5

Basic Work Problems

For exercises 1 - 6, find the work done.

1) Find the work done when a constant force $F = 12$ lb moves a chair from $x = 0.9$ to $x = 1.1$ ft.

2) How much work is done when a person lifts a 50 lb box of comics onto a truck that is 3 ft off the ground?

Answer

$W = 150$ ft-lb

3) What is the work done lifting a 20 kg child from the floor to a height of 2 m? (Note that 1 kg equates to 9.8 N)

4) Find the work done when you push a box along the floor 2 m, when you apply a constant force of $F = 100$ N.

Answer

$W = 200$ J

5) Compute the work done for a force $F = \dfrac{12}{x^2}$ N from $x = 1$ to $x = 2$ m.

6) What is the work done moving a particle from $x = 0$ to $x = 1$ m if the force acting on it is $F = 3x^2$ N?

Answer

$W = 1$ J

Density Problems

In exercises 7 - 11, find the mass of the one-dimensional object.

7) A wire that is 2 ft long (starting at $x = 0$) and has a density function of $\rho(x) = x^2 + 2x$ lb/ft

8) A car antenna that is 3 ft long (starting at $x = 0$) and has a density function of $\rho(x) = 3x + 2$ lb/ft

Answer

$\frac{39}{2}$

9) A metal rod that is 8 in. long (starting at $x = 0$) and has a density function of $\rho(x) = e^{1/2x}$ lb/in.

10) A pencil that is 4 in. long (starting at $x = 2$) and has a density function of $\rho(x) = \dfrac{5}{x}$ oz/in.

Answer

$\ln(243)$

11) A ruler that is 12 in. long (starting at $x = 5$) and has a density function of $\rho(x) = \ln(x) + (1/2)x^2$ oz/in.

In exercises 12 - 16, find the mass of the two-dimensional object that is centered at the origin.

12) An oversized hockey puck of radius 2 in. with density function $\rho(x) = x^3 - 2x + 5$

Answer

$\frac{332\pi}{15}$

13) A frisbee of radius 6 in. with density function $\rho(x) = e^{-x}$

14) A plate of radius 10 in. with density function $\rho(x) = 1 + \cos(\pi x)$

Answer

100π

15) A jar lid of radius 3 in. with density function $\rho(x) = \ln(x+1)$

16) A disk of radius 5 cm with density function $\rho(x) = \sqrt{3x}$

Answer

$20\pi\sqrt{15}$

Spring Work Problems

17) A 12-in. spring is stretched to 15 in. by a force of 75 lb. What is the spring constant?

18) A spring has a natural length of 10 cm. It takes 2 J to stretch the spring to 15 cm. How much work would it take to stretch the spring from 15 cm to 20 cm?

Answer

$W = 6$ J

19) A 1-m spring requires 10 J to stretch the spring to 1.1 m. How much work would it take to stretch the spring from 1 m to 1.2 m?

20) A spring requires 5 J to stretch the spring from 8 cm to 12 cm, and an additional 4 J to stretch the spring from 12 cm to 14 cm. What is the natural length of the spring?

Answer

The natural length is5 cm.

21) A shock absorber is compressed 1 in. by a weight of 1 ton. What is the spring constant?

22) A force of $F = (20x - x^3)$ N stretches a nonlinear spring by x meters. What work is required to stretch the spring from $x = 0$ to $x = 2$ m?

Answer

$W = 36$ J

Cable and Chain Work Problems

23) Find the work done by winding up a hanging cable of length 100 ft and weight-density 5 lb/ft.

24) For the cable in the preceding exercise, how much work is done to lift the cable 50 ft?

Answer

$W = 18,750$ ft-lb

25) For the cable in the preceding exercise, how much additional work is done by hanging a 200 lb weight at the end of the cable?

Pyramid & Satellite/Rocket Work Problems

26) [T] A pyramid of height 500 ft has a square base 800 ft by 800 ft. Find the area A at height h. If the rock used to build the pyramid weighs approximately $w = 100\,\text{lb/ft}^3$, how much work did it take to lift all the rock?

Answer

$W = \frac{32}{3} \times 10^9$ ft-lb

27) [T] For the pyramid in the preceding exercise, assume there were 1000 workers each working 10 hours a day, 5 days a week, 50 weeks a year. If each of the workers, on average, lifted ten 100-lb rocks 2 ft/hr, how long did it take to build the pyramid?

28) [T] The force of gravity on a mass m is $F = -((GMm)/x^2)$ newtons. For a rocket of mass $m = 1000$ kg, compute the work to lift the rocket from $x = 6400$ to $x = 6500$ km. (Note: $G = 6 \times 10^{-17}$ N m^2/kg^2 and $M = 6 \times 10^{24}$ kg.)

Answer

$W = 8.65 \times 10^5$ J

29) [T] For the rocket in the preceding exercise, find the work to lift the rocket from $x = 6400$ to $x = \infty$.

Hydrostatic Force and Pressure

30) [T] A rectangular dam is 40 ft high and 60 ft wide. Compute the total force F on the dam when

a. the surface of the water is at the top of the dam and

b. the surface of the water is halfway down the dam.

Answer

a. $3,000,000$ lb,
b. $749,000$ lb

Pumping Work Problems

31) [T] Find the work required to pump all the water out of a cylinder that has a circular base of radius 5ft and height 200 ft. Use the fact that the density of water is 62 lb/ft^3.

32) [T] Find the work required to pump all the water out of the cylinder in the preceding exercise if the cylinder is only half full.

Answer

$W = 23.25\pi$ million ft-lb

33) [T] How much work is required to pump out a swimming pool if the area of the base is $800\,\text{ft}^2$, the water is 4 ft deep, and the top is 1 ft above the water level? Assume that the density of water is 62 lb/ft^3.

34) A cylinder of depth H and cross-sectional area A stands full of water at density ρ. Compute the work to pump all the water to the top.

Answer

$$W = \frac{A\rho H^2}{2}$$

35) For the cylinder in the preceding exercise, compute the work to pump all the water to the top if the cylinder is only half full.

36) A cone-shaped tank has a cross-sectional area that increases with its depth: $A = \frac{\pi r^2 h^2}{H^3}$. Show that the work to empty it is half the work for a cylinder with the same height and base.

Answer

Answers may vary.

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 6.5E: Exercises for Section 6.5 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.6: Moments and Centers of Mass

Learning Objectives

- Find the center of mass of objects distributed along a line.
- Locate the center of mass of a thin plate.
- Use symmetry to help locate the centroid of a thin plate.
- Apply the theorem of Pappus for volume.

In this section, we consider centers of mass (also called *centroids*, under certain conditions) and moments. The basic idea of the center of mass is the notion of a balancing point. Many of us have seen performers who spin plates on the ends of sticks. The performers try to keep several of them spinning without allowing any of them to drop. If we look at a single plate (without spinning it), there is a sweet spot on the plate where it balances perfectly on the stick. If we put the stick anywhere other than that sweet spot, the plate does not balance and it falls to the ground. (That is why performers spin the plates; the spin helps keep the plates from falling even if the stick is not exactly in the right place.) Mathematically, that sweet spot is called the *center of mass of the plate.*

In this section, we first examine these concepts in a one-dimensional context, then expand our development to consider centers of mass of two-dimensional regions and symmetry. Last, we use centroids to find the volume of certain solids by applying the theorem of Pappus.

Center of Mass and Moments

Let's begin by looking at the center of mass in a one-dimensional context. Consider a long, thin wire or rod of negligible mass resting on a fulcrum, as shown in Figure 6.6.1a. Now suppose we place objects having masses m_1 and m_2 at distances d_1 and d_2 from the fulcrum, respectively, as shown in Figure 6.6.1b.

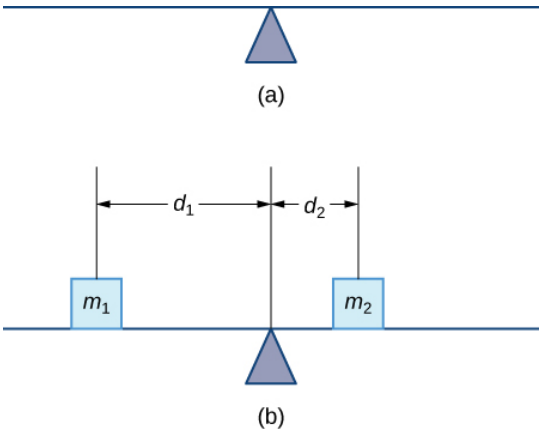


Figure 6.6.1: (a) A thin rod rests on a fulcrum. (b) Masses are placed on the rod.

The most common real-life example of a system like this is a playground seesaw, or teeter-totter, with children of different weights sitting at different distances from the center. On a seesaw, if one child sits at each end, the heavier child sinks down and the lighter child is lifted into the air. If the heavier child slides in toward the center, though, the seesaw balances. Applying this concept to the masses on the rod, we note that the masses balance each other if and only if

$$m_1 d_1 = m_2 d_2.$$

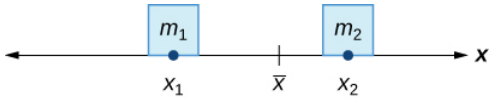


Figure 6.6.2: The center of mass $\bar{x}$ is the balance point of the system.

In the seesaw example, we balanced the system by moving the masses (children) with respect to the fulcrum. However, we are really interested in systems in which the masses are not allowed to move, and instead we balance the system by moving the

fulcrum. Suppose we have two point masses, m_1 and m_2, located on a number line at points x_1 and x_2, respectively (Figure 6.6.2). The center of mass, $\bar{x}$, is the point where the fulcrum should be placed to make the system balance.

Thus, we have

$$\begin{aligned} m_1|x_1 - \bar{x}| &= m_2|x_2 - \bar{x}| \\ m_1(\bar{x} - x_1) &= m_2(x_2 - \bar{x}) \\ m_1\bar{x} - m_1x_1 &= m_2x_2 - m_2\bar{x} \\ \bar{x}(m_1 + m_2) &= m_1x_1 + m_2x_2 \end{aligned}$$

or

$$\bar{x} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} \tag{6.6.1}$$

The expression in the numerator of Equation 6.6.1, $m_1x_1 + m_2x_2$, is called the *first moment of the system* with respect to the origin. If the context is clear, we often drop the word first and just refer to this expression as the **moment** of the system. The expression in the denominator, $m_1 + m_2$, is the total mass of the system. Thus, the **center of mass** of the system is the point at which the total mass of the system could be concentrated without changing the moment.

This idea is not limited just to two point masses. In general, if n masses, $m_1, m_2, \ldots, m_n$, are placed on a number line at points $x_1, x_2, \ldots, x_n$, respectively, then the center of mass of the system is given by

$$\bar{x} = \frac{\sum_{i=1}^{n} m_ix_i}{\sum_{i=1}^{n} m_i}$$

Center of Mass of Objects on a Line

Let $m_1, m_2, \ldots, m_n$ be point masses placed on a number line at points $x_1, x_2, \ldots, x_n$, respectively, and let $m = \sum_{i=1}^{n} m_i$ denote the total mass of the system. Then, the *moment of the system* with respect to the origin is given by

$$M = \sum_{i=1}^{n} m_ix_i \tag{6.6.2}$$

and the *center of mass of the system* is given by

$$\bar{x} = \frac{M}{m}. \tag{6.6.3}$$

We apply this theorem in the following example.

Example 6.6.1: Finding the Center of Mass of Objects along a Line

Suppose four point masses are placed on a number line as follows:

- $m_1 = 30\,kg$, placed at $x_1 = -2m$
- $m_2 = 5\,kg$, placed at $x_2 = 3m$
- $m_3 = 10\,kg$, placed at $x_3 = 6m$
- $m_4 = 15\,kg$, placed at $x_4 = -3m$.

Solution

Find the moment of the system with respect to the origin and find the center of mass of the system.

First, we need to calculate the moment of the system (Equation 6.6.2):

$$\begin{aligned} M &= \sum_{i=1}^{4} m_i x_i \\ &= -60 + 15 + 60 - 45 \\ &= -30. \end{aligned}$$

Now, to find the center of mass, we need the total mass of the system:

$$\begin{aligned} m &= \sum_{i=1}^{4} m_i \\ &= 30 + 5 + 10 + 15 \\ &= 60\, kg \end{aligned}$$

Then we have (from Equation 6.6.3)

$$\bar{x} = \frac{M}{m} = -\frac{30}{60} = -\frac{1}{2}.$$

The center of mass is located 1/2 m to the left of the origin.

? Exercise 6.6.1

Suppose four point masses are placed on a number line as follows:

- $m_1 = 12\, kg$ placed at $x_1 = -4m$
- $m_2 = 12\, kg$ placed at $x_2 = 4m$
- $m_3 = 30\, kg$ placed at $x_3 = 2m$
- $m_4 = 6\, kg$, placed at $x_4 = -6m$.

Find the moment of the system with respect to the origin and find the center of mass of the system.

Hint

Use the process from the previous example.

Answer

$M = 24, \bar{x} = \frac{2}{5}m$

We can generalize this concept to find the center of mass of a system of point masses in a plane. Let m_1 be a point mass located at point (x_1, y_1) in the plane. Then the moment M_x of the mass with respect to the x-axis is given by $M_x = m_1 y_1$. Similarly, the moment M_y with respect to the y-axis is given by

$$M_y = m_1 x_1.$$

Notice that the x-coordinate of the point is used to calculate the moment with respect to the y-axis, and vice versa. The reason is that the x-coordinate gives the distance from the point mass to the y-axis, and the y-coordinate gives the distance to the x-axis (see the following figure).

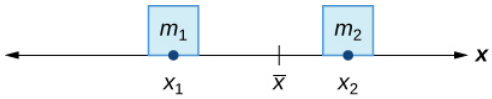


Figure 6.6.3: Point mass m_1 is located at point (x_1, y_1) in the plane.

If we have several point masses in the xy-plane, we can use the moments with respect to the x- and y-axes to calculate the x- and y-coordinates of the center of mass of the system.

Center of Mass of Objects in a Plane

Let m_1, m_2, …, m_n be point masses located in the xy-plane at points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, respectively, and let $m = \sum_{i=1}^{n} m_i$ denote the total mass of the system. Then the moments M_x and M_y of the system with respect to the x- and y-axes, respectively, are given by

$$M_x = \sum_{i=1}^{n} m_i y_i \tag{6.6.4}$$

and

$$M_y = \sum_{i=1}^{n} m_i x_i. \tag{6.6.5}$$

Also, the coordinates of the center of mass $(\bar{x}, \bar{y})$ of the system are

$$\bar{x} = \frac{M_y}{m} \tag{6.6.6}$$

and

$$\bar{y} = \frac{M_x}{m}. \tag{6.6.7}$$

The next example demonstrates how to the center of mass formulas (Equations 6.6.4 - 6.6.7) may be applied.

Example 6.6.2: Finding the Center of Mass of Objects in a Plane

Suppose three point masses are placed in the xy-plane as follows (assume coordinates are given in meters):

- $m_1 = 2\,kg$ placed at $(-1, 3)$,
- $m_2 = 6\,kg$ placed at $(1, 1)$,
- $m_3 = 4\,kg$ placed at $(2, -2)$.

Find the center of mass of the system.

Solution

First we calculate the total mass of the system:

$$m = \sum_{i=1}^{3} m_i = 2 + 6 + 4 = 12\,kg.$$

Next we find the moments with respect to the x- and y-axes:

$$M_y = \sum_{i=1}^{3} m_i x_i = -2 + 6 + 8 = 12,$$

$$M_x = \sum_{i=1}^{3} m_i y_i = 6 + 6 - 8 = 4.$$

Then we have

$$\bar{x} = \frac{M_y}{m} = \frac{12}{12} = 1$$

and

$$\bar{y} = \frac{M_x}{m} = \frac{4}{12} = \frac{1}{3}.$$

The center of mass of the system is $(1, 1/3)$, in meters.

Exercise 6.6.2

Suppose three point masses are placed on a number line as follows (assume coordinates are given in meters):

- $m_1 = 5\,kg$, placed at $(-2, -3)$,
- $m_2 = 3\,kg$, placed at $(2, 3)$,
- $m_3 = 2\,kg$, placed at $(-3, -2)$.

Find the center of mass of the system.

Hint

Use the process from the previous example.

Answer

$(-1, -1)$ m

Center of Mass of Thin Plates

So far we have looked at systems of point masses on a line and in a plane. Now, instead of having the mass of a system concentrated at discrete points, we want to look at systems in which the mass of the system is distributed continuously across a thin sheet of material. For our purposes, we assume the sheet is thin enough that it can be treated as if it is two-dimensional. Such a sheet is called a **lamina**. Next we develop techniques to find the center of mass of a lamina. In this section, we also assume the density of the lamina is constant.

Laminas are often represented by a two-dimensional region in a plane. The geometric center of such a region is called its **centroid**. Since we have assumed the density of the lamina is constant, the center of mass of the lamina depends only on the shape of the corresponding region in the plane; it does not depend on the density. In this case, the center of mass of the lamina corresponds to the centroid of the delineated region in the plane. As with systems of point masses, we need to find the total mass of the lamina, as well as the moments of the lamina with respect to the x- and y-axes.

We first consider a lamina in the shape of a rectangle. Recall that the center of mass of a lamina is the point where the lamina balances. For a rectangle, that point is both the horizontal and vertical center of the rectangle. Based on this understanding, it is clear that the center of mass of a rectangular lamina is the point where the diagonals intersect, which is a result of the **symmetry principle**, and it is stated here without proof.

The Symmetry Principle

If a region R is symmetric about a line l, then the centroid of R lies on l.

Let's turn to more general laminas. Suppose we have a lamina bounded above by the graph of a continuous function $f(x)$, below by the x-axis, and on the left and right by the lines $x = a$ and $x = b$, respectively, as shown in the following figure.

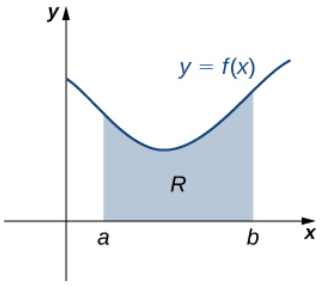


Figure 6.6.4: A region in the plane representing a lamina.

As with systems of point masses, to find the center of mass of the lamina, we need to find the total mass of the lamina, as well as the moments of the lamina with respect to the x- and y-axes. As we have done many times before, we approximate these quantities by partitioning the interval $[a, b]$ and constructing rectangles.

For $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of $[a, b]$. Recall that we can choose any point within the interval $[x_{i-1}, x_i]$ as our x_i^*. In this case, we want x_i^* to be the x-coordinate of the centroid of our rectangles. Thus, for $i = 1, 2, \ldots, n$, we select $x_i^* \in [x_{i-1}, x_i]$ such that x_i^* is the midpoint of the interval. That is, $x_i^* = (x_{i-1} + x_i)/2$. Now, for $i = 1, 2, \ldots, n$, construct a rectangle of height $f(x_i^*)$ on $[x_{i-1}, x_i]$. The center of mass of this rectangle is $(x_i^*, (f(x_i^*))/2)$, as shown in the following figure.

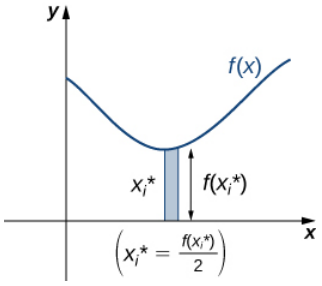


Figure 6.6.5: A representative rectangle of the lamina.

Next, we need to find the total mass of the rectangle. Let ρ represent the density of the lamina (note that ρ is a constant). In this case, ρ is expressed in terms of mass per unit area. Thus, to find the total mass of the rectangle, we multiply the area of the rectangle by ρ. Then, the mass of the rectangle is given by $\rho f(x_i^*)\Delta x$.

To get the approximate mass of the lamina, we add the masses of all the rectangles to get

$$m \approx \sum_{i=1}^{n} \rho f(x_i^*)\Delta x. \tag{6.6.8}$$

Equation 6.6.8 is a Riemann sum. Taking the limit as $n \to \infty$ gives the exact mass of the lamina:

$$\begin{aligned} m &= \lim_{n\to\infty} \sum_{i=1}^{n} \rho f(x_i^*)\Delta x \\ &= \rho \int_a^b f(x)dx. \end{aligned}$$

Next, we calculate the moment of the lamina with respect to the x-axis. Returning to the representative rectangle, recall its center of mass is $(x_i^*, (f(x_i^*))/2)$. Recall also that treating the rectangle as if it is a point mass located at the center of mass does not change the moment. Thus, the moment of the rectangle with respect to the x-axis is given by the mass of the rectangle, $\rho f(x_i^*)\Delta x$, multiplied by the distance from the center of mass to the x-axis: $(f(x_i^*))/2$. Therefore, the moment with respect to the x-axis of the rectangle is $\rho([f(x_i^*)]^2/2)\Delta x$. Adding the moments of the rectangles and taking the limit of the resulting Riemann sum, we see that the moment of the lamina with respect to the x-axis is

$$\begin{aligned} M_x &= \lim_{n\to\infty} \sum_{i=1}^{n} \rho \frac{[f(x_i^*)]^2}{2}\Delta x \\ &= \rho \int_a^b \frac{[f(x)]^2}{2}dx. \end{aligned}$$

We derive the moment with respect to the y-axis similarly, noting that the distance from the center of mass of the rectangle to the y-axis is x_i^*. Then the moment of the lamina with respect to the y-axis is given by

$$\begin{aligned} M_y &= \lim_{n\to\infty} \sum_{i=1}^{n} \rho x_i^* f(x^*)i)\Delta x \\ &= \rho \int_a^b x f(x)dx. \end{aligned}$$

We find the coordinates of the center of mass by dividing the moments by the total mass to give $\bar{x} = M_y/m$ and $\bar{y} = M_x/m$. If we look closely at the expressions for M_x, M_y, and m, we notice that the constant ρ cancels out when $\bar{x}$ and $\bar{y}$ are calculated.

We summarize these findings in the following theorem.

Center of Mass of a Thin Plate in the xy-Plane

Let R denote a region bounded above by the graph of a continuous function $f(x)$, below by the x-axis, and on the left and right by the lines $x = a$ and $x = b$, respectively. Let ρ denote the density of the associated lamina. Then we can make the following statements:

i. The mass of the lamina is

$$m = \rho \int_a^b f(x)dx. \tag{6.6.9}$$

ii. The moments M_x and M_y of the lamina with respect to the x- and y-axes, respectively, are

$$M_x = \rho \int_a^b \frac{[f(x)]^2}{2} dx \tag{6.6.10}$$

and

$$M_y = \rho \int_a^b x f(x)dx. \tag{6.6.11}$$

iii. The coordinates of the center of mass $(\bar{x}, \bar{y})$ are

$$\bar{x} = \frac{M_y}{m} \tag{6.6.12}$$

and

$$\bar{y} = \frac{M_x}{m}. \tag{6.6.13}$$

In the next example, we use this theorem to find the center of mass of a lamina.

Example 6.6.3: Finding the Center of Mass of a Lamina

Let R be the region bounded above by the graph of the function $f(x) = \sqrt{x}$ and below by the x-axis over the interval $[0, 4]$. Find the centroid of the region.

Solution

The region is depicted in the following figure.

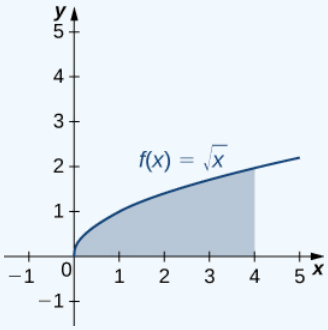


Figure 6.6.6: Finding the center of mass of a lamina.

Since we are only asked for the centroid of the region, rather than the mass or moments of the associated lamina, we know the density constant ρ cancels out of the calculations eventually. Therefore, for the sake of convenience, let's assume $\rho = 1$.

First, we need to calculate the total mass (Equation 6.6.9):

$$
\begin{aligned}
m &= \rho \int_a^b f(x)dx \\
&= \int_0^4 \sqrt{x}dx \\
&= \frac{2}{3}x^{3/2} \,|_0^4 \\
&= \frac{2}{3}[8-0] \\
&= \frac{16}{3}.
\end{aligned}
$$

Next, we compute the moments (Equation 6.6.12):

$$
\begin{aligned}
M_x &= \rho \int_a^b \frac{[f(x)]^2}{2}dx \\
&= \int_0^4 \frac{x}{2}dx \\
&= \frac{1}{4}x^2 \,|_0^4 \\
&= 4
\end{aligned}
$$

and (Equation 6.6.11):

$$
\begin{aligned}
M_y &= \rho \int_a^b xf(x)dx \\
&= \int_0^4 x\sqrt{x}dx \\
&= \int_0^4 x^{3/2}dx \\
&= \frac{2}{5}x^{5/2} \,|_0^4 \\
&= \frac{2}{5}[32-0] \\
&= \frac{64}{5}.
\end{aligned}
$$

Thus, we have (Equation 6.6.12):

$$
\begin{aligned}
\bar{x} &= \frac{M_y}{m} \\
&= \frac{64/5}{16/3} \\
&= \frac{64}{5} \cdot \frac{3}{16} \\
&= \frac{12}{5}
\end{aligned}
$$

and (Equation 6.6.13):

$$
\begin{aligned}
\bar{y} &= \frac{M_x}{y} \\
&= \frac{4}{16/3} \\
&= 4 \cdot \frac{3}{16} \\
&= \frac{3}{4}.
\end{aligned}
$$

The centroid of the region is $(12/5, 3/4)$.

Exercise 6.6.3

Let R be the region bounded above by the graph of the function $f(x) = x^2$ and below by the x-axis over the interval $[0, 2]$. Find the centroid of the region.

Hint

Use the process from the previous example.

Answer

The centroid of the region is $(3/2, 6/5)$.

We can adapt this approach to find centroids of more complex regions as well. Suppose our region is bounded above by the graph of a continuous function $f(x)$, as before, but now, instead of having the lower bound for the region be the x-axis, suppose the region is bounded below by the graph of a second continuous function, $g(x)$, as shown in Figure 6.6.7.

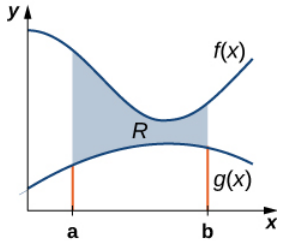


Figure 6.6.7: A region between two functions.

Again, we partition the interval $[a, b]$ and construct rectangles. A representative rectangle is shown in Figure 6.6.8.

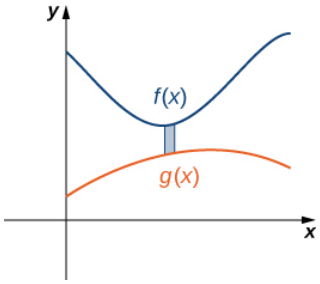


Figure 6.6.8: A representative rectangle of the region between two functions.

Note that the centroid of this rectangle is $(x_i^*, (f(x_i^*) + g(x_i^*))/2)$. We won't go through all the details of the Riemann sum development, but let's look at some of the key steps. In the development of the formulas for the mass of the lamina and the moment with respect to the y-axis, the height of each rectangle is given by $f(x_i^*) - g(x_i^*)$, which leads to the expression $f(x) - g(x)$ in the integrands.

In the development of the formula for the moment with respect to the x-axis, the moment of each rectangle is found by multiplying the area of the rectangle, $\rho[f(x_i^*) - g(x_i^*)]\Delta x$, by the distance of the centroid from the x-axis, $(f(x_i^*) + g(x_i^*))/2$, which gives $\rho(1/2)[f(x_i^*)]^2 - [g(x_i^*)]^2\Delta x$. Summarizing these findings, we arrive at the following theorem.

Center of Mass of a Lamina Bounded by Two Functions

Let R denote a region bounded above by the graph of a continuous function $f(x)$, below by the graph of the continuous function $g(x)$, and on the left and right by the lines $x = a$ and $x = b$, respectively. Let ρ denote the density of the associated lamina. Then we can make the following statements:

i. The mass of the lamina is

$$m = \rho \int_a^b [f(x) - g(x)]dx.$$

ii. The moments M_x and M_y of the lamina with respect to the x- and y-axes, respectively, are

$$M_x = \rho \int_a^b \frac{1}{2}([f(x)]^2 - [g(x)]^2)dx$$

and

$$M_y = \rho \int_a^b x[f(x) - g(x)]dx.$$

iii. The coordinates of the center of mass $\bar{x}, \bar{y})$ are

$$\bar{x} = \frac{M_y}{m}$$

and

$$\bar{y} = \frac{M_x}{m}$$

We illustrate this theorem in the following example.

Example 6.6.4: Finding the Centroid of a Region Bounded by Two Functions

Let *R* be the region bounded above by the graph of the function $f(x) = 1 - x^2$ and below by the graph of the function $g(x) = x - 1$. Find the centroid of the region.

Solution

The region is depicted in the following figure.

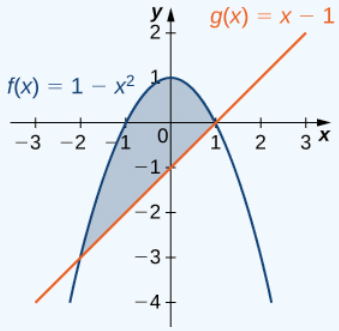


Figure 6.6.9: Finding the centroid of a region between two curves.

The graphs of the functions intersect at $(-2, -3)$ and $(1, 0)$, so we integrate from −2 to 1. Once again, for the sake of convenience, assume $\rho = 1$.

First, we need to calculate the total mass:

$$\begin{aligned} m &= \rho \int_a^b [f(x) - g(x)]dx \\ &= \int_{-2}^1 [1 - x^2 - (x - 1)]dx \\ &= \int_{-2}^1 (2 - x^2 - x)dx \\ &= \left[2x - \frac{1}{3}x^3 - \frac{1}{2}x^2\right] |_{-2}^1 \\ &= \left[2 - \frac{1}{3} - \frac{1}{2}\right] - \left[-4 + \frac{8}{3} - 2\right] \\ &= \frac{9}{2}. \end{aligned}$$

Next, we compute the moments:

$$
\begin{aligned}
M_x &= \rho \int_a^b \frac{1}{2}([f(x)]^2 - [g(x)]^2)dx \\
&= \frac{1}{2}\int_{-2}^{1}((1-x^2)^2 - (x-1)^2)dx \\
&= \frac{1}{2}\int_{-2}^{1}(x^4 - 3x^2 + 2x)dx \\
&= \frac{1}{2}\left[\frac{x^5}{5} - x^3 + x^2\right]|_{-2}^{1} \\
&= -\frac{27}{10}
\end{aligned}
$$

and

$$
\begin{aligned}
M_y &= \rho \int_a^b x[f(x) - g(x)]dx \\
&= \int_{-2}^{1} x[(1-x^2) - (x-1)]dx \\
&= \int_{-2}^{1} x[2 - x^2 - x]dx \\
&= \int_{-2}^{1} (2x - x^3 - x^2)dx \\
&= \left[x^2 - \frac{x^4}{4} - \frac{x^3}{3}\right]|_{-2}^{1} \\
&= -\frac{9}{4}.
\end{aligned}
$$

Therefore, we have

$$
\begin{aligned}
\bar{x} &= \frac{M_y}{m} \\
&= -\frac{9}{4}\cdot\frac{2}{9} \\
&= -\frac{1}{2}
\end{aligned}
$$

and

$$
\begin{aligned}
\bar{y} &= \frac{M_x}{y} \\
&= -\frac{27}{10}\cdot\frac{2}{9} \\
&= -\frac{3}{5}.
\end{aligned}
$$

The centroid of the region is $(-(1/2), -(3/5))$.

Exercise 6.6.4

Let R be the region bounded above by the graph of the function $f(x) = 6 - x^2$ and below by the graph of the function $g(x) = 3 - 2x$. Find the centroid of the region.

Hint

Use the process from the previous example.

Answer

The centroid of the region is $(1, 13/5)$.

The Symmetry Principle

We stated the symmetry principle earlier, when we were looking at the centroid of a rectangle. The symmetry principle can be a great help when finding centroids of regions that are symmetric. Consider the following example.

Example 6.6.5: Finding the Centroid of a Symmetric Region

Let R be the region bounded above by the graph of the function $f(x) = 4 - x^2$ and below by the x-axis. Find the centroid of the region.

Solution

The region is depicted in the following figure

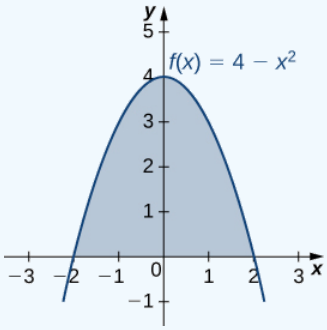


Figure 6.6.10: We can use the symmetry principle to help find the centroid of a symmetric region.

The region is symmetric with respect to the *y*-axis. Therefore, the x-coordinate of the centroid is zero. We need only calculate $\bar{y}$. Once again, for the sake of convenience, assume $\rho = 1$.

First, we calculate the total mass:

$$\begin{aligned} m &= \rho \int_a^b f(x)dx \\ &= \int_{-2}^{2} (4 - x^2)dx \\ &= \left[4x - \frac{x^3}{3}\right] |_{-2}^{2} \\ &= \frac{32}{3}. \end{aligned}$$

Next, we calculate the moments. We only need M_x:

$$\begin{aligned} M_x &= \rho \int_a^b \frac{[f(x)]^2}{2} dx \\ &= \frac{1}{2} \int_{-2}^{2} \left[4 - x^2\right]^2 dx = \frac{1}{2} \int_{-2}^{2} (16 - 8x^2 + x^4)dx \\ &= \frac{1}{2} \left[\frac{x^5}{5} - \frac{8x^3}{3} + 16x\right] |_{-2}^{2} = \frac{256}{15} \end{aligned}$$

Then we have

$$\bar{y} = \frac{M_x}{y} = \frac{256}{15} \cdot \frac{3}{32} = \frac{8}{5}.$$

The centroid of the region is $(0, 8/5)$.

Exercise 6.6.5

Let R be the region bounded above by the graph of the function $f(x) = 1 - x^2$ and below by x-axis. Find the centroid of the region.

Hint

Use the process from the previous example.

Answer

The centroid of the region is $(0, 2/5)$.

The Grand Canyon Skywalk

The Grand Canyon Skywalk opened to the public on March 28, 2007. This engineering marvel is a horseshoe-shaped observation platform suspended 4000 ft above the Colorado River on the West Rim of the Grand Canyon. Its crystal-clear glass floor allows stunning views of the canyon below (see the following figure).



Figure 6.6.11: The Grand Canyon Skywalk offers magnificent views of the canyon. (credit: 10da_ralta, Wikimedia Commons)

The Skywalk is a cantilever design, meaning that the observation platform extends over the rim of the canyon, with no visible means of support below it. Despite the lack of visible support posts or struts, cantilever structures are engineered to be very stable and the Skywalk is no exception. The observation platform is attached firmly to support posts that extend 46 ft down into bedrock. The structure was built to withstand 100-mph winds and an 8.0-magnitude earthquake within 50 mi, and is capable of supporting more than 70,000,000 lb.

One factor affecting the stability of the Skywalk is the center of gravity of the structure. We are going to calculate the center of gravity of the Skywalk, and examine how the center of gravity changes when tourists walk out onto the observation platform.

The observation platform is U-shaped. The legs of the U are 10 ft wide and begin on land, under the visitors' center, 48 ft from the edge of the canyon. The platform extends 70 ft over the edge of the canyon.

To calculate the center of mass of the structure, we treat it as a lamina and use a two-dimensional region in the xy-plane to represent the platform. We begin by dividing the region into three subregions so we can consider each subregion separately. The first region, denoted R_1, consists of the curved part of the U. We model R_1 as a semicircular annulus, with inner radius 25 ft and outer radius 35 ft, centered at the origin (Figure 6.6.12).

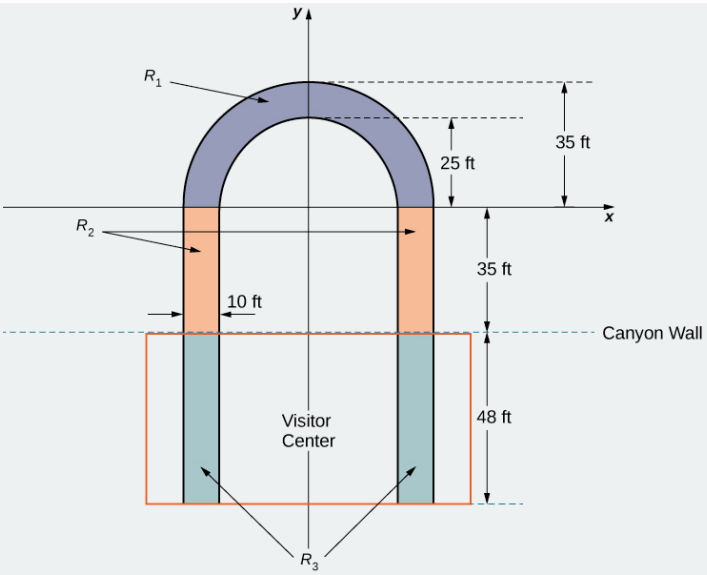


Figure 6.6.12: We model the Skywalk with three sub-regions.

The legs of the platform, extending 35 ft between R_1 and the canyon wall, comprise the second sub-region, R_2. Last, the ends of the legs, which extend 48 ft under the visitor center, comprise the third sub-region, R_3. Assume the density of the lamina is constant and assume the total weight of the platform is 1,200,000 lb (not including the weight of the visitor center; we will consider that later). Use $g = 32\ ft/sec^2$.

1. Compute the area of each of the three sub-regions. Note that the areas of regions R_2 and R_3 should include the areas of the legs only, not the open space between them. Round answers to the nearest square foot.
2. Determine the mass associated with each of the three sub-regions.
3. Calculate the center of mass of each of the three sub-regions.
4. Now, treat each of the three sub-regions as a point mass located at the center of mass of the corresponding sub-region. Using this representation, calculate the center of mass of the entire platform.
5. Assume the visitor center weighs 2,200,000 lb, with a center of mass corresponding to the center of mass of R_3.Treating the visitor center as a point mass, recalculate the center of mass of the system. How does the center of mass change?
6. Although the Skywalk was built to limit the number of people on the observation platform to 120, the platform is capable of supporting up to 800 people weighing 200 lb each. If all 800 people were allowed on the platform, and all of them went to the farthest end of the platform, how would the center of gravity of the system be affected? (Include the visitor center in the calculations and represent the people by a point mass located at the farthest edge of the platform, 70 ft from the canyon wall.)

Theorem of Pappus

This section ends with a discussion of the *theorem of Pappus for volume*, which allows us to find the volume of particular kinds of solids by using the centroid. (There is also a theorem of Pappus for surface area, but it is much less useful than the theorem for volume.)

Theorem of Pappus for Volume

Let R be a region in the plane and let l be a line in the plane that does not intersect R. Then the volume of the solid of revolution formed by revolving R around l is equal to the area of R multiplied by the distance d traveled by the centroid of R.

Proof

We can prove the case when the region is bounded above by the graph of a function $f(x)$ and below by the graph of a function $g(x)$ over an interval $[a, b]$, and for which the axis of revolution is the y-axis. In this case, the area of the region is $A = \int_a^b [f(x)$

$-g(x)]\,dx$. Since the axis of rotation is the y-axis, the distance traveled by the centroid of the region depends only on the x-coordinate of the centroid, $\bar{x}$, which is

$$x = \frac{M_y}{m},$$

where

$$m = \rho \int_a^b [f(x) - g(x)]dx$$

and

$$M_y = \rho \int_a^b x[f(x) - g(x)]dx.$$

Then,

$$d = 2\pi \frac{\rho \int_a^b x[f(x) - g(x)]dx}{\rho \int_a^b [f(x) - g(x)]dx}$$

and thus

$$d \cdot A = 2\pi \int_a^b x[f(x) - g(x)]dx.$$

However, using the method of cylindrical shells, we have

$$V = 2\pi \int_a^b x[f(x) - g(x)]dx.$$

So,

$$V = d \cdot A$$

and the proof is complete.

□

Example 6.6.6: Using the Theorem of Pappus for Volume

Let R be a circle of radius 2 centered at $(4, 0)$. Use the theorem of Pappus for volume to find the volume of the torus generated by revolving R around the y-axis.

Solution

The region and torus are depicted in the following figure.

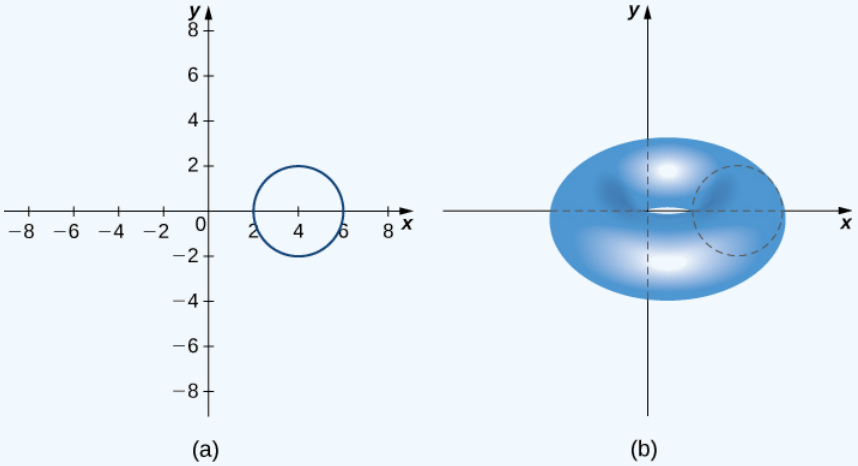


Figure 6.6.13: Determining the volume of a torus by using the theorem of Pappus. (a) A circular region R in the plane; (b) the torus generated by revolving R about the y-axis.

The region R is a circle of radius 2, so the area of R is $A = 4\pi$ units^2. By the symmetry principle, the centroid of R is the center of the circle. The centroid travels around the y-axis in a circular path of radius 4, so the centroid travels $d = 8\pi$ units. Then, the volume of the torus is $A \cdot d = 32\pi^2$ units^3.

? Exercise 6.6.6

Let R be a circle of radius 1 centered at $(3, 0)$. Use the theorem of Pappus for volume to find the volume of the torus generated by revolving R around the y-axis.

Hint

Use the process from the previous example.

Answer

$6\pi^2$ units^3

Key Concepts

- Mathematically, the center of mass of a system is the point at which the total mass of the system could be concentrated without changing the moment. Loosely speaking, the center of mass can be thought of as the balancing point of the system.
- For point masses distributed along a number line, the moment of the system with respect to the origin is $M = \sum_{i=1}^{n} m_i x_i$. For point masses distributed in a plane, the moments of the system with respect to the x- and y-axes, respectively, are $M_x = \sum_{i=1}^{n} m_i y_i$ and $M_y = \sum_{i=}^{n} m_i x_i$, respectively.
- For a lamina bounded above by a function $f(x)$, the moments of the system with respect to the x- and y-axes, respectively, are $M_x = \rho \int_a^b \frac{[f(x)]^2}{2}\,dx$ and $M_y = \rho \int_a^b x f(x)\,dx$.
- The x- and y-coordinates of the center of mass can be found by dividing the moments around the y-axis and around the x-axis, respectively, by the total mass. The symmetry principle says that if a region is symmetric with respect to a line, then the centroid of the region lies on the line.
- The theorem of Pappus for volume says that if a region is revolved around an external axis, the volume of the resulting solid is equal to the area of the region multiplied by the distance traveled by the centroid of the region.

Key Equations

- **Mass of a lamina**

$$m = \rho \int_a^b f(x)dx$$

- **Moments of a lamina**

$$M_x = \rho \int_a^b \frac{[f(x)]^2}{2}\, dx \text{ and } M_y = \rho \int_a^b x f(x)\, dx$$

- **Center of mass of a lamina**

$$\bar{x} = \frac{M_y}{m} \text{ and } \bar{y} = \frac{M_x}{m}$$

Glossary

center of mass

the point at which the total mass of the system could be concentrated without changing the moment

centroid

the centroid of a region is the geometric center of the region; laminas are often represented by regions in the plane; if the lamina has a constant density, the center of mass of the lamina depends only on the shape of the corresponding planar region; in this case, the center of mass of the lamina corresponds to the centroid of the representative region

lamina

a thin sheet of material; laminas are thin enough that, for mathematical purposes, they can be treated as if they are two-dimensional

moment

if n masses are arranged on a number line, the moment of the system with respect to the origin is given by $M = \sum_{i=1}^{n} m_i x_i$; if, instead, we consider a region in the plane, bounded above by a function $f(x)$ over an interval $[a, b]$, then the moments of the region with respect to the x- and y-axes are given by $M_x = \rho \int_a^b \frac{[f(x)]^2}{2}\, dx$ and $M_y = \rho \int_a^b x f(x)\, dx$, respectively

symmetry principle

the symmetry principle states that if a region R is symmetric about a line I, then the centroid of R lies on I

theorem of Pappus for volume

this theorem states that the volume of a solid of revolution formed by revolving a region around an external axis is equal to the area of the region multiplied by the distance traveled by the centroid of the region

---

This page titled 6.6: Moments and Centers of Mass is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.6E: Exercises for Section 6.6

In exercises 1 - 6, calculate the center of mass for the collection of masses given.

1) $m_1 = 2$ at $x_1 = 1$ and $m_2 = 4$ at $x_2 = 2$

2) $m_1 = 1$ at $x_1 = -1$ and $m_2 = 3$ at $x_2 = 2$

Answer

$x = \frac{5}{4}$

3) $m = 3$ at $x = 0, 1, 2, 6$

4) Unit masses at $(x, y) = (1, 0), (0, 1), (1, 1)$

Answer

$\left(\frac{2}{3}, \frac{2}{3}\right)$

5) $m_1 = 1$ at $(1, 0)$ and $m_2 = 4$ at $(0, 1)$

6) $m_1 = 1$ at $(1, 0)$ and $m_2 = 3$ at $(2, 2)$

Answer

$\left(\frac{7}{4}, \frac{3}{2}\right)$

In exercises 7 - 16, compute the center of mass $\bar{x}$.

7) $\rho = 1$ for $x \in (-1, 3)$

8) $\rho = x^2$ for $x \in (0, L)$

Answer

$$\frac{3L}{4}$$

9) $\rho = 1$ for $x \in (0, 1)$ and $\rho = 2$ for $x \in (1, 2)$

10) $\rho = \sin x$ for $x \in (0, \pi)$

Answer

$\frac{\pi}{2}$

11) $\rho = \cos x$ for $x \in (0, \frac{\pi}{2})$

12) $\rho = e^x$ for $x \in (0, 2)$

Answer

$$\frac{e^2 + 1}{e^2 - 1}$$

13) $\rho = x^3 + xe^{-x}$ for $x \in (0, 1)$

14) $\rho = x \sin x$ for $x \in (0, \pi)$

Answer

$$\frac{\pi^2 - 4}{\pi}$$

15) $\rho = \sqrt{x}$ for $x \in (1, 4)$

16) $\rho = \ln x$ for $x \in (1, e)$

Answer

Answer

$\frac{1}{4}(1+e^2)$

In exercises 17 - 19, compute the center of mass $(\bar{x},\bar{y})$. Use symmetry to help locate the center of mass whenever possible.

17) $\rho = 7$ in the square $0 \le x \le 1,\ 0 \le y \le 1$

18) $\rho = 3$ in the triangle with vertices $(0,0),\ (a,0)$, and $(0,b)$

Answer

$\left(\frac{a}{3},\ \frac{b}{3}\right)$

19) $\rho = 2$ for the region bounded by $y = \cos(x),\ y = -\cos(x),\ x = -\frac{\pi}{2}$, and $x = \frac{\pi}{2}$

In exercises 20 - 26, use a calculator to draw the region, then compute the center of mass $(\bar{x},\bar{y})$. Use symmetry to help locate the center of mass whenever possible.

20) [T] The region bounded by $y = \cos(2x),\ x = -\frac{\pi}{4}$, and $x = \frac{\pi}{4}$

Answer

$\left(0,\ \frac{\pi}{8}\right)$

21) [T] The region between $y = 2x^2,\ y = 0,\ x = 0$, and $x = 1$

22) [T] The region between $y = \frac{5}{4}x^2$ and $y = 5$

Answer

$(0,3)$

23) [T] Region between $y = \sqrt{x},\ y = \ln x,\ x = 1$, and $x = 4$

24) [T] The region bounded by $y = 0$ and $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$

Answer

$\left(0,\ \frac{4}{\pi}\right)$

25) [T] The region bounded by $y = 0,\ x = 0$, and $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$

26) [T] The region bounded by $y = x^2$ and $y = x^4$ in the first quadrant

Answer

$\left(\frac{5}{8},\ \frac{1}{3}\right)$

In exercises 27 - 31, use the theorem of Pappus to determine the volume of the shape.

27) Rotating $y = mx$ around the x-axis between $x = 0$ and $x = 1$

28) Rotating $y = mx$ around the y-axis between $x = 0$ and $x = 1$

Answer

$V = \frac{m\pi}{3}$ units^3

29) A general cone created by rotating a triangle with vertices $(0,0),\ (a,0)$, and $(0,b)$ around the y-axis. Does your answer agree with the volume of a cone?

30) A general cylinder created by rotating a rectangle with vertices $(0,0),\ (a,0),\ (0,b)$, and (a,b) around the y-axis. Does your answer agree with the volume of a cylinder?

Answer

$V = \pi a^2 b$ units^3

31) A sphere created by rotating a semicircle with radius a around the y-axis. Does your answer agree with the volume of a sphere?

In exercises 32 - 36, use a calculator to draw the region enclosed by the curve. Find the area M and the centroid $(\bar{x}, \bar{y})$ for the given shapes. Use symmetry to help locate the center of mass whenever possible.

32) [T] Quarter-circle: $y = \sqrt{1 - x^2},\ y = 0$, and $x = 0$

Answer

$\left(\frac{4}{3\pi}, \frac{4}{3\pi}\right)$

33) [T] Triangle: $y = x,\ y = 2 - x$, and $y = 0$

34) [T] Lens: $y = x^2$ and $y = x$

Answer

$\left(\frac{1}{2}, \frac{2}{5}\right)$

35) [T] Ring: $y^2 + x^2 = 1$ and $y^2 + x^2 = 4$

36) [T] Half-ring: $y^2 + x^2 = 1,\ y^2 + x^2 = 4$, and $y = 0$

Answer

$\left(0, \frac{28}{9\pi}\right)$

37) Find the generalized center of mass in the sliver between $y = x^a$ and $y = x^b$ with $a > b$. Then, use the Pappus theorem to find the volume of the solid generated when revolving around the y-axis.

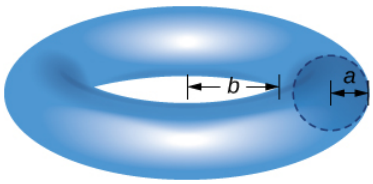
38) Find the generalized center of mass between $y = a^2 - x^2,\ x = 0$, and $y = 0$. Then, use the Pappus theorem to find the volume of the solid generated when revolving around the y-axis.

Answer

Center of mass: $\left(\frac{a}{6}, \frac{4a^2}{5}\right)$,
Volume: $\frac{2\pi a^4}{9}$ units^3

39) Find the generalized center of mass between $y = b\sin(ax),\ x = 0$, and $x = \frac{\pi}{a}$. Then, use the Pappus theorem to find the volume of the solid generated when revolving around the y-axis.

40) Use the theorem of Pappus to find the volume of a torus (pictured here). Assume that a disk of radius a is positioned with the left end of the circle at $x = b$, $b > 0$, and is rotated around the y-axis.



Answer

Volume: $V = 2\pi^2 a^2(b + a)$

41) Find the center of mass $(\bar{x}, \bar{y})$ for a thin wire along the semicircle $y = \sqrt{1 - x^2}$ with unit mass. (Hint: Use the theorem of Pappus.)

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 6.6E: Exercises for Section 6.6 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.7: Integrals, Exponential Functions, and Logarithms

Learning Objectives

- Write the definition of the natural logarithm as an integral.
- Recognize the derivative of the natural logarithm.
- Integrate functions involving the natural logarithmic function.
- Define the number e through an integral.
- Recognize the derivative and integral of the exponential function.
- Prove properties of logarithms and exponential functions using integrals.
- Express general logarithmic and exponential functions in terms of natural logarithms and exponentials.

We already examined exponential functions and logarithms in earlier chapters. However, we glossed over some key details in the previous discussions. For example, we did not study how to treat exponential functions with exponents that are irrational. The definition of the number e is another area where the previous development was somewhat incomplete. We now have the tools to deal with these concepts in a more mathematically rigorous way, and we do so in this section.

For purposes of this section, assume we have not yet defined the natural logarithm, the number e, or any of the integration and differentiation formulas associated with these functions. By the end of the section, we will have studied these concepts in a mathematically rigorous way (and we will see they are consistent with the concepts we learned earlier). We begin the section by defining the natural logarithm in terms of an integral. This definition forms the foundation for the section. From this definition, we derive differentiation formulas, define the number e, and expand these concepts to logarithms and exponential functions of any base.

The Natural Logarithm as an Integral

Recall the power rule for integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1.$$

Clearly, this does not work when $n = -1$, as it would force us to divide by zero. So, what do we do with $\int \frac{1}{x}\,dx$? Recall from the Fundamental Theorem of Calculus that $\int_1^x \frac{1}{t}dt$ is an antiderivative of $\frac{1}{x}$. Therefore, we can make the following definition.

Definition: The Natural Logarithm

For $x > 0$, define the natural logarithm function by

$$\ln x = \int_1^x \frac{1}{t}\,dt.$$

For $x > 1$, this is just the area under the curve $y = \frac{1}{t}$ from 1 to x. For $x < 1$, we have

$$\int_1^x \frac{1}{t}\,dt = -\int_x^1 \frac{1}{t}\,dt,$$

so in this case it is the negative of the area under the curve from x to 1 (see the following figure).

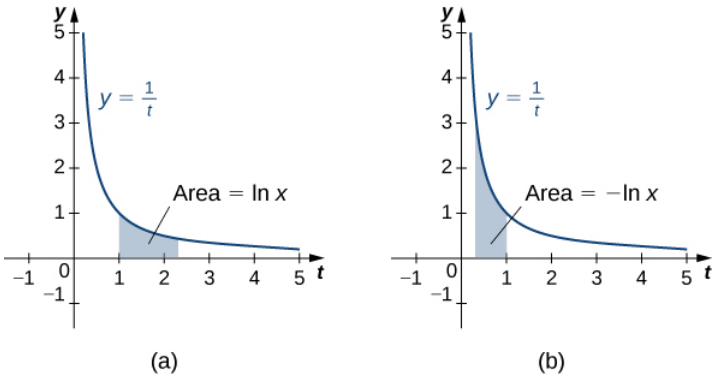


Figure 6.7.1: (a) When $x > 1$, the natural logarithm is the area under the curve $y = 1/t$ from 1 to x. (b) When $x < 1$, the natural logarithm is the negative of the area under the curve from x to 1.

Notice that $\ln 1 = 0$. Furthermore, the function $y = \dfrac{1}{t} > 0$ for $x > 0$. Therefore, by the properties of integrals, it is clear that $\ln x$ is increasing for $x > 0$.

Properties of the Natural Logarithm

Because of the way we defined the natural logarithm, the following differentiation formula falls out immediately as a result of to the Fundamental Theorem of Calculus.

Definition: Derivative of the Natural Logarithm

For $x > 0$, the derivative of the natural logarithm is given by

$$\frac{d}{dx}\left(\ln x\right) = \frac{1}{x}.$$

Corollary to the Derivative of the Natural Logarithm

The function $\ln x$ is differentiable; therefore, it is continuous.

A graph of $\ln x$ is shown in Figure. Notice that it is continuous throughout its domain of $(0, \infty)$.

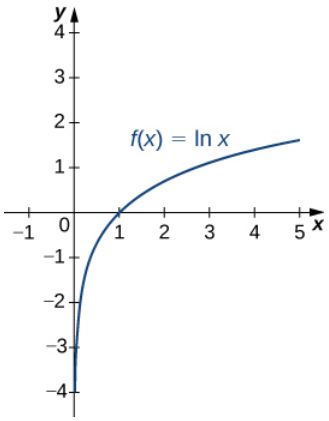


Figure 6.7.2: The graph of $f(x) = \ln x$ shows that it is a continuous function.

Example 6.7.1: Calculating Derivatives of Natural Logarithms

Calculate the following derivatives:

a. $\dfrac{d}{dx}\left(\ln(5x^3 - 2)\right)$
b. $\dfrac{d}{dx}\left((\ln(3x))^2\right)$

Solution

We need to apply the chain rule in both cases.

a. $\frac{d}{dx}\Big(\ln(5x^3-2)\Big) = \frac{15x^2}{5x^3-2}$

b. $\frac{d}{dx}\Big((\ln(3x))^2\Big) = \frac{2(\ln(3x))\cdot 3}{3x} = \frac{2(\ln(3x))}{x}$

Exercise 6.7.1

Calculate the following derivatives:

a. $\frac{d}{dx}\Big(\ln(2x^2+x)\Big)$

b. $\frac{d}{dx}\Big((\ln(x^3))^2\Big)$

Hint

Apply the differentiation formula just provided and use the chain rule as necessary.

Answer

a. $\frac{d}{dx}\Big(\ln(2x^2+x)\Big) = \frac{4x+1}{2x^2+x}$

b. $\frac{d}{dx}\Big((\ln(x^3))^2\Big) = \frac{6\ln(x^3)}{x}$

Note that if we use the absolute value function and create a new function $\ln|x|$, we can extend the domain of the natural logarithm to include $x<0$. Then $\frac{d}{dx}\Big(\ln x\Big) = \frac{1}{x}$. This gives rise to the familiar integration formula.

Integral of $\frac{1}{u}\,du$

The natural logarithm is the antiderivative of the function $f(u)=\frac{1}{u}$:

$$\int \frac{1}{u}\,du = \ln|u| + C.$$

Example 6.7.2: Calculating Integrals Involving Natural Logarithms

Calculate the integral $\int \frac{x}{x^2+4}\,dx$.

Solution

Using u-substitution, let $u=x^2+4$. Then $du = 2x\,dx$ and we have

$$\int \frac{x}{x^2+4}\,dx = \frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln|u| + C = \frac{1}{2}\ln|x^2+4| + C = \frac{1}{2}\ln(x^2+4) + C.$$

Exercise 6.7.2

Calculate the integral $\int \frac{x^2}{x^3+6}\,dx$.

Hint

Apply the integration formula provided earlier and use u-substitution as necessary.

Answer

$$\int \frac{x^2}{x^3+6}\,dx = \frac{1}{3}\ln\,|\,x^3+6\,|\,+C$$

Although we have called our function a "logarithm," we have not actually proved that any of the properties of logarithms hold for this function. We do so here.

Properties of the Natural Logarithm

If $a,\ b > 0$ and r is a rational number, then

i. $\ln 1 = 0$
ii. $\ln(ab) = \ln a + \ln b$
iii. $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$
iv. $\ln\left(a^r\right) = r \ln a$

Proof

i. By definition, $\ln 1 = \int_1^1 \frac{1}{t}\, dt = 0.$

ii. We have

$$\ln(ab) = \int_1^{ab} \frac{1}{t}\, dt = \int_1^a \frac{1}{t}\, dt + \int_a^{ab} \frac{1}{t}\, dt.$$

Use $u - substitution$ on the last integral in this expression. Let $u = t/a$. Then $du = (1/a)dt$. Furthermore, when $t = a,\ u = 1$, and when $t = ab,\ u = b$. So we get

$$\ln(ab) = \int_1^a \frac{1}{t}\, dt + \int_a^{ab} \frac{1}{t}\, dt = \int_1^a \frac{1}{t}\, dt + \int_1^{ab} \frac{a}{t} \cdot \frac{1}{a}\, dt = \int_1^a \frac{1}{t}\, dt + \int_1^b \frac{1}{u}\, du = \ln a + \ln b.$$

iii. Note that

$$\frac{d}{dx}\left(\ln(x^r)\right) = \frac{rx^{r-1}}{x^r} = \frac{r}{x}.$$

Furthermore,

$$\frac{d}{dx}\Big((r \ln x)\Big) = \frac{r}{x}.$$

Since the derivatives of these two functions are the same, by the Fundamental Theorem of Calculus, they must differ by a constant. So we have

$$\ln(x^r) = r \ln x + C$$

for some constant C. Taking $x = 1$, we get

$$\ln(1^r) = r \ln(1) + C$$

$$0 = r(0) + C$$

$$C = 0.$$

Thus $\ln(x^r) = r \ln x$ and the proof is complete. Note that we can extend this property to irrational values of r later in this section.

Part iii. follows from parts ii. and iv. and the proof is left to you.

□

Example 6.7.3: Using Properties of Logarithms

Use properties of logarithms to simplify the following expression into a single logarithm:

$$\ln 9 - 2 \ln 3 + \ln\left(\tfrac{1}{3}\right).$$

Solution

We have

$$\ln 9 - 2\ln 3 + \ln\left(\tfrac{1}{3}\right) = \ln(3^2) - 2\ln 3 + \ln(3^{-1}) = 2\ln 3 - 2\ln 3 - \ln 3 = -\ln 3.$$

Exercise 6.7.3

Use properties of logarithms to simplify the following expression into a single logarithm:

$$\ln 8 - \ln 2 - \ln\left(\tfrac{1}{4}\right)$$

Hint

Apply the properties of logarithms.

Answer

$4\ln 2$

Defining the Number e

Now that we have the natural logarithm defined, we can use that function to define the number e.

Definition: e

The number e is defined to be the real number such that

$$\ln e = 1$$

To put it another way, the area under the curve $y = 1/t$ between $t = 1$ and $t = e$ is 1 (Figure). The proof that such a number exists and is unique is left to you. (Hint: Use the Intermediate Value Theorem to prove existence and the fact that $\ln x$ is increasing to prove uniqueness.)

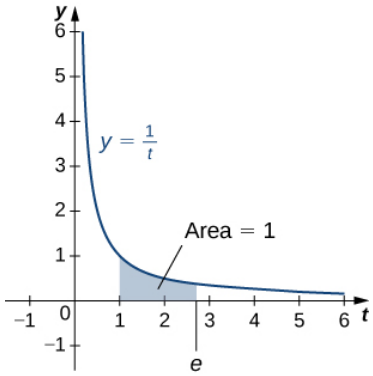


Figure 6.7.3:The area under the curve from 1 to e is equal to one.

The number e can be shown to be *irrational*, although we won't do so here (see the Student Project in Taylor and Maclaurin Series). Its approximate value is given by

$$e \approx 2.71828182846.$$

The Exponential Function

We now turn our attention to the function e^x. Note that the natural logarithm is one-to-one and therefore has an inverse function. For now, we denote this inverse function by $\exp x$. Then,

$$\exp(\ln x) = x$$

for $x > 0$ and

$$\ln(\exp x) = x$$

for all x.

The following figure shows the graphs of $\exp x$ and $\ln x$.

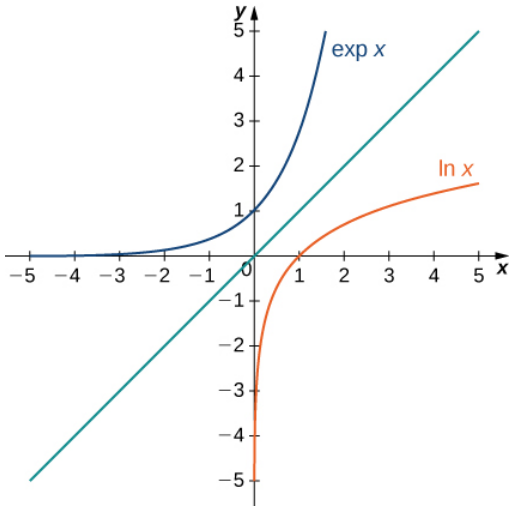


Figure 6.7.4: The graphs of $\ln x$ and $\exp x$.

We hypothesize that $\exp x = e^x$. For rational values of x, this is easy to show. If x is rational, then we have $\ln(e^x) = x \ln e = x$. Thus, when x is rational, $e^x = \exp x$. For irrational values of x, we simply define e^x as the inverse function of $\ln x$.

Definition

For any real number x, define $y = e^x$ to be the number for which

$$\ln y = \ln(e^x) = x.$$

Then we have $e^x = \exp x$ for all x, and thus

$$e^{\ln x} = x \text{ for } x > 0 \text{ and } \ln(e^x) = x$$

for all x.

Properties of the Exponential Function

Since the exponential function was defined in terms of an inverse function, and not in terms of a power of e we must verify that the usual laws of exponents hold for the function e^x.

Properties of the Exponential Function

If p and q are any real numbers and r is a rational number, then

i. $e^p e^q = e^{p+q}$
ii. $\dfrac{e^p}{e^q} = e^{p-q}$
iii. $(e^p)^r = e^{pr}$

Proof

Note that if p and q are rational, the properties hold. However, if p or q are irrational, we must apply the inverse function definition of e^x and verify the properties. Only the first property is verified here; the other two are left to you. We have

$$\ln(e^p e^q) = \ln(e^p) + \ln(e^q) = p + q = \ln(e^{p+q}).$$

Since $\ln x$ is one-to-one, then

$$e^p e^q = e^{p+q}.$$

□

As with part iv. of the logarithm properties, we can extend property iii. to irrational values of r, and we do so by the end of the section.

We also want to verify the differentiation formula for the function $y = e^x$. To do this, we need to use implicit differentiation. Let $y = e^x$. Then

$$\begin{aligned} \ln y &= x \\ \frac{d}{dx}\left(\ln y\right) &= \frac{d}{dx}\left(x\right) \\ \frac{1}{y}\frac{dy}{dx} &= 1 \\ \frac{dy}{dx} &= y. \end{aligned}$$

Thus, we see

$$\frac{d}{dx}\left(e^x\right) = e^x$$

as desired, which leads immediately to the integration formula

$$\int e^x\,dx = e^x + C.$$

We apply these formulas in the following examples.

Example 6.7.4: Using Properties of Exponential Functions

Evaluate the following derivatives:

a. $\frac{d}{dt}\left(e^{3t}e^{t^2}\right)$
b. $\frac{d}{dx}\left(e^{3x^2}\right)$

Solution

We apply the chain rule as necessary.

a. $\frac{d}{dt}\left(e^{3t}e^{t^2}\right) = \frac{d}{dt}\left(e^{3t+t^2}\right) = e^{3t+t^2}(3+2t)$
b. $\frac{d}{dx}\left(e^{3x^2}\right) = e^{3x^2}6x$

Exercise 6.7.4

Evaluate the following derivatives:

a. $\frac{d}{dx}\left(\frac{e^{x^2}}{e^{5x}}\right)$
b. $\frac{d}{dt}\left((e^{2t})^3\right)$

Hint

Use the properties of exponential functions and the chain rule as necessary.

Answer

a. $\frac{d}{dx}\left(\frac{e^{x^2}}{e^{5x}}\right) = e^{x^{2-5x}}(2x-5)$
b. $\frac{d}{dt}\left((e^{2t})^3\right) = 6e^{6t}$

Example 6.7.5: Using Properties of Exponential Functions

Evaluate the following integral: $\int 2xe^{-x^2}\,dx$.

Solution

Using u-substitution, let $u = -x^2$. Then $du = -2x\,dx$, and we have

$$\int 2xe^{-x^2}\,dx = -\int e^u\,du = -e^u + C = -e^{-x^2} + C.$$

Exercise 6.7.5

Evaluate the following integral: $\int \frac{4}{e^{3x}}\,dx$.

Hint

Use the properties of exponential functions and $u - substitution$ as necessary.

Answer

$$\int \frac{4}{e^{3x}}\,dx = -\frac{4}{3}e^{-3x} + C$$

General Logarithmic and Exponential Functions

We close this section by looking at exponential functions and logarithms with bases other than e. Exponential functions are functions of the form $f(x) = a^x$. Note that unless $a = e$, we still do not have a mathematically rigorous definition of these functions for irrational exponents. Let's rectify that here by defining the function $f(x) = a^x$ in terms of the exponential function e^x. We then examine logarithms with bases other than e as inverse functions of exponential functions.

Definition: Exponential Function

For any $a > 0$, and for any real number x, define $y = a^x$ as follows:

$$y = a^x = e^{x \ln a}.$$

Now a^x is defined rigorously for all values of x. This definition also allows us to generalize property iv. of logarithms and property iii. of exponential functions to apply to both rational and irrational values of r. It is straightforward to show that properties of exponents hold for general exponential functions defined in this way.

Let's now apply this definition to calculate a differentiation formula for a^x. We have

$$\frac{d}{dx}\left(a^x\right) = \frac{d}{dx}\left(e^{x \ln a}\right) = e^{x \ln a} \ln a = a^x \ln a.$$

The corresponding integration formula follows immediately.

Derivatives and Integrals Involving General Exponential Functions

Let $a > 0$. Then,

$$\frac{d}{dx}\left(a^x\right) = a^x \ln a$$

and

$$\int a^x\,dx = \frac{1}{\ln a}a^x + C.$$

If $a \neq 1$, then the function a^x is one-to-one and has a well-defined inverse. Its inverse is denoted by $\log_a x$. Then,

$$y = \log_a x \text{ if and only if } x = a^y.$$

Note that general logarithm functions can be written in terms of the natural logarithm. Let $y = \log_a x$. Then, $x = a^y$. Taking the natural logarithm of both sides of this second equation, we get

$$\begin{aligned} \ln x &= \ln(a^y) \\ \ln x &= y \ln a \\ y &= \frac{\ln x}{\ln a} \\ \log_a x &= \frac{\ln x}{\ln a}. \end{aligned}$$

Thus, we see that all logarithmic functions are constant multiples of one another. Next, we use this formula to find a differentiation formula for a logarithm with base a. Again, let $y = \log_a x$. Then,

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\log_a x\right) \\ &= \frac{d}{dx}\left(\frac{\ln x}{\ln a}\right) \\ &= (\frac{1}{\ln a})\frac{d}{dx}\left(\ln x\right) \\ &= \frac{1}{\ln a} \cdot \frac{1}{x} = \frac{1}{x \ln a} \end{aligned}$$

Derivatives of General Logarithm Functions

Let $a > 0$. Then,

$$\frac{d}{dx}\left(\log_a x\right) = \frac{1}{x \ln a}.$$

Example 6.7.6: Calculating Derivatives of General Exponential and Logarithm Functions

Evaluate the following derivatives:

a. $\frac{d}{dt}\left(4^t \cdot 2^{t^2}\right)$
b. $\frac{d}{dx}\left(\log_8(7x^2+4)\right)$

Solution: We need to apply the chain rule as necessary.

a. $\frac{d}{dt}\left(4^t \cdot 2^{t^2}\right) = \frac{d}{dt}\left(2^{2t} \cdot 2^{t^2}\right) = \frac{d}{dt}\left(2^{2t+t^2}\right) = 2^{2t+t^2}\ln(2)(2+2t)$
b. $\frac{d}{dx}\left(\log_8(7x^2+4)\right) = \frac{1}{(7x^2+4)(\ln 8)}(14x)$

Exercise 6.7.6

Evaluate the following derivatives:

a. $\frac{d}{dt}\left(4^{t^4}\right)$
b. $\frac{d}{dx}\left(\log_3(\sqrt{x^2+1})\right)$

Hint

Use the formulas and apply the chain rule as necessary.

Answer

a. $\frac{d}{dt}\left(4^{t^4}\right) = 4^{t^4}(\ln 4)(4t^3)$

b. $\frac{d}{dx}\left(\log_3(\sqrt{x^2+1})\right) = \frac{x}{(\ln 3)(x^2+1)}$

Example 6.7.7: Integrating General Exponential Functions

Evaluate the following integral: $\int \frac{3}{2^{3x}}\,dx.$

Solution

Use $u-substitution$ and let $u=-3x$. Then $du=-3\,dx$ and we have

$$\int \frac{3}{2^{3x}}\,dx = \int 3\cdot 2^{-3x}\,dx = -\int 2^u\,du = -\frac{1}{\ln 2}2^u + C = -\frac{1}{\ln 2}2^{-3x} + C.$$

Exercise 6.7.7

Evaluate the following integral: $\int x^2 2^{x^3}\,dx.$

Hint

Use the properties of exponential functions and u-substitution

Answer

$$\int x^2 2^{x^3}\,dx = \frac{1}{3\ln 2}2^{x^3} + C$$

Key Concepts

- The earlier treatment of logarithms and exponential functions did not define the functions precisely and formally. This section develops the concepts in a mathematically rigorous way.
- The cornerstone of the development is the definition of the natural logarithm in terms of an integral.
- The function e^x is then defined as the inverse of the natural logarithm. General exponential functions are defined in terms of e^x, and the corresponding inverse functions are general logarithms.
- Familiar properties of logarithms and exponents still hold in this more rigorous context.

Key Equations

- **Natural logarithm function**
- $\ln x = \int_1^x \frac{1}{t}\,dt$
- **Exponential function** $y = e^x$
- $\ln y = \ln(e^x) = x$

This page titled 6.7: Integrals, Exponential Functions, and Logarithms is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.7E: Exercises for Section 6.7

In exercises 1 - 3, find the derivative $\frac{dy}{dx}$.

1) $y = \ln(2x)$

Answer

$$\frac{dy}{dx} = \frac{1}{x}$$

2) $y = \ln(2x + 1)$

3) $y = \frac{1}{\ln x}$

Answer

$$\frac{dy}{dx} = -\frac{1}{x(\ln x)^2}$$

In exercises 4 - 5, find the indefinite integral.

4) $\int \frac{dt}{3t}$

5) $\int \frac{dx}{1+x}$

Answer

$$\int \frac{dx}{1+x} = \ln|x+1| + C$$

In exercises 6 - 15, find the derivative $\frac{dy}{dx}$. (You can use a calculator to plot the function and the derivative to confirm that it is correct.)

6) [T] $y = \frac{\ln x}{x}$

7) [T] $y = x \ln x$

Answer

$$\frac{dy}{dx} = \ln(x) + 1$$

8) [T] $y = \log_{10} x$

9) [T] $y = \ln(\sin x)$

Answer

$$\frac{dy}{dx} = \cot x$$

10) [T] $y = \ln(\ln x)$

11) [T] $y = 7\ln(4x)$

Answer

$$\frac{dy}{dx} = \frac{7}{x}$$

12) [T] $y = \ln\left((4x)^7\right)$

13) [T] $y = \ln(\tan x)$

Answer

$$\frac{dy}{dx} = \csc x \sec x$$

14) [T] $y = \ln(\tan 3x)$

15) [T] $y = \ln(\cos^2 x)$

Answer

$$\frac{dy}{dx} = -2\tan x$$

In exercises 16 - 25, find the definite or indefinite integral.

16) $\int_0^1 \frac{dx}{3+x}$

17) $\int_0^1 \frac{dt}{3+2t}$

Answer

$$\int_0^1 \frac{dt}{3+2t} = \tfrac{1}{2}\ln\left(\tfrac{5}{3}\right)$$

18) $\int_0^2 \frac{x}{x^2+1}\,dx$

19) $\int_0^2 \frac{x^3}{x^2+1}\,dx$

Answer

$$\int_0^2 \frac{x^3}{x^2+1}\,dx = 2 - \tfrac{1}{2}\ln(5)$$

20) $\int_2^e \frac{dx}{x\ln x}$

21) $\int_2^e \frac{dx}{(x\ln x)^2}$

Answer

$$\int_2^e \frac{dx}{(x\ln x)^2} = \frac{1}{\ln(2)} - 1$$

22) $\int \frac{\cos x}{\sin x}\,dx$

23) $\int_0^{\pi/4} \tan x\,dx$

Answer

$$\int_0^{\pi/4} \tan x\,dx = \tfrac{1}{2}\ln(2)$$

24) $\int \cot(3x)\,dx$

25) $\int \frac{(\ln x)^2}{x}\,dx$

Answer

$$\int \frac{(\ln x)^2}{x}\,dx = \tfrac{1}{3}(\ln x)^3$$

In exercises 26 - 35, compute $\frac{dy}{dx}$ by differentiating $\ln y$.

26) $y = \sqrt{x^2+1}$

27) $y = \sqrt{x^2+1}\sqrt{x^2-1}$

Answer

$$\frac{dy}{dx} = \frac{2x^3}{\sqrt{x^2+1}\sqrt{x^2-1}}$$

28) $y = e^{\sin x}$

29) $y = x^{-1/x}$

Answer

$$\frac{dy}{dx} = x^{-2-(1/x)}(\ln x - 1)$$

30) $y = e^{ex}$

31) $y = x^e$

Answer

$$\frac{dy}{dx} = ex^{e-1}$$

32) $y = x^{(ex)}$

33) $y = \sqrt{x}\sqrt[3]{x}\sqrt[6]{x}$

Answer

$$\frac{dy}{dx} = 1$$

34) $y = x^{-1/\ln x}$

35) $y = e^{-\ln x}$

Answer

$$\frac{dy}{dx} = -\frac{1}{x^2}$$

In exercises 36 - 40, evaluate by any method.

36) $\int_5^{10} \frac{dt}{t} - \int_{5x}^{10x} \frac{dt}{t}$

37) $\int_1^{e^\pi} \frac{dx}{x} + \int_{-2}^{-1} \frac{dx}{x}$

Answer

$\pi - \ln(2)$

38) $\frac{d}{dx}\left[\int_x^1 \frac{dt}{t}\right]$

39) $\frac{d}{dx}\left[\int_x^{x^2} \frac{dt}{t}\right]$

Answer

$$\frac{1}{x}$$

40) $\frac{d}{dx}\left[\ln(\sec x + \tan x)\right]$

In exercises 41 - 44, use the function $\ln x$. If you are unable to find intersection points analytically, use a calculator.

41) Find the area of the region enclosed by $x = 1$ and $y = 5$ above $y = \ln x$.

Answer

$(e^5 - 6)$ units^2

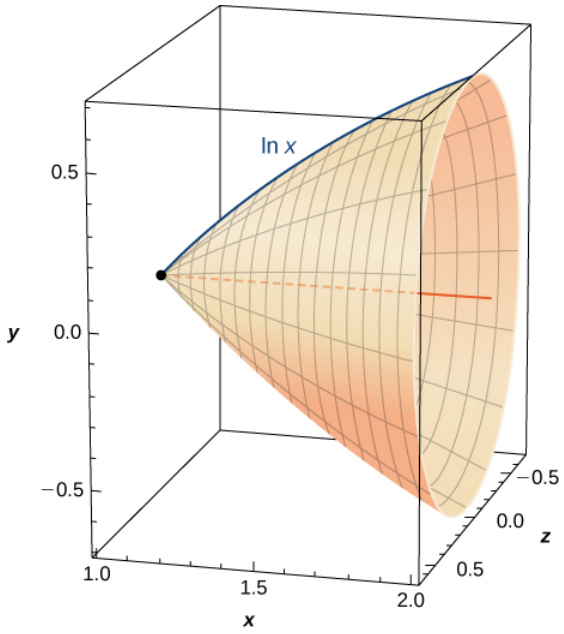
42) [T] Find the arc length of $\ln x$ from $x = 1$ to $x = 2$.

43) Find the area between $\ln x$ and the x-axis from $x = 1$ to $x = 2$.

Answer

$\ln(4) - 1)$ units^2

44) Find the volume of the shape created when rotating this curve from $x = 1$ to $x = 2$ around the x-axis, as pictured here.



45) [T] Find the surface area of the shape created when rotating the curve in the previous exercise from $x = 1$ to $x = 2$ around the x-axis.

Answer

2.8656 units^2

If you are unable to find intersection points analytically in the following exercises, use a calculator.

46) Find the area of the hyperbolic quarter-circle enclosed by $x = 2$ and $y = 2$ above $y = 1/x$.

47) [T] Find the arc length of $y = 1/x$ from $x = 1$ to $x = 4$.

Answer

$s = 3.1502$ units

48) Find the area under $y = 1/x$ and above the x-axis from $x = 1$ to $x = 4$.

In exercises 49 - 53, verify the derivatives and antiderivatives.

49) $\frac{d}{dx}\left[\ln(x + \sqrt{x^2 + 1})\right] = \frac{1}{\sqrt{1 + x^2}}$

50) $\frac{d}{dx}\left[\ln\left(\frac{x-a}{x+a}\right)\right] = \frac{2a}{(x^2 - a^2)}$

51) $\frac{d}{dx}\left[\ln\left(\frac{1+\sqrt{1-x^2}}{x}\right)\right] = -\frac{1}{x\sqrt{1 - x^2}}$

52) $\frac{d}{dx}\left[\ln(x+\sqrt{x^2-a^2})\right] = \frac{1}{\sqrt{x^2-a^2}}$

53) $\int \frac{dx}{x\ln(x)\ln(\ln x)} = \ln|\ln(\ln x)| + C$

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 6.7E: Exercises for Section 6.7 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.8: Exponential Growth and Decay

Learning Objectives

- Use the exponential growth model in applications, including population growth and compound interest.
- Explain the concept of doubling time.
- Use the exponential decay model in applications, including radioactive decay and Newton's law of cooling.
- Explain the concept of half-life.

One of the most prevalent applications of exponential functions involves growth and decay models. Exponential growth and decay show up in a host of natural applications. From population growth and continuously compounded interest to radioactive decay and Newton's law of cooling, exponential functions are ubiquitous in nature. In this section, we examine exponential growth and decay in the context of some of these applications.

Exponential Growth Model

Many systems exhibit exponential growth. These systems follow a model of the form $y = y_0 e^{kt}$, where y_0 represents the initial state of the system and k is a positive constant, called the growth constant. Notice that in an exponential growth model, we have

$$y' = ky_0 e^{kt} = ky. \tag{6.8.1}$$

That is, the rate of growth is proportional to the current function value. This is a key feature of exponential growth. Equation 6.8.1 involves derivatives and is called *a differential equation*.

Exponential Growth

Systems that exhibit *exponential growth* increase according to the mathematical model

$$y = y_0 e^{kt}$$

where y_0 represents the initial state of the system and $k > 0$ is a constant, called the growth constant.

Population growth is a common example of exponential growth. Consider a population of bacteria, for instance. It seems plausible that the rate of population growth would be proportional to the size of the population. After all, the more bacteria there are to reproduce, the faster the population grows. Figure 6.8.1 and Table 6.8.1 represent the growth of a population of bacteria with an initial population of 200 bacteria and a growth constant of 0.02. Notice that after only 2 hours (120 minutes), the population is 10 times its original size!

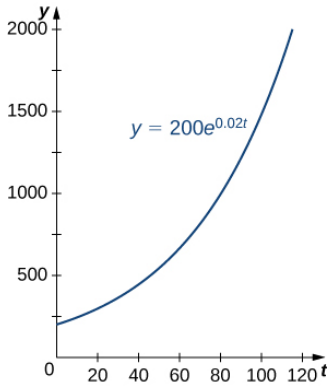


Figure 6.8.1: An example of exponential growth for bacteria.

Table 6.8.1: Exponential Growth of a Bacterial Population

Time(min)	Population Size (no. of bacteria)
10	244
20	298
30	364
40	445
50	544
60	664
70	811
80	991
90	1210
100	1478
110	1805
120	2205

Note that we are using a continuous function to model what is inherently discrete behavior. At any given time, the real-world population contains a whole number of bacteria, although the model takes on noninteger values. When using exponential growth models, we must always be careful to interpret the function values in the context of the phenomenon we are modeling.

Example 6.8.1: Population Growth

Consider the population of bacteria described earlier. This population grows according to the function $f(t) = 200e^{0.02t}$, where t is measured in minutes. How many bacteria are present in the population after 5 hours (300 minutes)? When does the population reach $100,000$ bacteria?

Solution

We have $f(t) = 200e^{0.02t}$. Then

$$f(300) = 200e^{0.02(300)} \approx 80,686.$$

There are $80,686$ bacteria in the population after 5 hours.

To find when the population reaches $100,000$ bacteria, we solve the equation

$$\begin{aligned} 100,000 &= 200e^{0.02t} \\ 500 &= e^{0.02t} \\ \ln 500 &= 0.02t \\ t &= \frac{\ln 500}{0.02} \approx 310.73. \end{aligned}$$

The population reaches $100,000$ bacteria after 310.73 minutes.

Exercise 6.8.1

Consider a population of bacteria that grows according to the function $f(t) = 500e^{0.05t}$, where t is measured in minutes. How many bacteria are present in the population after 4 hours? When does the population reach 100 million bacteria?

Answer

Use the process from the previous example.

Answer

There are $81,377,396$ bacteria in the population after 4 hours. The population reaches 100 million bacteria after 244.12 minutes.

Let's now turn our attention to a financial application: **compound interest**. Interest that is not compounded is called simple interest. ***Simple interest*** is paid once, at the end of the specified time period (usually 1 year). So, if we put $\$1000$ in a savings account earning 2 simple interest per year, then at the end of the year we have

$$1000(1+0.02) = \$1020.$$

Compound interest is paid multiple times per year, depending on the compounding period. Therefore, if the bank compounds the interest every 6 months, it credits half of the year's interest to the account after 6 months. During the second half of the year, the account earns interest not only on the initial $\$1000$, but also on the interest earned during the first half of the year. Mathematically speaking, at the end of the year, we have

$$1000\left(1+\frac{0.02}{2}\right)^2 = \$1020.10.$$

Similarly, if the interest is compounded every 4 months, we have

$$1000\left(1+\frac{0.02}{3}\right)^3 = \$1020.13,$$

and if the interest is compounded daily (365 times per year), we have $\$1020.20$. If we extend this concept, so that the interest is compounded continuously, after t years we have

$$1000\lim_{n\to\infty}\left(1+\frac{0.02}{n}\right)^{nt}.$$

Now let's manipulate this expression so that we have an exponential growth function. Recall that the number e can be expressed as a limit:

$$e = \lim_{m\to\infty}\left(1+\frac{1}{m}\right)^m.$$

Based on this, we want the expression inside the parentheses to have the form $(1+1/m)$. Let $n = 0.02m$. Note that as $n \to \infty, m \to \infty$ as well. Then we get

$$1000\lim_{n\to\infty}\left(1+\frac{0.02}{n}\right)^{nt} = 1000\lim_{m\to\infty}\left(1+\frac{0.02}{0.02m}\right)^{0.02mt} = 1000\left[\lim_{m\to\infty}\left(1+\frac{1}{m}\right)^m\right]^{0.02t}.$$

We recognize the limit inside the brackets as the number e. So, the balance in our bank account after t years is given by $1000e^{0.02t}$. Generalizing this concept, we see that if a bank account with an initial balance of $\$P$ earns interest at a rate of r, compounded continuously, then the balance of the account after t years is

$$\text{Balance } = Pe^{rt}.$$

Example 6.8.2: Compound Interest

A 25-year-old student is offered an opportunity to invest some money in a retirement account that pays 5 annual interest compounded continuously. How much does the student need to invest today to have $\$1$ million when she retires at age 65? What if she could earn 6 annual interest compounded continuously instead?

Solution

We have

$$1,000,000 = Pe^{0.05(40)}$$

$$P = 135,335.28.$$

She must invest $\$135,335.28$ at 5 interest.

If, instead, she is able to earn 6 then the equation becomes

$$1,000,000 = Pe^{0.06(40)}$$

$$P = 90,717.95.$$

In this case, she needs to invest only $\$90,717.95$. This is roughly two-thirds the amount she needs to invest at 5. The fact that the interest is compounded continuously greatly magnifies the effect of the 1 increase in interest rate.

Exercise 6.8.2

Suppose instead of investing at age $25\sqrt{b^2-4ac}$, the student waits until age 35. How much would she have to invest at 5? At 6?

Hint

Use the process from the previous example.

Answer

At 5 interest, she must invest $\$223,130.16$. At 6 interest, she must invest $\$165,298.89$.

If a quantity grows exponentially, the time it takes for the quantity to double remains constant. In other words, it takes the same amount of time for a population of bacteria to grow from 100 to 200 bacteria as it does to grow from $10,000$ to $20,000$ bacteria. This time is called the doubling time. To calculate the doubling time, we want to know when the quantity reaches twice its original size. So we have

$$\begin{aligned} 2y_0 &= y_0 e^{kt} \\ 2 &= e^{kt} \\ \ln 2 &= kt \\ t &= \frac{\ln 2}{k}. \end{aligned}$$

Definition: Doubling Time

If a quantity grows exponentially, the **doubling time** is the amount of time it takes the quantity to double. It is given by

$$\text{Doubling time} = \frac{\ln 2}{k}.$$

Example 6.8.3: Using the Doubling Time

Assume a population of fish grows exponentially. A pond is stocked initially with 500 fish. After 6 months, there are 1000 fish in the pond. The owner will allow his friends and neighbors to fish on his pond after the fish population reaches $10,000$. When will the owner's friends be allowed to fish?

Solution

We know it takes the population of fish 6 months to double in size. So, if t represents time in months, by the doubling-time formula, we have $6 = (\ln 2)/k$. Then, $k = (\ln 2)/6$. Thus, the population is given by $y = 500e^{((\ln 2)/6)t}$. To figure out when the population reaches $10,000$ fish, we must solve the following equation:

$$\begin{aligned} 10,000 &= 500e^{(\ln 2/6)t} \\ 20 &= e^{(\ln 2/6)t} \\ \ln 20 &= \left(\frac{\ln 2}{6}\right) t \\ t &= \frac{6(\ln 20)}{\ln 2} \\ &\approx 25.93. \end{aligned}$$

The owner's friends have to wait 25.93 months (a little more than 2 years) to fish in the pond.

Exercise $6.8.3$

Suppose it takes 9 months for the fish population in Example $6.8.3$ to reach 1000 fish. Under these circumstances, how long do the owner's friends have to wait?

Hint

Use the process from the previous example.

Answer

38.90 months

Exponential Decay Model

Exponential functions can also be used to model populations that shrink (from disease, for example), or chemical compounds that break down over time. We say that such systems exhibit exponential decay, rather than exponential growth. The model is nearly the same, except there is a negative sign in the exponent. Thus, for some positive constant k, we have

$$y = y_0 e^{-kt}.$$

As with exponential growth, there is a differential equation associated with exponential decay. We have

$$y' = -ky_0 e^{-kt} = -ky.$$

Exponential Decay

Systems that exhibit exponential decay behave according to the model

$$y = y_0 e^{-kt},$$

where y_0 represents the initial state of the system and $k > 0$ is a constant, called the decay constant.

Figure $6.8.2$ shows a graph of a representative exponential decay function.

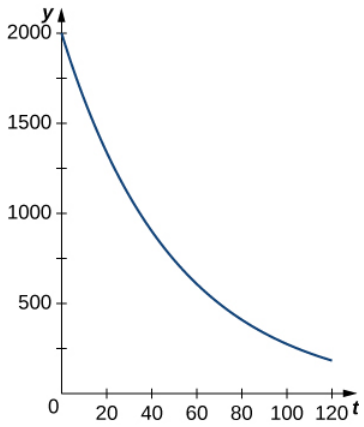


Figure $6.8.2$: An example of exponential decay.

Let's look at a physical application of exponential decay. *Newton's law of cooling* says that an object cools at a rate proportional to the difference between the temperature of the object and the temperature of the surroundings. In other words, if T represents the temperature of the object and T_a represents the ambient temperature in a room, then

$$T' = -k(T - T_a).$$

Note that this is not quite the right model for exponential decay. We want the derivative to be proportional to the function, and this expression has the additional T_a term. Fortunately, we can make a change of variables that resolves this issue. Let $y(t) = T(t) - T_a$. Then $y'(t) = T'(t) - 0 = T'(t)$, and our equation becomes

$$y' = -ky.$$

From our previous work, we know this relationship between y and its derivative leads to exponential decay. Thus,

$$y = y_0 e^{-kt},$$

and we see that

$$T - T_a = (T_0 - T_a)e^{-kt}$$

$$T = (T_0 - T_a)e^{-kt} + T_a$$

where T_0 represents the initial temperature. Let's apply this formula in the following example.

Example 6.8.4: Newton's Law of Cooling

According to experienced baristas, the optimal temperature to serve coffee is between $155°F$ and $175°F$. Suppose coffee is poured at a temperature of $200°F$, and after 2 minutes in a $70°F$ room it has cooled to $180°F$. When is the coffee first cool enough to serve? When is the coffee too cold to serve? Round answers to the nearest half minute.

Solution

We have

$$\begin{aligned} T &= (T_0 - T_a)e^{-kt} + T_a \\ 180 &= (200 - 70)e^{-k(2)} + 70 \\ 110 &= 130e^{-2k} \\ \frac{11}{13} &= e^{-2k} \\ \ln\frac{11}{13} &= -2k \\ \ln 11 - \ln 13 &= -2k \\ k &= \frac{\ln 13 - \ln 11}{2} \end{aligned}$$

Then, the model is

$$T = 130e^{(\ln 11 - \ln 13/2)t} + 70.$$

The coffee reaches $175°F$ when

$$\begin{aligned} 175 &= 130e^{(\ln 11 - \ln 13/2)t} + 70 \\ 105 &= 130e^{(\ln 11 - \ln 13/2)t} \\ \frac{21}{26} &= e^{(\ln 11 - \ln 13/2)t} \\ \ln\frac{21}{26} &= \frac{\ln 11 - \ln 13}{2}t \\ \ln 21 - \ln 26 &= \left(\frac{\ln 11 - \ln 13}{2}\right)t \\ t &= \frac{2(\ln 21 - \ln 26)}{\ln 11 - \ln 13} \\ &\approx 2.56. \end{aligned}$$

The coffee can be served about 2.5 minutes after it is poured. The coffee reaches $155°F$ at

$$155 = 130e^{(\ln 11 - \ln 13/2)t} + 70$$
$$85 = 130e^{(\ln 11 - \ln 13)t}$$
$$\frac{17}{26} = e^{(\ln 11 - \ln 13)t}$$
$$\ln 17 - \ln 26 = \left(\frac{\ln 11 - \ln 13}{2}\right) t$$
$$t = \frac{2(\ln 17 - \ln 26)}{\ln 11 - \ln 13}$$
$$\approx 5.09.$$

The coffee is too cold to be served about 5 minutes after it is poured.

? Exercise 6.8.4

Suppose the room is warmer $(75°F)$ and, after 2 minutes, the coffee has cooled only to $185°F$. When is the coffee first cool enough to serve? When is the coffee be too cold to serve? Round answers to the nearest half minute.

Hint

Use the process from the previous example.

Answer

The coffee is first cool enough to serve about 3.5 minutes after it is poured. The coffee is too cold to serve about 7 minutes after it is poured.

Just as systems exhibiting exponential growth have a constant doubling time, systems exhibiting exponential decay have a constant half-life. To calculate the half-life, we want to know when the quantity reaches half its original size. Therefore, we have

$$\frac{y_0}{2} = y_0 e^{-kt}$$
$$\frac{1}{2} = e^{-kt}$$
$$-\ln 2 = -kt$$
$$t = \frac{\ln 2}{k}.$$

Note: This is the same expression we came up with for doubling time.

Definition: Half-Life

If a quantity decays exponentially, the half-life is the amount of time it takes the quantity to be reduced by half. It is given by

$$\text{Half-life} = \frac{\ln 2}{k}.$$

✓ Example 6.8.5: Radiocarbon Dating

One of the most common applications of an exponential decay model is carbon dating. Carbon-14 decays (emits a radioactive particle) at a regular and consistent exponential rate. Therefore, if we know how much carbon-14 was originally present in an object and how much carbon-14 remains, we can determine the age of the object. The half-life of carbon-14 is approximately 5730 years—meaning, after that many years, half the material has converted from the original carbon-14 to the new nonradioactive nitrogen-14. If we have 100 g carbon-14 today, how much is left in 50 years? If an artifact that originally contained 100 g of carbon-14 now contains 10 g of carbon-14, how old is it? Round the answer to the nearest hundred years.

Solution

We have

$$5730 = \frac{\ln 2}{k}$$

$$k = \frac{\ln 2}{5730}.$$

So, the model says

$$y = 100e^{-(\ln 2/5730)t}.$$

In 50 years, we have

$$y = 100e^{-(\ln 2/5730)(50)} \approx 99.40$$

Therefore, in 50 years, 99.40 g of carbon-14 remains.

To determine the age of the artifact, we must solve

$$10 = 100e^{-(\ln 2/5730)t}$$

$$\frac{1}{10} = e^{-(\ln 2/5730)t}$$

$$t \approx 19035.$$

The artifact is about $19,000$ years old.

? Exercise 6.8.5: Carbon-14 Decay

If we have 100 g of carbon-14 , how much is left after 500 years? If an artifact that originally contained 100 g of carbon-14 now contains 20 g of carbon-14, how old is it? Round the answer to the nearest hundred years.

Hint

Use the process from the previous example.

Answer

A total of 94.13 g of carbon-14 remains after 500 years. The artifact is approximately 13,300 years old.

Key Concepts

- Exponential growth and exponential decay are two of the most common applications of exponential functions.
- Systems that exhibit exponential growth follow a model of the form $y = y_0e^{kt}$.
- In exponential growth, the rate of growth is proportional to the quantity present. In other words, $y' = ky$.
- Systems that exhibit exponential growth have a constant doubling time, which is given by $(\ln 2)/k$.
- Systems that exhibit exponential decay follow a model of the form $y = y_0e^{-kt}$.
- Systems that exhibit exponential decay have a constant half-life, which is given by $(\ln 2)/k$.

Glossary

doubling time

if a quantity grows exponentially, the doubling time is the amount of time it takes the quantity to double, and is given by $(\ln 2)/k$

exponential decay

systems that exhibit exponential decay follow a model of the form $y = y_0e^{-kt}$

exponential growth

systems that exhibit exponential growth follow a model of the form $y = y_0e^{kt}$

half-life

if a quantity decays exponentially, the half-life is the amount of time it takes the quantity to be reduced by half. It is given by $(\ln 2)/k$

This page titled 6.8: Exponential Growth and Decay is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.8E: Exercises for Section 6.8

In exercises 1 - 2, answer *True or False*? If true, prove it. If false, find the true answer.

1) The doubling time for $y = e^{ct}$ is $\frac{\ln 2}{\ln c}$.

2) If you invest $\$500$, an annual rate of interest of 3% yields more money in the first year than a 2.5% continuous rate of interest.

Answer

True

3) If you leave a $100°C$ pot of tea at room temperature ($25°C$) and an identical pot in the refrigerator $(5°C)$, with $k = 0.02$, the tea in the refrigerator reaches a drinkable temperature $(70°C)$ more than 5 minutes before the tea at room temperature.

4) If given a half-life of t years, the constant k for $y = e^{kt}$ is calculated by $k = \frac{\ln(\frac{1}{2})}{t}$.

Answer

False; $k = \frac{\ln 2}{t}$

In exercises 5 - 18, use $y = y_0 e^{kt}$.

5) If a culture of bacteria doubles in 3 hours, how many hours does it take to multiply by 10?

6) If bacteria increase by a factor of 10 in 10 hours, how many hours does it take to increase by 100?

Answer

20 hours

7) How old is a skull that contains one-fifth as much radiocarbon as a modern skull? Note that the half-life of radiocarbon is 5730 years.

8) If a relic contains 90% as much radiocarbon as new material, can it have come from the time of Christ (approximately 2000 years ago)? Note that the half-life of radiocarbon is 5730 years.

Answer

No. The relic is approximately 871 years old.

9) The population of Cairo grew from 5 million to 10 million in 20 years. Use an exponential model to find when the population was 8 million.

10) The populations of New York and Los Angeles are growing at 1% and 1.4% a year, respectively. Starting from 8 million (New York) and 6 million (Los Angeles), when are the populations equal?

Answer

71.92 years

11) Suppose the value of $\$1$ in Japanese yen decreases at 2% per year. Starting from $\$1 = ¥250$, when will $\$1 = ¥1$?

12) The effect of advertising decays exponentially. If 40% of the population remembers a new product after 3 days, how long will 20%remember it?

Answer

5 days 6 hours 27minutes

13) If $y = 1000$ at $t = 3$ and $y = 3000$ at $t = 4$, what was y_0 at $t = 0$?

14) If $y = 100$ at $t = 4$ and $y = 10$ at $t = 8$, when does $y = 1$?

Answer

At $t = 12$

15) If a bank offers annual interest of 7.5% or continuous interest of 7.25%, which has a better annual yield?

16) What continuous interest rate has the same yield as an annual rate of 9%?

Answer

8.618%

17) If you deposit $\$5000$at 8% annual interest, how many years can you withdraw $\$500$ (starting after the first year) without running out of money?

18) You are trying to save $\$50,000$ in 20 years for college tuition for your child. If interest is a continuous 10%, how much do you need to invest initially?

Answer

$6766.76

19) You are cooling a turkey that was taken out of the oven with an internal temperature of $165°F$. After 10 minutes of resting the turkey in a $70°F$ apartment, the temperature has reached $155°F$. What is the temperature of the turkey 20 minutes after taking it out of the oven?

20) You are trying to thaw some vegetables that are at a temperature of $1°F$. To thaw vegetables safely, you must put them in the refrigerator, which has an ambient temperature of $44°F$. You check on your vegetables 2 hours after putting them in the refrigerator to find that they are now $12°F$. Plot the resulting temperature curve and use it to determine when the vegetables reach $33°$.

Answer

9hours 13minutes

21) You are an archeologist and are given a bone that is claimed to be from a Tyrannosaurus Rex. You know these dinosaurs lived during the Cretaceous Era (146 million years to 65 million years ago), and you find by radiocarbon dating that there is 0.000001% the amount of radiocarbon. Is this bone from the Cretaceous?

22) The spent fuel of a nuclear reactor contains plutonium-239, which has a half-life of $24,000$ years. If 1 barrel containing 10 kg of plutonium-239 is sealed, how many years must pass until only 10 g of plutonium-239 is left?

Answer

$239,179$ years

For exercises 23 - 26, use the following table, which features the world population by decade.

Years since 1950	Population (millions)
0	2,556
10	3,039
20	3,706
30	4,453
40	5,279
50	6,083
60	6,849

Source: http:/www.factmonster.com/ipka/A0762181.html.

23) [T] The best-fit exponential curve to the data of the form $P(t) = ae^{bt}$ is given by $P(t) = 2686e^{0.01604t}$. Use a graphing calculator to graph the data and the exponential curve together.

24) [T] Find and graph the derivative y' of your equation. Where is it increasing and what is the meaning of this increase?

Answer

$P'(t) = 43e^{0.01604t}$. The population is always increasing.

25) [T] Find and graph the second derivative of your equation. Where is it increasing and what is the meaning of this increase?

26) [T] Find the predicted date when the population reaches 10 billion. Using your previous answers about the first and second derivatives, explain why exponential growth is unsuccessful in predicting the future.

Answer

The population reaches 10 billion people in 2027.

For exercises 27 - 29, use the following table, which shows the population of San Francisco during the 19th century.

Years since 1850	Population (thousands)
0	21.00
10	56.80
20	149.5
30	234.0

Source: http:/www.sfgenealogy.com/sf/history/hgpop.htm.

27) [T] The best-fit exponential curve to the data of the form $P(t) = ae^{bt}$ is given by $P(t) = 35.26e^{0.06407t}$. Use a graphing calculator to graph the data and the exponential curve together.

28) [T] Find and graph the derivative y' of your equation. Where is it increasing? What is the meaning of this increase? Is there a value where the increase is maximal?

Answer

$P'(t) = 2.259e^{0.06407t}$. The population is always increasing.

29) [T] Find and graph the second derivative of your equation. Where is it increasing? What is the meaning of this increase?

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 6.8E: Exercises for Section 6.8 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.9: Calculus of the Hyperbolic Functions

Learning Objectives

- Apply the formulas for derivatives and integrals of the hyperbolic functions.
- Apply the formulas for the derivatives of the inverse hyperbolic functions and their associated integrals.
- Describe the common applied conditions of a catenary curve.

We were introduced to hyperbolic functions previously, along with some of their basic properties. In this section, we look at differentiation and integration formulas for the hyperbolic functions and their inverses.

Derivatives and Integrals of the Hyperbolic Functions

Recall that the hyperbolic sine and hyperbolic cosine are defined as

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

and

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

The other hyperbolic functions are then defined in terms of $\sinh x$ and $\cosh x$. The graphs of the hyperbolic functions are shown in Figure 6.9.1.

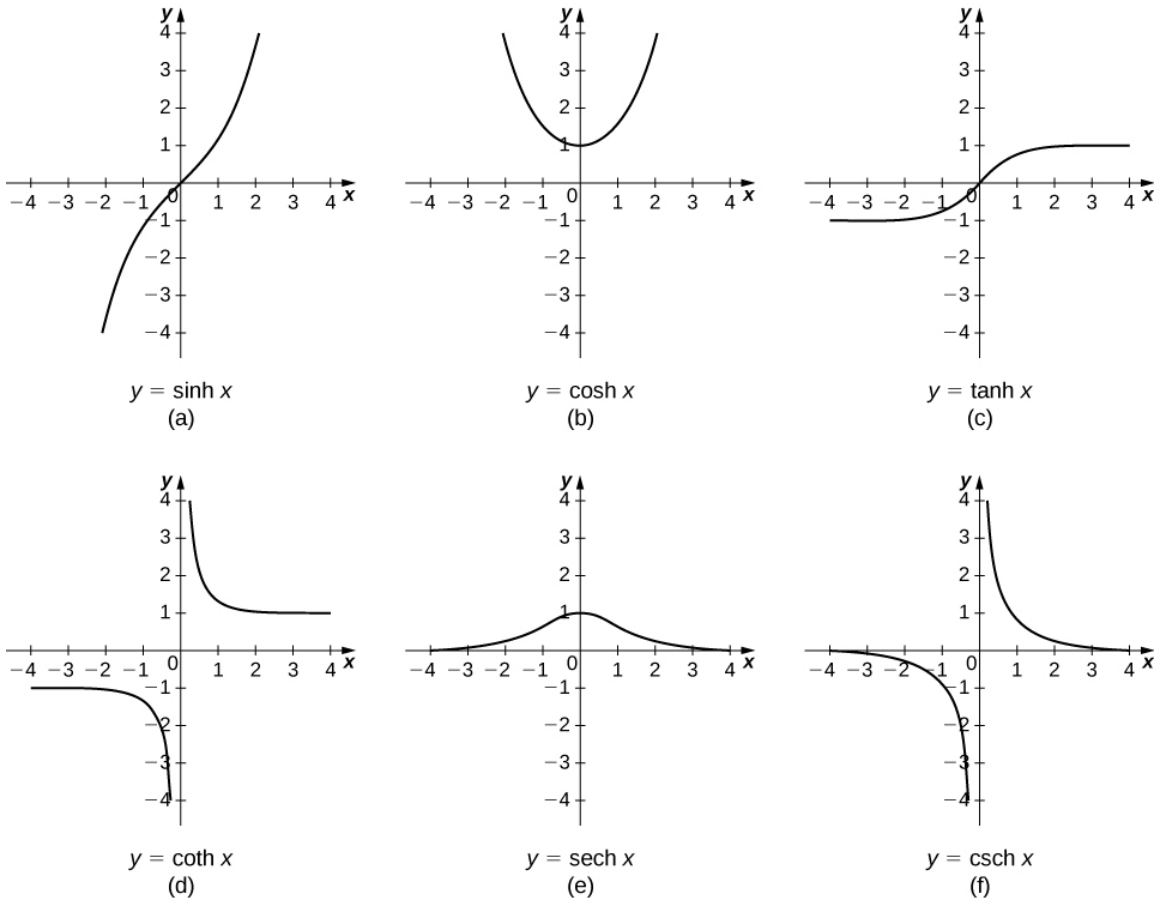


Figure 6.9.1: Graphs of the hyperbolic functions.

It is easy to develop differentiation formulas for the hyperbolic functions. For example, looking at $\sinh x$ we have

$$\begin{aligned}\frac{d}{dx}(\sinh x) &= \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right) \\ &= \frac{1}{2}\left[\frac{d}{dx}(e^x) - \frac{d}{dx}(e^{-x})\right] \\ &= \frac{1}{2}[e^x + e^{-x}] \\ &= \cosh x.\end{aligned}$$

Similarly,

$$\frac{d}{dx}\cosh x = \sinh x.$$

We summarize the differentiation formulas for the hyperbolic functions in Table 6.9.1.

Table 6.9.1: Derivatives of the Hyperbolic Functions

$f(x)$	$\frac{d}{dx}f(x)$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$
$\coth x$	$-\operatorname{csch}^2 x$
$\operatorname{sech} x$	$-\operatorname{sech} x \tanh x$
$\operatorname{csch} x$	$-\operatorname{csch} x \coth x$

Let's take a moment to compare the derivatives of the hyperbolic functions with the derivatives of the standard trigonometric functions. There are a lot of similarities, but differences as well. For example, the derivatives of the sine functions match:

$$\frac{d}{dx}\sin x = \cos x$$

and

$$\frac{d}{dx}\sinh x = \cosh x.$$

The derivatives of the cosine functions, however, differ in sign:

$$\frac{d}{dx}\cos x = -\sin x,$$

but

$$\frac{d}{dx}\cosh x = \sinh x.$$

As we continue our examination of the hyperbolic functions, we must be mindful of their similarities and differences to the standard trigonometric functions. These differentiation formulas for the hyperbolic functions lead directly to the following integral formulas.

$$\int \sinh u\, du = \cosh u + C \tag{6.9.1}$$

$$\int \operatorname{csch}^2 u\, du = -\coth u + C \tag{6.9.2}$$

$$\int \cosh u\, du = \sinh u + C \tag{6.9.3}$$

$$\int \operatorname{sech}\, u \tanh u\, du = -\operatorname{sech}\ u + C - \operatorname{csch} u + C \tag{6.9.4}$$

$$\int \operatorname{sech}\ ^2 u\, du = \tanh u + C \tag{6.9.5}$$

$$\int \operatorname{csch}\, u \coth u\, du = -\operatorname{csch} u + C \tag{6.9.6}$$

Example 6.9.1: Differentiating Hyperbolic Functions

Evaluate the following derivatives:

a. $\frac{d}{dx}(\sinh(x^2))$
b. $\frac{d}{dx}(\cosh x)^2$

Solution:

Using the formulas in Table 6.9.1 and the chain rule, we get

a. $\frac{d}{dx}(\sinh(x^2)) = \cosh(x^2) \cdot 2x$
b. $\frac{d}{dx}(\cosh x)^2 = 2\cosh x \sinh x$

Exercise 6.9.1

Evaluate the following derivatives:

a. $\frac{d}{dx}(\tanh(x^2 + 3x))$
b. $\frac{d}{dx}\left(\frac{1}{(\sinh x)^2}\right)$

Hint

Use the formulas in Table 6.9.1 and apply the chain rule as necessary.

Answer a

$$\frac{d}{dx}(\tanh(x^2 + 3x)) = (\operatorname{sech}^2(x^2 + 3x))(2x + 3)$$

Answer b

$$\frac{d}{dx}\left(\frac{1}{(\sinh x)^2}\right) = \frac{d}{dx}(\sinh x)^{-2} = -2(\sinh x)^{-3}\cosh x$$

Example 6.9.2: Integrals Involving Hyperbolic Functions

Evaluate the following integrals:

a. $\int x\cosh(x^2)dx$
b. $\int \tanh x\, dx$

Solution

We can use u-substitution in both cases.

a. Let $u = x^2$. Then, $du = 2x\, dx$ and

$$\int x\cosh(x^2)dx = \int \frac{1}{2}\cosh u\, du$$
$$= \frac{1}{2}\sinh u + C$$
$$= \frac{1}{2}\sinh(x^2) + C.$$

b. Let $u = \cosh x$. Then, $du = \sinh x\, dx$ and

$$\int \tanh x\, dx = \int \frac{\sinh x}{\cosh x}\, dx$$
$$= \int \frac{1}{u}du$$
$$= \ln|u| + C$$
$$= \ln|\cosh x| + C.$$

Note that $\cosh x > 0$ for all x, so we can eliminate the absolute value signs and obtain

$$\int \tanh x\, dx = \ln(\cosh x) + C.$$

? Exercise 6.9.2

Evaluate the following integrals:

a. $\int \sinh^3 x \cosh x\, dx$

b. $\int \text{sech}^2(3x)\, dx$

Hint

Use the formulas above and apply u-substitution as necessary.

Answer a

$$\int \sinh^3 x \cosh x\, dx = \frac{\sinh^4 x}{4} + C$$

Answer b

$$\int \text{sech}^2(3x)\, dx = \frac{\tanh(3x)}{3} + C$$

Calculus of Inverse Hyperbolic Functions

Looking at the graphs of the hyperbolic functions, we see that with appropriate range restrictions, they all have inverses. Most of the necessary range restrictions can be discerned by close examination of the graphs. The domains and ranges of the inverse hyperbolic functions are summarized in Table 6.9.2.

Table 6.9.2: Domains and Ranges of the Inverse Hyperbolic Functions

Function	Domain	Range
$\sinh^{-1} x$	$(-\infty,\infty)$	$(-\infty,\infty)$
$\cosh^{-1} x$	$(1,\infty)$	$[0,\infty)$
$\tanh^{-1} x$	$(-1,1)$	$(-\infty,\infty)$
$\coth^{-1} x$	$(-\infty,1)\cup(1,\infty)$	$(-\infty,0)\cup(0,\infty)$
$\text{sech}^{-1} x$	$(0,1)$	$[0,\infty)$
$\text{csch}^{-1} x$	$(-\infty,0)\cup(0,\infty)$	$(-\infty,0)\cup(0,\infty)$

The graphs of the inverse hyperbolic functions are shown in the following figure.

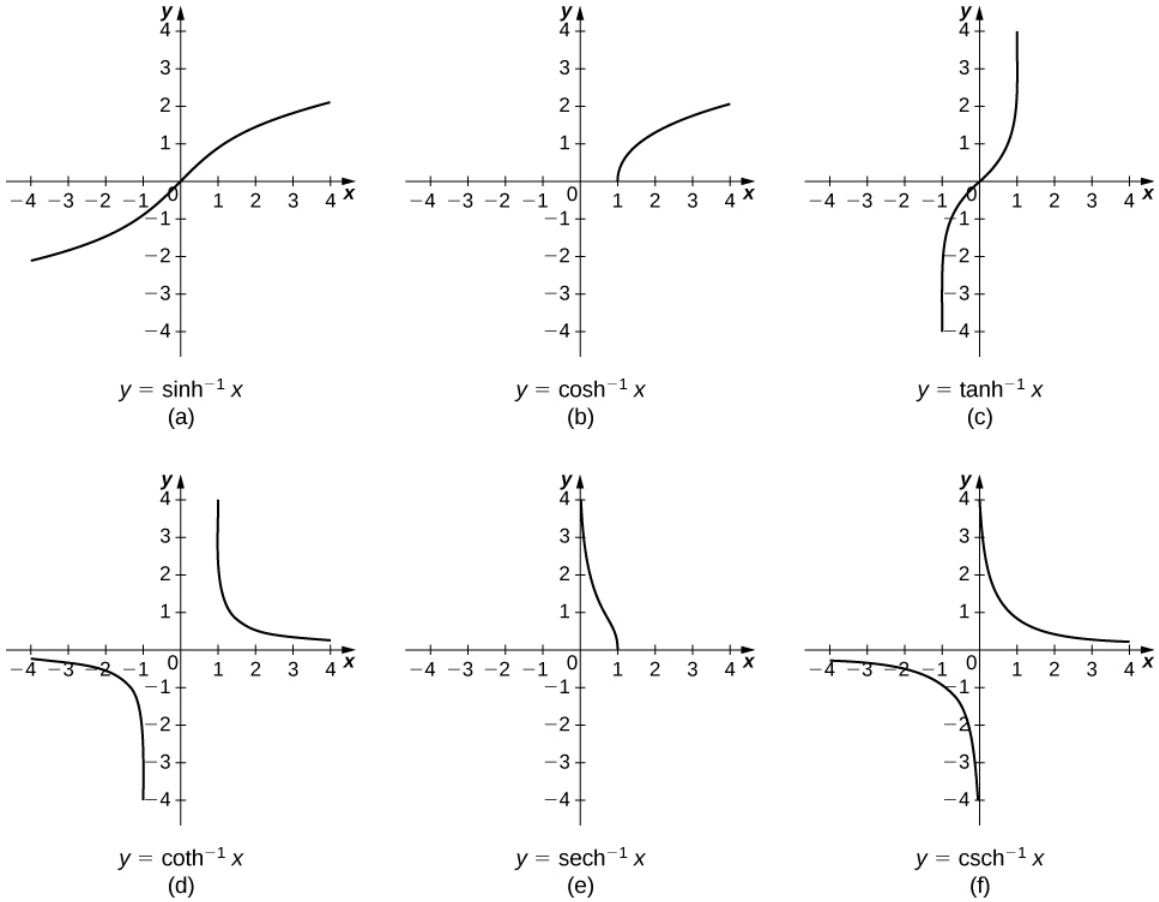


Figure 6.9.3: Graphs of the inverse hyperbolic functions.

To find the derivatives of the inverse functions, we use implicit differentiation. We have

$$y = \sinh^{-1} x \tag{6.9.7}$$

$$\sinh y = x \tag{6.9.8}$$

$$\frac{d}{dx}\sinh y = \frac{d}{dx}x \tag{6.9.9}$$

$$\cosh y \frac{dy}{dx} = 1. \tag{6.9.10}$$

Recall that $\cosh^2 y - \sinh^2 y = 1$, so $\cosh y = \sqrt{1 + \sinh^2 y}$. Then,

$$\frac{dy}{dx} = \frac{1}{\cosh y} = \frac{1}{\sqrt{1 + \sinh^2 y}} = \frac{1}{\sqrt{1 + x^2}}.$$

We can derive differentiation formulas for the other inverse hyperbolic functions in a similar fashion. These differentiation formulas are summarized in Table 6.9.3.

Table 6.9.3: Derivatives of the Inverse Hyperbolic Functions

$f(x)$	$\frac{d}{dx}f(x)$
$\sinh^{-1} x$	$\frac{1}{\sqrt{1+x^2}}$
$\cosh^{-1} x$	$\frac{1}{\sqrt{x^2-1}}$
$\tanh^{-1} x$	$\frac{1}{1-x^2}$
$\coth^{-1} x$	$\frac{1}{1-x^2}$
$\text{sech}^{-1} x$	$\frac{-1}{x\sqrt{1-x^2}}$
$\text{csch}^{-1} x$	$\frac{-1}{\lvert x\rvert\sqrt{1+x^2}}$

Note that the derivatives of $\tanh^{-1} x$ and $\coth^{-1} x$ are the same. Thus, when we integrate $1/(1-x^2)$, we need to select the proper antiderivative based on the domain of the functions and the values of x. Integration formulas involving the inverse hyperbolic functions are summarized as follows.

$$\int \frac{1}{\sqrt{1+u^2}}du = \sinh^{-1} u + C$$

$$\int \frac{1}{u\sqrt{1-u^2}}du = -\text{sech}^{-1}|u| + C$$

$$\int \frac{1}{\sqrt{u^2-1}}du = \cosh^{-1} u + C$$

$$\int \frac{1}{u\sqrt{1+u^2}}du = -\text{csch}^{-1}|u| + C$$

$$\int \frac{1}{1-u^2}du = \begin{cases} \tanh^{-1} u + C & \text{if } |u| < 1 \\ \coth^{-1} u + C & \text{if } |u| > 1 \end{cases}$$

Example 6.9.3: Differentiating Inverse Hyperbolic Functions

Evaluate the following derivatives:

a. $\frac{d}{dx}\left(\sinh^{-1}\left(\frac{x}{3}\right)\right)$
b. $\frac{d}{dx}\left(\tanh^{-1} x\right)^2$

Solution

Using the formulas in Table 6.9.3 and the chain rule, we obtain the following results:

a. $\frac{d}{dx}(\sinh^{-1}(\frac{x}{3})) = \frac{1}{3\sqrt{1+\frac{x^2}{9}}} = \frac{1}{\sqrt{9+x^2}}$
b. $\frac{d}{dx}(\tanh^{-1} x)^2 = \frac{2(\tanh^{-1} x)}{1-x^2}$

Exercise 6.9.3

Evaluate the following derivatives:

a. $\frac{d}{dx}(\cosh^{-1}(3x))$
b. $\frac{d}{dx}(\coth^{-1} x)^3$

Hint

Use the formulas in Table 6.9.3 and apply the chain rule as necessary.

Answer a

$$\frac{d}{dx}(\cosh^{-1}(3x)) = \frac{3}{\sqrt{9x^2-1}}$$

Answer b

$$\frac{d}{dx}(\coth^{-1}x)^3 = \frac{3(\coth^{-1}x)^2}{1-x^2}$$

Example 6.9.4: Integrals Involving Inverse Hyperbolic Functions

Evaluate the following integrals:

a. $\int \frac{1}{\sqrt{4x^2-1}}dx$

b. $\int \frac{1}{2x\sqrt{1-9x^2}}dx$

Solution

We can use u-**substitution** in both cases.

Let $u = 2x$. Then, $du = 2\,dx$ and we have

$$\begin{aligned}\int \frac{1}{\sqrt{4x^2-1}}\,dx &= \int \frac{1}{2\sqrt{u^2-1}}\,du \\ &= \frac{1}{2}\cosh^{-1}u + C \\ &= \frac{1}{2}\cosh^{-1}(2x) + C.\end{aligned}$$

Let $u = 3x$. Then, $du = 3\,dx$ and we obtain

$$\begin{aligned}\int \frac{1}{2x\sqrt{1-9x^2}}dx &= \frac{1}{2}\int \frac{1}{u\sqrt{1-u^2}}du \\ &= -\frac{1}{2}\text{sech}^{-1}|u| + C \\ &= -\frac{1}{2}\text{sech}^{-1}|3x| + C\end{aligned}$$

Exercise 6.9.4

Evaluate the following integrals:

a. $\int \frac{1}{\sqrt{x^2-4}}dx, x > 2$

b. $\int \frac{1}{\sqrt{1-e^{2x}}}dx$

Hint

Use the formulas above and apply u-substitution as necessary.

Answer a

$$\int \frac{1}{\sqrt{x^2-4}}dx = \cosh^{-1}(\frac{x}{2}) + C$$

Answer b

$$\int \frac{1}{\sqrt{1-e^{2x}}}dx = -\text{sech}^{-1}(e^x) + C$$

Applications

One physical application of hyperbolic functions involves hanging cables. If a cable of uniform density is suspended between two supports without any load other than its own weight, the cable forms a curve called a **catenary**. High-voltage power lines, chains hanging between two posts, and strands of a spider's web all form catenaries. The following figure shows chains hanging from a row of posts.



Figure 6.9.3: Chains between these posts take the shape of a catenary. (credit: modification of work by OKFoundryCompany, Flickr)

Hyperbolic functions can be used to model catenaries. Specifically, functions of the form $y = a \cdot \cosh(x/a)$ are catenaries. Figure 6.9.4 shows the graph of $y = 2\cosh(x/2)$.

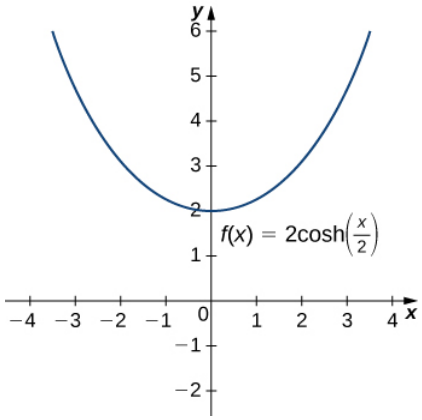


Figure 6.9.4: A hyperbolic cosine function forms the shape of a catenary.

Example 6.9.5: Using a Catenary to Find the Length of a Cable

Assume a hanging cable has the shape $10\cosh(x/10)$ for $-15 \le x \le 15$, where x is measured in feet. Determine the length of the cable (in feet).

Solution

Recall from Section 6.4 that the formula for arc length is

$$\underbrace{\int_a^b \sqrt{1+[f'(x)]^2}dx}_{\text{Arc Length}}.$$

We have $f(x) = 10\cosh(x/10)$, so $f'(x) = \sinh(x/10)$. Then the arc length is

$$\int_a^b \sqrt{1+[f'(x)]^2}dx = \int_{-15}^{15} \sqrt{1+\sinh^2\left(\frac{x}{10}\right)}dx.$$

Now recall that

$$1+\sinh^2 x = \cosh^2 x,$$

so we have

$$\begin{aligned}
\text{Arc Length} &= \int_{-15}^{15} \sqrt{1+\sinh^2\left(\frac{x}{10}\right)}dx \\
&= \int_{-15}^{15} \cosh\left(\frac{x}{10}\right)dx \\
&= 10\sinh\left(\frac{x}{10}\right)\Big|_{-15}^{15} \\
&= 10\left[\sinh\left(\frac{3}{2}\right) - \sinh\left(-\frac{3}{2}\right)\right] \\
&= 20\sinh\left(\frac{3}{2}\right) \\
&\approx 42.586\,\text{ft}.
\end{aligned}$$

? Exercise 6.9.5:

Assume a hanging cable has the shape $15\cosh(x/15)$ for $-20 \le x \le 20$. Determine the length of the cable (in feet).

Answer

52.95 ft

Key Concepts

- Hyperbolic functions are defined in terms of exponential functions.
- Term-by-term differentiation yields differentiation formulas for the hyperbolic functions. These differentiation formulas give rise, in turn, to integration formulas.
- With appropriate range restrictions, the hyperbolic functions all have inverses.
- Implicit differentiation yields differentiation formulas for the inverse hyperbolic functions, which in turn give rise to integration formulas.
- The most common physical applications of hyperbolic functions are calculations involving catenaries.

Glossary

catenary

a curve in the shape of the function $y = a \cdot \cosh(x/a)$ is a catenary; a cable of uniform density suspended between two supports assumes the shape of a catenary

This page titled 6.9: Calculus of the Hyperbolic Functions is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.9E: Exercises for Section 6.9

1) [T] Find expressions for $\cosh x + \sinh x$ and $\cosh x - \sinh x$. Use a calculator to graph these functions and ensure your expression is correct.

Answer

e^x and e^{-x}

2) From the definitions of $\cosh(x)$ and $\sinh(x)$, find their antiderivatives.

3) Show that $\cosh(x)$ and $\sinh(x)$ satisfy $y'' = y$.

Answer

Answers may vary

4) Use the quotient rule to verify that $\frac{d}{dx}\big(\tanh(x)\big) = \operatorname{sech}^2(x)$.

5) Derive $\cosh^2(x) + \sinh^2(x) = \cosh(2x)$ from the definition.

Answer

Answers may vary

6) Take the derivative of the previous expression to find an expression for $\sinh(2x)$.

7) Prove $\sinh(x + y) = \sinh(x)\cosh(y) + \cosh(x)\sinh(y)$ by changing the expression to exponentials.

Answer

Answers may vary

8) Take the derivative of the previous expression to find an expression for $\cosh(x + y)$.

In exercises 9 - 18, find the derivatives of the given functions and graph along with the function to ensure your answer is correct.

9) [T] $\cosh(3x + 1)$

Answer

$3\sinh(3x + 1)$

10) [T] $\sinh(x^2)$

11) [T] $\frac{1}{\cosh(x)}$

Answer

$-\tanh(x)\operatorname{sech}(x)$

12) [T] $\sinh(\ln(x))$

13) [T] $\cosh^2(x) + \sinh^2(x)$

Answer

$4\cosh(x)\sinh(x)$

14) [T] $\cosh^2(x) - \sinh^2(x)$

15) [T] $\tanh(\sqrt{x^2 + 1})$

Answer

$$\frac{x\text{sech}^2(\sqrt{x^2+1})}{\sqrt{x^2+1}}$$

16) [T] $\frac{1+\tanh(x)}{1-\tanh(x)}$

17) [T] $\sinh^6(x)$

Answer

$6\sinh^5(x)\cosh(x)$

18) [T] $\ln(\text{sech}(x)+\tanh(x))$

In exercises 19 - 28, find the antiderivatives for the given functions.

19) $\cosh(2x+1)$

Answer

$\frac{1}{2}\sinh(2x+1)+C$

20) $\tanh(3x+2)$

21) $x\cosh(x^2)$

Answer

$\frac{1}{2}\sinh^2(x^2)+C$

22) $3x^3\tanh(x^4)$

23) $\cosh^2(x)\sinh(x)$

Answer

$\frac{1}{3}\cosh^3(x)+C$

24) $\tanh^2(x)\text{sech}^2(x)$

25) $\frac{\sinh(x)}{1+\cosh(x)}$

Answer

$\ln(1+\cosh(x))+C$

26) $\coth(x)$

27) $\cosh(x)+\sinh(x)$

Answer

$\cosh(x)+\sinh(x)+C$

28) $(\cosh(x)+\sinh(x))^n$

In exercises 29 - 35, find the derivatives for the functions.

29) $\tanh^{-1}(4x)$

Answer

$\frac{4}{1-16x^2}$

30) $\sinh^{-1}(x^2)$

31) $\sinh^{-1}(\cosh(x))$

Answer

$$\frac{\sinh(x)}{\sqrt{\cosh^2(x)+1}}$$

32) $\cosh^{-1}(x^3)$

33) $\tanh^{-1}(\cos(x))$

Answer

$$-\csc(x)$$

34) $e^{\sinh^{-1}(x)}$

35) $\ln(\tanh^{-1}(x))$

Answer

$$-\frac{1}{(x^2-1)\tanh^{-1}(x)}$$

In exercises 36 - 42, find the antiderivatives for the functions.

36) $\int \frac{dx}{4-x^2}$

37) $\int \frac{dx}{a^2-x^2}$

Answer

$$\frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right)+C$$

38) $\int \frac{dx}{\sqrt{x^2+1}}$

39) $\int \frac{xdx}{\sqrt{x^2+1}}$

Answer

$$\sqrt{x^2+1}+C$$

40) $\int -\frac{dx}{x\sqrt{1-x^2}}$

41) $\int \frac{e^x}{\sqrt{e^{2x}-1}}$

Answer

$$\cosh^{-1}(e^x)+C$$

42) $\int -\frac{2x}{x^4-1}$

In exercises 43 - 45, use the fact that a falling body with friction equal to velocity squared obeys the equation $\frac{dv}{dt}=g-v^2$.

43) Show that $v(t)=\sqrt{g}\tanh(\sqrt{g}t)$ satisfies this equation.

Answer

Answers may vary

44) Derive the previous expression for $v(t)$ by integrating $\frac{dv}{g-v^2}=dt$.

45) [T] Estimate how far a body has fallen in 12seconds by finding the area underneath the curve of $v(t)$.

Answer

37.30

In exercises 46 - 48, use this scenario: A cable hanging under its own weight has a slope $S = \frac{dy}{dx}$ that satisfies $\frac{dS}{dx} = c\sqrt{1+S^2}$. The constant c is the ratio of cable density to tension.

46) Show that $S = \sinh(cx)$ satisfies this equation.

47) Integrate $\frac{dy}{dx} = \sinh(cx)$ to find the cable height $y(x)$ if $y(0) = 1/c$.

Answer

$y = \frac{1}{c}\cosh(cx)$

48) Sketch the cable and determine how far down it sags at $x = 0$.

In exercises 49 - 52, solve each problem.

49) [T] A chain hangs from two posts 2m apart to form a catenary described by the equation $y = 2\cosh(x/2) - 1$. Find the slope of the catenary at the left fence post.

Answer

-0.521095

50) [T] A chain hangs from two posts four meters apart to form a catenary described by the equation $y = 4\cosh(x/4) - 3$. Find the total length of the catenary (arc length).

51) [T] A high-voltage power line is a catenary described by $y = 10\cosh(x/10)$. Find the ratio of the area under the catenary to its arc length. What do you notice?

Answer

10

52) A telephone line is a catenary described by $y = a\cosh(x/a)$. Find the ratio of the area under the catenary to its arc length. Does this confirm your answer for the previous question?

53) Prove the formula for the derivative of $y = \sinh^{-1}(x)$ by differentiating $x = \sinh(y)$.

(Hint: Use hyperbolic trigonometric identities.)

54) Prove the formula for the derivative of $y = \cosh^{-1}(x)$ by differentiating $x = \cosh(y)$.

(Hint: Use hyperbolic trigonometric identities.)

55) Prove the formula for the derivative of $y = \operatorname{sech}^{-1}(x)$ by differentiating $x = \operatorname{sech}(y)$.

(Hint: Use hyperbolic trigonometric identities.)

56) Prove that $\cosh(x) + \sinh(x))^n = \cosh(nx) + \sinh(nx)$.

57) Prove the expression for $\sinh^{-1}(x)$. Multiply $x = \sinh(y) = \frac{e^y - e^{-y}}{2}$ by $2e^y$ and solve for y. Does your expression match the textbook?

58) Prove the expression for $\cosh^{-1}(x)$. Multiply $x = \cosh(y) = \frac{e^y + e^{-y}}{2}$ by $2e^y$ and solve for y. Does your expression match the textbook?

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

This page titled 6.9E: Exercises for Section 6.9 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

6.R: CHAPTER 6 REVIEW EXERCISES

***True or False?* Justify your answer with a proof or a counterexample.**

1) The amount of work to pump the water out of a half-full cylinder is half the amount of work to pump the water out of the full cylinder.

Answer

False

2) If the force is constant, the amount of work to move an object from $x = a$ to $x = b$ is $F(b - a)$.

3) The disk method can be used in any situation in which the washer method is successful at finding the volume of a solid of revolution.

Answer

False

4) If the half-life of $seaborgium - 266$ is 360 ms, then $k = \dfrac{\ln 2}{360}$.

For exercises 5 - 8, use the requested method to determine the volume of the solid.

5) The volume that has a base of the ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$ and cross-sections of an equilateral triangle perpendicular to the y-axis. Use the method of slicing.

Answer

$V = 32\sqrt{3}\,\text{units}^3$

6) $y = x^2 - x$, from $x = 1$ to $x = 4$, rotated around the y-axis using the washer method

7) $x = y^2$ and $x = 3y$ rotated around the y-axis using the washer method

Answer

$V = \frac{162\pi}{5}\,\text{units}^3$

8) $x = 2y^2 - y^3,\ x = 0$,and $y = 0$ rotated around the x-axis using cylindrical shells

For exercises 9 - 14, find

a. the area of the region,

b.the volume of the solid when rotated around the x-axis, and

c. the volume of the solid when rotated around the y-axis. Use whichever method seems most appropriate to you.

9) $y = x^3, x = 0, y = 0$, and $x = 2$

Answer

a. $A = 4$ units^2
b. $V = \frac{128\pi}{7}$ units^3
c. $V = \frac{64\pi}{5}$ units^3

10) $y = x^2 - x$ and $x = 0$

11) [T] $y = \ln(x) + 2$ and $y = x$

Answer

a. $A \approx 1.949$ units^2
b. $V \approx 21.952$ units^3
c. $V = \approx 17.099$ units^3

12) $y = x^2$ and $y = \sqrt{x}$

13) $y = 5 + x, y = x^2, x = 0$, and $x = 1$

Answer

https://math.libretexts.org/@go/page/70408

a. $A = \frac{31}{6}$ units^2
b. $V = \frac{452\pi}{15}$ units^3
c. $V = \frac{31\pi}{6}$ units^3

14) Below $x^2 + y^2 = 1$ and above $y = 1 - x$

15) Find the mass of $\rho = e^{-x}$ on a disk centered at the origin with radius 4.

Answer

$m \approx 245.282$

16) Find the center of mass for $\rho = \tan^2 x$ on $x \in (-\frac{\pi}{4}, \frac{\pi}{4})$.

17) Find the mass and the center of mass of $\rho = 1$ on the region bounded by $y = x^5$ and $y = \sqrt{x}$.

Answer

Mass: $\frac{1}{2}$,
Center of mass: $(\frac{18}{35}, \frac{9}{11})$

For exercises 18 - 19, find the requested arc lengths.

18) The length of x for $y = \cosh(x)$ from $x = 0$ to $x = 2$.

19) The length of y for $x = 3 - \sqrt{y}$ from $y = 0$ to $y = 4$

Answer

$s = [\sqrt{17} + \frac{1}{8}\ln(33 + 8\sqrt{17})]$ units

For exercises 20 - 21, find the surface area and volume when the given curves are revolved around the specified axis.

20) The shape created by revolving the region between $y = 4 + x$, $y = 3 - x$, $x = 0$, and $x = 2$ rotated around the y-axis.

21) The loudspeaker created by revolving $y = \dfrac{1}{x}$ from $x = 1$ to $x = 4$ around the x-axis.

Answer

Volume: $V = \frac{3\pi}{4}$ units^3
Surface area: $A = \pi\left(\sqrt{2} - \sinh^{-1}(1) + \sinh^{-1}(16) - \frac{\sqrt{257}}{16}\right)$ units^2

For exercise 22, consider the Karun-3 dam in Iran. Its shape can be approximated as an isosceles triangle with height 205 m and width 388 m. Assume the current depth of the water is 180 m. The density of water is 1000 kg/m^3.

22) Find the total force on the wall of the dam.

23) You are a crime scene investigator attempting to determine the time of death of a victim. It is noon and 45 °F outside and the temperature of the body is 78 °F. You know the cooling constant is $k = 0.00824$ °F/min. When did the victim die, assuming that a human's temperature is 98 °F?

Answer

11:02 a.m.

For the following exercise, consider the stock market crash in 1929 in the United States. The table lists the Dow Jones industrial average per year leading up to the crash.

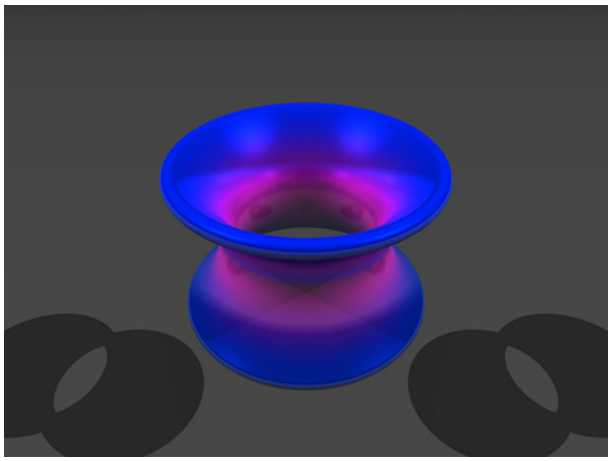
Year after 1920	Value ($)
1	63.90
3	100
5	110
7	160
9	381.17

Source: http:/stockcharts.com/freecharts/hi...a19201940.html

24) [T] The best-fit exponential curve to these data is given by $y = 40.71 + 1.224^x$. Why do you think the gains of the market were unsustainable? Use first and second derivatives to help justify your answer. What would this model predict the Dow Jones industrial average to be in **2014** ?

For exercises 25 - 26, consider the catenoid, the only solid of revolution that has a minimal surface, or zero mean curvature. A catenoid in nature can be found when stretching soap between two rings.

25) Find the volume of the catenoid $y = \cosh(x)$ from $x = -1$ to $x = 1$ that is created by rotating this curve around the x-**axis,** as shown here.



Answer

$V = \pi\left(1 + \sinh(1)\cosh(1)\right)$ units^3

26) Find surface area of the catenoid $y = \cosh(x)$ from $x = -1$ to $x = 1$ that is created by rotating this curve around the x-**axis.**

6.R: Chapter 6 Review Exercises is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

CHAPTER OVERVIEW

7: Techniques of Integration

We saw in the previous chapter how important integration can be for all kinds of different topics—from calculations of volumes to flow rates, and from using a velocity function to determine a position to locating centers of mass. It is no surprise, then, that techniques for finding antiderivatives (or indefinite integrals) are important to know for everyone who uses them. We have already discussed some basic integration formulas and the method of integration by substitution. In this chapter, we study some additional techniques, including some ways of approximating definite integrals when normal techniques do not work.

7.0: Prelude to Techniques of Integration

7.1: Integration by Parts

7.1E: Exercises for Section 7.1

7.2: Trigonometric Integrals

7.2E: Exercises for Section 7.2

7.3: Trigonometric Substitution

7.3E: Exercises for Section 7.3

7.4: Partial Fractions

7.4E: Exercises for Section 7.4

7.5: Other Strategies for Integration

7.5E: Exercises for Section 7.5

7.6: Numerical Integration

7.6E: Exercises for Section 7.6

7.7: Improper Integrals

7.7E: Exercises for Section 7.7

7.R: Chapter 7 Review Exercises

This page titled 7: Techniques of Integration is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

7.0: Prelude to Techniques of Integration

In a large city, accidents occurred at an average rate of one every three months at a particularly busy intersection. After residents complained, changes were made to the traffic lights at the intersection. It has now been eight months since the changes were made and there have been no accidents. Were the changes effective or is the eight-month interval without an accident a result of chance? We explore this question later in this chapter and see that integration is an essential part of determining the answer.



Figure 7.0.1: Careful planning of traffic signals can prevent or reduce the number of accidents at busy intersections. (credit: modification of work by David McKelvey, Flickr)

We saw in the previous chapter how important integration can be for all kinds of different topics—from calculations of volumes to flow rates, and from using a velocity function to determine a position to locating centers of mass. It is no surprise, then, that techniques for finding antiderivatives (or indefinite integrals) are important to know for everyone who uses them. We have already discussed some basic integration formulas and the method of integration by substitution. In this chapter, we study some additional techniques, including some ways of approximating definite integrals when normal techniques do not work.

Template:HideTOC

This page titled 7.0: Prelude to Techniques of Integration is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

7.1: Integration by Parts

Learning Objectives

- Recognize when to use integration by parts.
- Use the integration-by-parts formula to solve integration problems.
- Use the integration-by-parts formula for definite integrals.

By now we have a fairly thorough procedure for how to evaluate many basic integrals. However, although we can integrate $\int x\sin(x^2)\,dx$ by using the substitution, $u = x^2$, something as simple looking as $\int x\sin x\;dx$ defies us. Many students want to know whether there is a product rule for integration. There is not, but there is a technique based on the product rule for differentiation that allows us to exchange one integral for another. We call this technique **integration by parts.**

The Integration-by-Parts Formula

If, $h(x) = f(x)g(x)$, then by using the product rule, we obtain

$$h'(x) = f'(x)g(x) + g'(x)f(x). \tag{7.1.1}$$

Although at first it may seem counterproductive, let's now integrate both sides of Equation 7.1.1:

$$\int h'(x)\;dx = \int (g(x)f'(x) + f(x)g'(x))\;dx.$$

This gives us

$$h(x) = f(x)g(x) = \int g(x)f'(x)\,dx + \int f(x)g'(x)\;dx.$$

Now we solve for $\int f(x)g'(x)\;dx$:

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\;dx.$$

By making the substitutions $u = f(x)$ and $v = g(x)$, which in turn make $du = f'(x)\,dx$ and $dv = g'(x)\,dx$, we have the more compact form

$$\int u\,dv = uv - \int v\,du.$$

Integration by Parts

Let $u = f(x)$ and $v = g(x)$ be functions with continuous derivatives. Then, the integration-by-parts formula for the integral involving these two functions is:

$$\int u\,dv = uv - \int v\,du. \tag{7.1.2}$$

The advantage of using the integration-by-parts formula is that we can use it to exchange one integral for another, possibly easier, integral. The following example illustrates its use.

Example 7.1.1: Using Integration by Parts

Use integration by parts with $u = x$ and $dv = \sin x\;dx$ to evaluate

$$\int x\sin x\;dx.$$

Solution

By choosing $u = x$, we have $du = 1\;dx$. Since $dv = \sin x\;dx$, we get

$$v = \int \sin x \; dx = -\cos x.$$

It is handy to keep track of these values as follows:

- $u = x$
- $dv = \sin x \; dx$
- $du = 1 \, dx$
- $v = \int \sin x \; dx = -\cos x.$

Applying the integration-by-parts formula (Equation 7.1.2) results in

$$\begin{aligned} \int x \sin x \; dx &= (x)(-\cos x) - \int (-\cos x)(1 \; dx) && \text{(Substitute)} \\ &= -x \cos x + \int \cos x \; dx && \text{(Simplify)} \end{aligned}$$

Then use

$$\int \cos x \; dx = \sin x + C.$$

to obtain

$$\int x \sin x \; dx = -x \cos x + \sin x + C.$$

Analysis

At this point, there are probably a few items that need clarification. First of all, you may be curious about what would have happened if we had chosen $u = \sin x$ and $dv = x \, dx$. If we had done so, then we would have $du = \cos x$ and $v = \frac{1}{2}x^2$. Thus, after applying integration by parts (Equation 7.1.2), we have

$$\int x \sin x \; dx = \frac{1}{2}x^2 \sin x - \int \frac{1}{2}x^2 \cos x \; dx.$$

Unfortunately, with the new integral, we are in no better position than before. It is important to keep in mind that when we apply integration by parts, we may need to try several choices for u and dv before finding a choice that works.

Second, you may wonder why, when we find $v = \int \sin x \; dx = -\cos x$, we do not use $v = -\cos x + K$. To see that it makes no difference, we can rework the problem using $v = -\cos x + K$:

$$\begin{aligned} \int x \sin x \; dx &= (x)(-\cos x + K) - \int (-\cos x + K)(1 \; dx) \\ &= -x \cos x + Kx + \int \cos x \; dx - \int K \; dx \\ &= -x \cos x + Kx + \sin x - Kx + C \\ &= -x \cos x + \sin x + C. \end{aligned}$$

As you can see, it makes no difference in the final solution.

Last, we can check to make sure that our antiderivative is correct by differentiating $-x \cos x + \sin x + C$:

$$\begin{aligned} \frac{d}{dx}(-x \cos x + \sin x + C) &= \cancel{(-1)\cos x} + (-x)(-\sin x) + \cancel{\cos x} \\ &= x \sin x \end{aligned}$$

Therefore, the antiderivative checks out.

? Exercise 7.1.1

Evaluate $\int xe^{2x}\,dx$ using the integration-by-parts formula (Equation 7.1.2) with $u = x$ and $dv = e^{2x}\,dx$.

Hint

Find du and v, and use the previous example as a guide.

Answer

$$\int xe^{2x}\,dx = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$$

The natural question to ask at this point is: How do we know how to choose u and dv? Sometimes it is a matter of trial and error; however, the acronym **LIATE** can often help to take some of the guesswork out of our choices. This acronym stands for **L**ogarithmic Functions, **I**nverse Trigonometric Functions, **A**lgebraic Functions, **T**rigonometric Functions, and **E**xponential Functions. This mnemonic serves as an aid in determining an appropriate choice for u. The type of function in the integral that appears first in the list should be our first choice of u.

For example, if an integral contains a **logarithmic function** and an **algebraic function,** we should choose u to be the logarithmic function, because L comes before A in LIATE. The integral in Example 7.1.1 has a trigonometric function ($\sin x$) and an algebraic function (x). Because A comes before T in LIATE, we chose u to be the algebraic function. When we have chosen u, dv is selected to be the remaining part of the function to be integrated, together with dx.

Why does this mnemonic work? Remember that whatever we pick to be dv must be something we can integrate. Since we do not have integration formulas that allow us to integrate simple logarithmic functions and inverse trigonometric functions, it makes sense that they should not be chosen as values for dv. Consequently, they should be at the head of the list as choices for u. Thus, we put LI at the beginning of the mnemonic. (We could just as easily have started with IL, since these two types of functions won't appear together in an integration-by-parts problem.) The exponential and trigonometric functions are at the end of our list because they are fairly easy to integrate and make good choices for dv. Thus, we have TE at the end of our mnemonic. (We could just as easily have used ET at the end, since when these types of functions appear together it usually doesn't really matter which one is u and which one is dv.) Algebraic functions are generally easy both to integrate and to differentiate, and they come in the middle of the mnemonic.

✓ Example 7.1.2: Using Integration by Parts

Evaluate

$$\int \frac{\ln x}{x^3}\,dx.$$

Solution

Begin by rewriting the integral:

$$\int \frac{\ln x}{x^3}\,dx = \int x^{-3}\ln x\,dx.$$

Since this integral contains the algebraic function x^{-3} and the logarithmic function $\ln x$, choose $u = \ln x$, since L comes before A in LIATE. After we have chosen $u = \ln x$, we must choose $dv = x^{-3}\,dx$.

Next, since $u = \ln x$, we have $du = \frac{1}{x}\,dx$. Also, $v = \int x^{-3}\,dx = -\frac{1}{2}x^{-2}$. Summarizing,

- $u = \ln x$
- $du = \frac{1}{x}\,dx$
- $dv = x^{-3}\,dx$
- $v = \int x^{-3}\,dx = -\frac{1}{2}x^{-2}$.

Substituting into the integration-by-parts formula (Equation 7.1.2) gives

$$\int \frac{\ln x}{x^3}\,dx = \int x^{-3}\ln x\,dx = (\ln x)(-\frac{1}{2}x^{-2}) - \int (-\frac{1}{2}x^{-2})(\frac{1}{x}\,dx)$$
$$= -\frac{1}{2}x^{-2}\ln x + \int \frac{1}{2}x^{-3}\,dx$$
$$= -\frac{1}{2}x^{-2}\ln x - \frac{1}{4}x^{-2} + C$$
$$= -\frac{1}{2x^2}\ln x - \frac{1}{4x^2} + C$$

? Exercise 7.1.2

Evaluate

$$\int x\ln x\ dx.$$

Hint

Use $u = \ln x$ and $dv = x\ dx$.

Answer

$$\int x\ln x\ dx = \frac{1}{2}x^2\ln x - \frac{1}{4}x^2 + C$$

In some cases, as in the next two examples, it may be necessary to apply integration by parts more than once.

✓ Example 7.1.3A: Applying Integration by Parts More Than Once

Evaluate

$$\int x^2e^{3x}\,dx.$$

Solution

Using LIATE, choose $u = x^2$ and $dv = e^{3x}\,dx$. Thus, $du = 2x\,dx$ and $v = \int e^{3x}\,dx = \left(\frac{1}{3}\right)e^{3x}$. Therefore,

- $u = x^2$
- $du = 2x\,dx$
- $dv = e^{3x}\,dx$
- $v = \int e^{3x}\,dx = \frac{1}{3}e^{3x}$.

Substituting into Equation 7.1.2 produces

$$\int x^2e^{3x}\,dx = \frac{1}{3}x^2e^{3x} - \int \frac{2}{3}xe^{3x}\,dx. \tag{7.1.3}$$

We still cannot integrate $\int \frac{2}{3}xe^{3x}\,dx$ directly, but the integral now has a lower power on x. We can evaluate this new integral by using integration by parts again. To do this, choose

$$u = x$$

and

$$dv = \frac{2}{3}e^{3x}\,dx.$$

Thus,

$$du = dx$$

and

$$v = \int \left(\frac{2}{3}\right) e^{3x}\, dx = \left(\frac{2}{9}\right) e^{3x}.$$

Now we have

- $u = x$
- $du = dx$
- $dv = \frac{2}{3} e^{3x}\, dx$
- $v = \int \frac{2}{3} e^{3x}\, dx = \frac{2}{9} e^{3x}$.

Substituting back into Equation 7.1.3 yields

$$\int x^2 e^{3x}\, dx = \frac{1}{3} x^2 e^{3x} - \left(\frac{2}{9} x e^{3x} - \int \frac{2}{9} e^{3x}\, dx\right).$$

After evaluating the last integral and simplifying, we obtain

$$\int x^2 e^{3x}\, dx = \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + C.$$

Example 7.1.3B: Applying Integration by Parts When LIATE Does not Quite Work

Evaluate

$$\int t^3 e^{t^2} dt.$$

Solution

If we use a strict interpretation of the mnemonic LIATE to make our choice of u, we end up with $u = t^3$ and $dv = e^{t^2} dt$. Unfortunately, this choice won't work because we are unable to evaluate $\int e^{t^2} dt$. However, since we can evaluate $\int t e^{t^2}\, dx$, we can try choosing $u = t^2$ and $dv = t e^{t^2} dt$. With these choices we have

- $u = t^2$
- $du = 2t dt$
- $dv = t e^{t^2} dt$
- $v = \int t e^{t^2} dt = \frac{1}{2} e^{t^2}$.

Thus, we obtain

$$\begin{aligned} \int t^3 e^{t^2} dt &= \frac{1}{2} t^2 e^{t^2} - \int \frac{1}{2} e^{t^2} 2t\, dt \\ &= \frac{1}{2} t^2 e^{t^2} - \frac{1}{2} e^{t^2} + C. \end{aligned}$$

Example 7.1.3C: Applying Integration by Parts More Than Once

Evaluate

$$\int \sin(\ln x)\, dx.$$

Solution

This integral appears to have only one function—namely, $\sin(\ln x)$—however, we can always use the constant function 1 as the other function. In this example, let's choose $u = \sin(\ln x)$ and $dv = 1\, dx$. (The decision to use $u = \sin(\ln x)$ is easy. We can't choose $dv = \sin(\ln x)\, dx$ because if we could integrate it, we wouldn't be using integration by parts in the first place!) Consequently, $du = (1/x)\cos(\ln x)\, dx$ and $v = \int 1\, dx = x$. After applying integration by parts to the integral and simplifying, we have

$$\int \sin(\ln x)\, dx = x\sin(\ln x) - \int \cos(\ln x)\, dx.$$

Unfortunately, this process leaves us with a new integral that is very similar to the original. However, let's see what happens when we apply integration by parts again. This time let's choose $u = \cos(\ln x)$ and $dv = 1\, dx$, making $du = -(1/x)\sin(\ln x)\, dx$ and $v = \int 1\, dx = x$.

Substituting, we have

$$\int \sin(\ln x)\, dx = x\sin(\ln x) - (x\cos(\ln x) - \int -\sin(\ln x)\, dx).$$

After simplifying, we obtain

$$\int \sin(\ln x)\, dx = x\sin(\ln x) - x\cos(\ln x) - \int \sin(\ln x)\, dx.$$

The last integral is now the same as the original. It may seem that we have simply gone in a circle, but now we can actually evaluate the integral. To see how to do this more clearly, substitute $I = \int \sin(\ln x)\, dx$. Thus, the equation becomes

$$I = x\sin(\ln x) - x\cos(\ln x) - I.$$

First, add I to both sides of the equation to obtain

$$2I = x\sin(\ln x) - x\cos(\ln x).$$

Next, divide by 2:

$$I = \frac{1}{2}x\sin(\ln x) - \frac{1}{2}x\cos(\ln x).$$

Substituting $I = \int \sin(\ln x)\, dx$ again, we have

$$\int \sin(\ln x)\, dx = \frac{1}{2}x\sin(\ln x) - \frac{1}{2}x\cos(\ln x).$$

From this we see that $(1/2)x\sin(\ln x) - (1/2)x\cos(\ln x)$ is an antiderivative of $\sin(\ln x)\, dx$. For the most general antiderivative, add $+C$:

$$\int \sin(\ln x)\, dx = \frac{1}{2}x\sin(\ln x) - \frac{1}{2}x\cos(\ln x) + C.$$

Analysis

If this method feels a little strange at first, we can check the answer by differentiation:

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{2}x\sin(\ln x) - \frac{1}{2}x\cos(\ln x)\right) &\\ = \frac{1}{2}(\sin(\ln x)) + \cos(\ln x)\cdot\frac{1}{x}\cdot\frac{1}{2}x &\\ - \left(\frac{1}{2}\cos(\ln x) - \sin(\ln x)\cdot\frac{1}{x}\cdot\frac{1}{2}x\right) &\\ = \sin(\ln x). &\end{aligned}$$

? Exercise 7.1.3

Evaluate

$$\int x^2 \sin x\, dx.$$

Hint

This is similar to Examples 7.1.3A - 7.1.3C.

Answer

$$\int x^2 \sin x \, dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

Integration by Parts for Definite Integrals

Now that we have used integration by parts successfully to evaluate *indefinite integrals*, we turn our attention to definite integrals. The integration technique is really the same, only we add a step to evaluate the integral at the upper and lower limits of integration.

Integration by Parts for Definite Integrals

Let $u = f(x)$ and $v = g(x)$ be functions with continuous derivatives on $[a, b]$. Then

$$\int_a^b u \, dv = uv\Big|_a^b - \int_a^b v \, du$$

Example 7.1.4A: Finding the Area of a Region

Find the area of the region bounded above by the graph of $y = \tan^{-1} x$ and below by the x-axis over the interval $[0, 1]$.

Solution

This region is shown in Figure 7.1.1. To find the area, we must evaluate

$$\int_0^1 \tan^{-1} x \; dx.$$

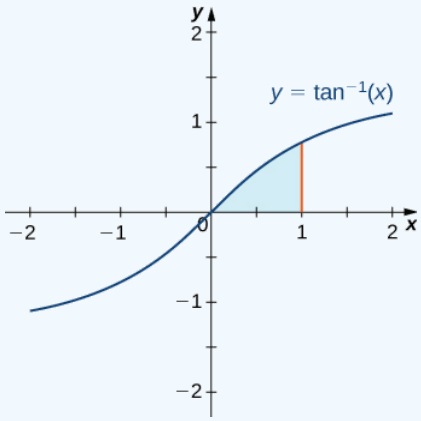


Figure 7.1.1: To find the area of the shaded region, we have to use integration by parts.

For this integral, let's choose $u = tan^{-1}x$ and $dv = dx$, thereby making $du = \dfrac{1}{x^2+1}\, dx$ and $v = x$. After applying the integration-by-parts formula (Equation 7.1.2) we obtain

$$\text{Area} = x \tan^{-1} x\Big|_0^1 - \int_0^1 \frac{x}{x^2+1}\, dx.$$

Use u-substitution to obtain

$$\int_0^1 \frac{x}{x^2+1}\, dx = \frac{1}{2}\ln\left(x^2+1\right)\Big|_0^1.$$

Thus,

$$\text{Area} = x \tan^{-1} x\Big|_0^1 - \frac{1}{2}\ln\left(x^2+1\right)\Big|_0^1 = \left(\frac{\pi}{4} - \frac{1}{2}\ln 2\right) \text{ units}^2.$$

At this point it might not be a bad idea to do a "reality check" on the reasonableness of our solution. Since $\frac{\pi}{4} - \frac{1}{2}\ln 2 \approx 0.4388$ units^2, and from Figure 7.1.1 we expect our area to be slightly less than 0.5 units^2, this solution appears to be reasonable.

Example 7.1.4B: Finding a Volume of Revolution

Find the volume of the solid obtained by revolving the region bounded by the graph of $f(x) = e^{-x}$, the x-axis, the y-axis, and the line $x = 1$ about the y-axis.

Solution

The best option to solving this problem is to use the shell method. Begin by sketching the region to be revolved, along with a typical rectangle (Figure 7.1.2).

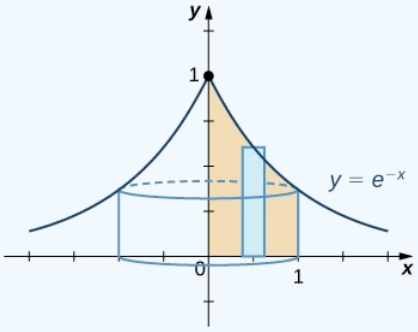


Figure 7.1.2: We can use the shell method to find a volume of revolution.

To find the volume using shells, we must evaluate

$$2\pi \int_0^1 xe^{-x}\,dx. \tag{7.1.4}$$

To do this, let $u = x$ and $dv = e^{-x}$. These choices lead to $du = dx$ and $v = \int e^{-x}\,dx = -e^{-x}$. Using the Shell Method formula, we obtain

$$\begin{aligned}
\text{Volume} &= 2\pi \int_0^1 xe^{-x}\,dx \\
&= 2\pi \left(-xe^{-x}\Big|_0^1 + \int_0^1 e^{-x}\,dx\right) && \text{(Use integration by parts)} \\
&= 2\pi \left(-e^{-1} + 0 - e^{-x}\Big|_0^1\right) \\
&= 2\pi \left(-e^{-1} - e^{-1} + 1\right) \\
&= 2\pi \left(1 - \frac{2}{e}\right) && \text{(Evaluate and simplify)} \\
&\quad \text{units}^3.
\end{aligned}$$

Analysis

Again, it is a good idea to check the reasonableness of our solution. We observe that the solid has a volume slightly less than that of a cylinder of radius 1 and height of $1/e$ added to the volume of a cone of base radius 1 and height of $1 - \frac{1}{e}$. Consequently, the solid should have a volume a bit less than

$$\pi(1)^2 \frac{1}{e} + \left(\frac{\pi}{3}\right)(1)^2 \left(1 - \frac{1}{e}\right) = \frac{2\pi}{3e} + \frac{\pi}{3} \approx 1.8177 \text{ units}^3.$$

Since $2\pi - \frac{4\pi}{e} \approx 1.6603$, we see that our calculated volume is reasonable.

? Exercise 7.1.4

Evaluate

$$\int_0^{\pi/2} x \cos x \, dx.$$

Hint

Use Equation 7.1.2 with $u = x$ and $dv = \cos x \, dx$.

Answer

$$\int_0^{\pi/2} x \cos x \, dx = \frac{\pi}{2} - 1$$

Key Concepts

- The integration-by-parts formula (Equation 7.1.2) allows the exchange of one integral for another, possibly easier, integral.
- Integration by parts applies to both definite and indefinite integrals.

Key Equations

- **Integration by parts formula**

  $$\int u \, dv = uv - \int v \, du$$

- **Integration by parts for definite integrals**

  $$\int_a^b u \, dv = uv\Big|_a^b - \int_a^b v \, du$$

Glossary

integration by parts

a technique of integration that allows the exchange of one integral for another using the formula $\int u \, dv = uv - \int v \, du$

This page titled 7.1: Integration by Parts is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

7.1E: EXERCISES FOR SECTION 7.1

In using the technique of integration by parts, you must carefully choose which expression is u. For each of the following problems, use the guidelines in this section to choose u. Do not evaluate the integrals.

1) $\int x^3 e^{2x}\,dx$

Answer

$u = x^3$

2) $\int x^3 \ln(x)\,dx$

3) $\int y^3 \cos y\,dy$

Answer

$u = y^3$

4) $\int x^2 \arctan x\,dx$

5) $\int e^{3x} \sin(2x)\,dx$

Answer

$u = \sin(2x)$

In exercises 6 - 37, find the integral by using the simplest method. Not all problems require integration by parts.

6) $\int v \sin v\,dv$

7) $\int \ln x\,dx$ (Hint: $\int \ln x\,dx$ is equivalent to $\int 1 \cdot \ln(x)\,dx$.)

Answer

$$\int \ln x\,dx \quad = \quad -x + x\ln x + C$$

8) $\int x \cos x\,dx$

9) $\int \tan^{-1} x\,dx$

Answer

$$\int \tan^{-1} x\,dx \quad = \quad x\tan^{-1} x - \tfrac{1}{2}\ln(1+x^2) + C$$

10) $\int x^2 e^x\,dx$

11) $\int x \sin(2x)\,dx$

Answer

$$\int x \sin(2x)\,dx \quad = \quad -\tfrac{1}{2}x\cos(2x) + \tfrac{1}{4}\sin(2x) + C$$

12) $\int x e^{4x}\,dx$

13) $\int x e^{-x}\,dx$

Answer

$$\int x e^{-x}\,dx \quad = \quad e^{-x}(-1-x) + C$$

14) $\int x \cos 3x\,dx$

15) $\int x^2 \cos x \, dx$

Answer

$$\int x^2 \cos x \, dx = 2x \cos x + (-2 + x^2) \sin x + C$$

16) $\int x \ln x \, dx$

17) $\int \ln(2x+1) \, dx$

Answer

$$\int \ln(2x+1) \, dx = \tfrac{1}{2}(1+2x)(-1+\ln(1+2x)) + C$$

18) $\int x^2 e^{4x} \, dx$

19) $\int e^x \sin x \, dx$

Answer

$$\int e^x \sin x \, dx = \tfrac{1}{2} e^x (-\cos x + \sin x) + C$$

20) $\int e^x \cos x \, dx$

21) $\int x e^{-x^2} \, dx$

Answer

$$\int x e^{-x^2} \, dx = -\frac{e^{-x^2}}{2} + C$$

22) $\int x^2 e^{-x} \, dx$

23) $\int \sin(\ln(2x)) \, dx$

Answer

$$\int \sin(\ln(2x)) \, dx = -\tfrac{1}{2} x \cos[\ln(2x)] + \tfrac{1}{2} x \sin[\ln(2x)] + C$$

24) $\int \cos(\ln x) \, dx$

25) $\int (\ln x)^2 \, dx$

Answer

$$\int (\ln x)^2 \, dx = 2x - 2x \ln x + x (\ln x)^2 + C$$

26) $\int \ln(x^2) \, dx$

27) $\int x^2 \ln x \, dx$

Answer

$$\int x^2 \ln x \, dx = -\frac{x^3}{9} + \tfrac{1}{3} x^3 \ln x + C$$

28) $\int \sin^{-1} x \, dx$

29) $\int \cos^{-1}(2x) \, dx$

Answer

$$\int \cos^{-1}(2x) \, dx = -\tfrac{1}{2} \sqrt{1 - 4x^2} + x \cos^{-1}(2x) + C$$

30) $\int x \arctan x \, dx$

31) $\int x^2 \sin x \, dx$

Answer

$$\int x^2 \sin x \, dx \quad = \quad -(-2+x^2)\cos x + 2x \sin x + C$$

32) $\int x^3 \cos x \, dx$

33) $\int x^3 \sin x \, dx$

Answer

$$\int x^3 \sin x \, dx \quad = \quad -x(-6+x^2)\cos x + 3(-2+x^2)\sin x + C$$

34) $\int x^3 e^x \, dx$

35) $\int x \sec^{-1} x \, dx$

Answer

$$\int x \sec^{-1} x \, dx \quad = \quad \tfrac{1}{2}x\left(-\sqrt{1-\frac{1}{x^2}} + x \cdot \sec^{-1} x\right) + C$$

36) $\int x \sec^2 x \, dx$

37) $\int x \cosh x \, dx$

Answer

$$\int x \cosh x \, dx \quad = \quad -\cosh x + x \sinh x + C$$

In exercises 38 - 46, compute the definite integrals. Use a graphing utility to confirm your answers.

38) $\int_{1/e}^{1} \ln x \, dx$

39) $\int_0^1 x e^{-2x} \, dx$ (Express the answer in exact form.)

Answer

$$\int_0^1 x e^{-2x} \, dx \quad = \quad \frac{1}{4} - \frac{3}{4e^2}$$

40) $\int_0^1 e^{\sqrt{x}} \, dx \quad (\text{let } u = \sqrt{x})$

41) $\int_1^e \ln(x^2) \, dx$

Answer

$$\int_1^e \ln(x^2) \, dx \quad = \quad 2$$

42) $\int_0^{\pi} x \cos x \, dx$

43) $\int_{-\pi}^{\pi} x \sin x \, dx$ (Express the answer in exact form.)

Answer

$$\int_{-\pi}^{\pi} x \sin x \, dx \quad = \quad 2\pi$$

44) $\int_0^3 \ln(x^2+1) \, dx$ (Express the answer in exact form.)

45) $\int_0^{\pi/2} x^2 \sin x \, dx$ (Express the answer in exact form.)

Answer

$$\int_0^{\pi/2} x^2 \sin x \, dx \quad = \quad -2 + \pi$$

46) $\int_0^1 x 5^x \, dx$ (Express the answer using five significant digits.)

47) Evaluate $\int \cos x \ln(\sin x) \, dx$

Answer

$$\int \cos x \ln(\sin x) \, dx \quad = \quad -\sin(x) + \ln[\sin(x)] \sin x + C$$

In exercises 48 - 50, derive the following formulas using the technique of integration by parts. Assume that n is a positive integer. These formulas are called *reduction formulas* because the exponent in the x term has been reduced by one in each case. The second integral is simpler than the original integral.

48) $\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx$

49) $\int x^n \cos x \, dx = x^n \sin x - n \int x^{n-1} \sin x \, dx$

Answer

Answers vary

50) $\int x^n \sin x \, dx =$_____

51) Integrate $\int 2x\sqrt{2x-3} \, dx$ using two methods:

a. Using parts, letting $dv = \sqrt{2x-3} \, dx$

b. Substitution, letting $u = 2x - 3$

Answer

a. $\int 2x\sqrt{2x-3} \, dx \quad = \quad \frac{2}{5}(1+x)(-3+2x)^{3/2} + C$

b. $\int 2x\sqrt{2x-3} \, dx \quad = \quad \frac{2}{5}(1+x)(-3+2x)^{3/2} + C$

In exercises 52 - 57, state whether you would use integration by parts to evaluate the integral. If so, identify u and dv. If not, describe the technique used to perform the integration without actually doing the problem.

52) $\int x \ln x \, dx$

53) $\int \frac{\ln^2 x}{x} \, dx$

Answer

Do not use integration by parts. Choose u to be $\ln x$, and the integral is of the form $\int u^2 \, du$.

54) $\int x e^x \, dx$

55) $\int x e^{x^2-3} \, dx$

Answer

Do not use integration by parts. Let $u = x^2 - 3$, and the integral can be put into the form $\int e^u \, du$.

56) $\int x^2 \sin x \, dx$

57) $\int x^2 \sin(3x^3 + 2) \, dx$

Answer

Do not use integration by parts. Choose u to be $u = 3x^3 + 2$ and the integral can be put into the form $\int \sin(u)\, du$.

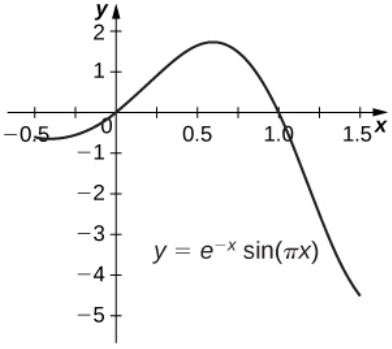
In exercises 58-59, sketch the region bounded above by the curve, the x-axis, and $x = 1$, and find the area of the region. Provide the exact form or round answers to the number of places indicated.

58) $y = 2xe^{-x}$ (Approximate answer to four decimal places.)

59) $y = e^{-x}\sin(\pi x)$ (Approximate answer to five decimal places.)

Answer

The area under graph is 0.39535 units^2.



In exercises 60 - 61, find the volume generated by rotating the region bounded by the given curves about the specified line. Express the answers in exact form or approximate to the number of decimal places indicated.

60) $y = \sin x$, $y = 0$, $x = 2\pi$, $x = 3\pi$; about the y-axis (Express the answer in exact form.)

61) $y = e^{-x}$, $y = 0$, $x = -1$, $x = 0$; about $x = 1$ (Express the answer in exact form.)

Answer

$V = 2\pi e \text{ units}^3$

62) A particle moving along a straight line has a velocity of $v(t) = t^2e^{-t}$ after t sec. How far does it travel in the first 2 sec? (Assume the units are in feet and express the answer in exact form.)

63) Find the area under the graph of $y = \sec^3 x$ from $x = 0$ to $x = 1$. (Round the answer to two significant digits.)

Answer

$A = 2.05 \text{ units}^2$

64) Find the area between $y = (x - 2)e^x$ and the x-axis from $x = 2$ to $x = 5$. (Express the answer in exact form.)

65) Find the area of the region enclosed by the curve $y = x\cos x$ and the x-axis for $\frac{11\pi}{2} \le x \le \frac{13\pi}{2}$. (Express the answer in exact form.)

Answer

$A = 12\pi \text{ units}^2$

66) Find the volume of the solid generated by revolving the region bounded by the curve $y = \ln x$, the x-axis, and the vertical line $x = e^2$ about the x-axis. (Express the answer in exact form.)

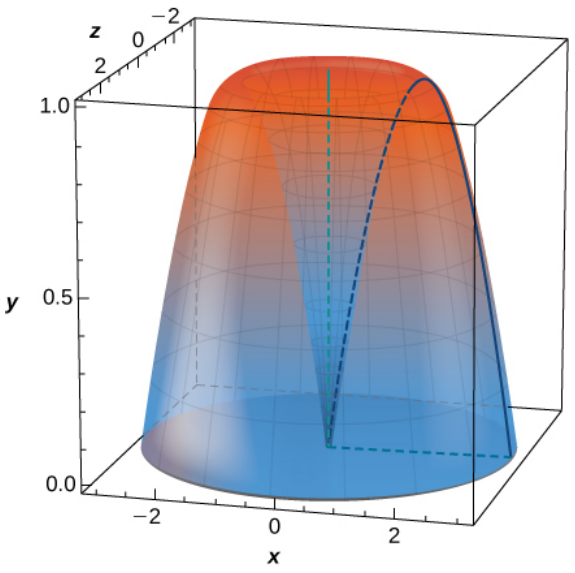
67) Find the volume of the solid generated by revolving the region bounded by the curve $y = 4\cos x$ and the x-axis, $\frac{\pi}{2} \le x \le \frac{3\pi}{2}$, about the x-axis. (Express the answer in exact form.)

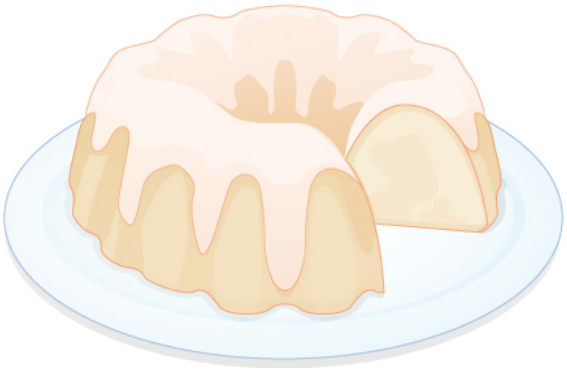
Answer

$V = 8\pi^2 \text{ units}^3$

68) Find the volume of the solid generated by revolving the region in the first quadrant bounded by $y = e^x$ and the x-axis, from $x = 0$ to $x = \ln(7)$, about the y-axis. (Express the answer in exact form.)

69) What is the volume of the Bundt cake that comes from rotating $y = \sin x$ around the y-axis from $x = 0$ to $x = \pi$?





Answer

$V = 2\pi^2$ units^3

7.1E: Exercises for Section 7.1 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

7.2: Trigonometric Integrals

Learning Objectives

- Solve integration problems involving products and powers of $\sin x$ and $\cos x$.
- Solve integration problems involving products and powers of $\tan x$ and $\sec x$.
- Use reduction formulas to solve trigonometric integrals.

In this section we look at how to integrate a variety of products of trigonometric functions. These integrals are called **trigonometric integrals**. They are an important part of the integration technique called *trigonometric substitution*, which is featured in Trigonometric Substitution. This technique allows us to convert algebraic expressions that we may not be able to integrate into expressions involving trigonometric functions, which we may be able to integrate using the techniques described in this section. In addition, these types of integrals appear frequently when we study polar, cylindrical, and spherical coordinate systems later. Let's begin our study with products of $\sin x$ and $\cos x$.

Integrating Products and Powers of sin x and cos x

A key idea behind the strategy used to integrate combinations of products and powers of $\sin x$ and $\cos x$ involves rewriting these expressions as sums and differences of integrals of the form $\int \sin^j x \cos x \, dx$ or $\int \cos^j x \sin x \, dx$. After rewriting these integrals, we evaluate them using u-substitution. Before describing the general process in detail, let's take a look at the following examples.

Example 7.2.1: Integrating $\int \cos^j x \sin x \, dx$

Evaluate $\int \cos^3 x \sin x \, dx$.

Solution

Use u-substitution and let $u = \cos x$. In this case, $du = -\sin x \, dx$.

Thus,

$$\int \cos^3 x \sin x \, dx = -\int u^3 \, du = -\frac{1}{4}u^4 + C = -\frac{1}{4}\cos^4 x + C.$$

Exercise 7.2.1

Evaluate $\int \sin^4 x \cos x \, dx$.

Hint

Let $u = \sin x$.

Answer

$$\int \sin^4 x \cos x \, dx = \frac{1}{5}\sin^5 x + C$$

Example 7.2.2: A Preliminary Example: Integrating $\int \cos^j x \sin^k x \, dx$ where k is Odd

Evaluate $\int \cos^2 x \sin^3 x \, dx$.

Solution

To convert this integral to integrals of the form $\int \cos^j x \sin x \, dx$, rewrite $\sin^3 x = \sin^2 x \sin x$ and make the substitution $\sin^2 x = 1 - \cos^2 x$.

Thus,

$$\int \cos^2 x \sin^3 x\, dx = \int \cos^2 x(1-\cos^2 x)\sin x\, dx \qquad \text{Let } u = \cos x;\ \text{then } du = -\sin x\, dx.$$

$$= -\int u^2(1-u^2)\, du$$

$$= \int (u^4 - u^2)\, du$$

$$= \frac{1}{5}u^5 - \frac{1}{3}u^3 + C$$

$$= \frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x + C.$$

? Exercise 7.2.2

Evaluate $\int \cos^3 x \sin^2 x\, dx$.

Hint

Write $\cos^3 x = \cos^2 x \cos x = (1-\sin^2 x)\cos x$ and let $u = \sin x$.

Answer

$$\int \cos^3 x \sin^2 x\, dx = \frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x + C$$

In the next example, we see the strategy that must be applied when there are only even powers of $\sin x$ and $\cos x$. For integrals of this type, the identities

$$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x) = \frac{1-\cos(2x)}{2}$$

and

$$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos(2x) = \frac{1+\cos(2x)}{2}$$

are invaluable. These identities are sometimes known as **power-reducing identities** and they may be derived from the double-angle identity $\cos(2x) = \cos^2 x - \sin^2 x$ and the Pythagorean identity $\cos^2 x + \sin^2 x = 1$.

✓ Example 7.2.3: Integrating an Even Power of $\sin x$

Evaluate $\int \sin^2 x\, dx$.

Solution

To evaluate this integral, let's use the trigonometric identity $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x)$. Thus,

$$\int \sin^2 x\, dx = \int \left(\frac{1}{2} - \frac{1}{2}\cos(2x)\right) dx = \frac{1}{2}x - \frac{1}{4}\sin(2x) + C.$$

? Exercise 7.2.3

Evaluate $\int \cos^2 x\, dx$.

Hint

$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos(2x)$

Answer

$$\int \cos^2 x\, dx = \frac{1}{2}x + \frac{1}{4}\sin(2x) + C$$

The general process for integrating products of powers of $\sin x$ and $\cos x$ is summarized in the following set of guidelines.

Problem-Solving Strategy: Integrating Products and Powers of $\sin x$ and $cosx$

To integrate $\int \cos^j x \sin^k x\, dx$ use the following strategies:

1. If k is odd, rewrite $\sin^k x = \sin^{k-1} x \sin x$ and use the identity $\sin^2 x = 1 - \cos^2 x$ to rewrite $\sin^{k-1} x$ in terms of $\cos x$. Integrate using the substitution $u = \cos x$. This substitution makes $du = -\sin x\, dx$.

2. If j is odd, rewrite $\cos^j x = \cos^{j-1} x \cos x$ and use the identity $\cos^2 x = 1 - \sin^2 x$ to rewrite $\cos^{j-1} x$ in terms of $\sin x$. Integrate using the substitution $u = \sin x$. This substitution makes $du = \cos x\, dx$. (Note: If both j and k are odd, either strategy 1 or strategy 2 may be used.)

3. If both j and k are even, use $\sin^2 x = \dfrac{1 - \cos(2x)}{2}$ and $\cos^2 x = \dfrac{1 + \cos(2x)}{2}$. After applying these formulas, simplify and reapply strategies 1 through 3 as appropriate.

Example 7.2.4: Integrating $\int \cos^j x \sin^k x\, dx$ where k is Odd

Evaluate $\int \cos^8 x \sin^5 x\, dx$.

Solution

Since the power on $\sin x$ is odd, use strategy 1. Thus,

$$\begin{aligned}
\int \cos^8 x \sin^5 x\, dx &= \int \cos^8 x \sin^4 x \sin x\, dx && \text{Break off } \sin x.\\
&= \int \cos^8 x (\sin^2 x)^2 \sin x\, dx && \text{Rewrite } \sin^4 x = (\sin^2 x)^2.\\
&= \int \cos^8 x (1 - \cos^2 x)^2 \sin x\, dx && \text{Substitute } \sin^2 x = 1 - \cos^2 x.\\
&= \int u^8 (1 - u^2)^2 (-du) && \text{Let } u = \cos x \text{ and } du = -\sin x\, dx.\\
&= \int (-u^8 + 2u^{10} - u^{12}) du && \text{Expand.}\\
&= -\frac{1}{9}u^9 + \frac{2}{11}u^{11} - \frac{1}{13}u^{13} + C && \text{Evaluate the integral.}\\
&= -\frac{1}{9}\cos^9 x + \frac{2}{11}\cos^{11} x - \frac{1}{13}\cos^{13} x + C && \text{Substitute } u = \cos x.
\end{aligned}$$

Example 7.2.5: Integrating $\int \cos^j x \sin^k x\, dx$ where k and j are Even

Evaluate $\int \sin^4 x\, dx$.

Solution: Since the power on $\sin x$ is even $(k = 4)$ and the power on $\cos x$ is even $(j = 0)$, we must use strategy 3. Thus,

$$\begin{aligned}
\int \sin^4 x\, dx &= \int \left(\sin^2 x\right)^2 dx && \text{Rewrite } \sin^4 x = \left(\sin^2 x\right)^2.\\
&= \int \left(\frac{1}{2} - \frac{1}{2}\cos(2x)\right)^2 dx && \text{Substitute } \sin^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x).\\
&= \int \left(\frac{1}{4} - \frac{1}{2}\cos(2x) + \frac{1}{4}\cos^2(2x)\right) dx && \text{Expand } \left(\frac{1}{2} - \frac{1}{2}\cos(2x)\right)^2.\\
&= \int \left(\frac{1}{4} - \frac{1}{2}\cos(2x) + \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2}\cos(4x)\right)\right) dx && \text{Use Power Reduction Formula on } \cos^2(2x).\\
&= \int \left(\frac{3}{8} - \frac{1}{2}\cos(2x) + \frac{1}{8}\cos(4x)\right) dx && \text{Simplify.}\\
&= \frac{3}{8}x - \frac{1}{4}\sin(2x) + \frac{1}{32}\sin(4x) + C && \text{Evaluate the integral.}
\end{aligned}$$

Exercise 7.2.4

Evaluate $\int \cos^3 x\, dx$.

Hint

Use strategy 2. Write $\cos^3 x = \cos^2 x \cos x$ and substitute $\cos^2 x = 1 - \sin^2 x$.

Answer

$$\int \cos^3 x\, dx = \sin x - \frac{1}{3}\sin^3 x + C$$

Exercise 7.2.5

Evaluate $\int \cos^2(3x)\, dx$.

Hint

Use strategy 3. Substitute $\cos^2(3x) = \frac{1}{2} + \frac{1}{2}\cos(6x)$

Answer

$$\int \cos^2(3x)\, dx = \frac{1}{2}x + \frac{1}{12}\sin(6x) + C$$

Product to Sum Formulas

In some areas of physics, such as quantum mechanics, signal processing, and the computation of Fourier series, it is often necessary to integrate products that include $sin(ax), sin(bx), cos(ax)$, and $cos(bx)$. These integrals are evaluated by applying trigonometric identities, as outlined in the following rule.

Rule: Integrating Products of Sines and Cosines of Different Angles

To integrate products involving $\sin(ax)$, $\sin(bx)$, $\cos(ax)$, and $\cos(bx)$, use the substitutions

$$\sin(ax)\sin(bx) = \frac{1}{2}\cos((a-b)x) - \frac{1}{2}\cos((a+b)x)$$

$$\sin(ax)\cos(bx) = \frac{1}{2}\sin((a-b)x) + \frac{1}{2}\sin((a+b)x)$$

$$\cos(ax)\cos(bx) = \frac{1}{2}\cos((a-b)x) + \frac{1}{2}\cos((a+b)x)$$

These formulas may be derived from the sum-of-angle formulas for sine and cosine.

Example 7.2.6: Evaluating $\int \sin(ax)\cos(bx)\, dx$

Evaluate $\int \sin(5x)\cos(3x)\, dx$.

Solution: Apply the identity $\sin(5x)\cos(3x) = \frac{1}{2}\sin(2x) + \frac{1}{2}\sin(8x)$. Thus,

$$\int \sin(5x)\cos(3x)\, dx = \int \frac{1}{2}\sin(2x) + \frac{1}{2}\sin(8x)\, dx = -\frac{1}{4}\cos(2x) - \frac{1}{16}\cos(8x) + C.$$

Exercise 7.2.6

Evaluate $\int \cos(6x)\cos(5x)\, dx$.

Hint

Substitute $\cos(6x)\cos(5x) = \frac{1}{2}\cos x + \frac{1}{2}\cos(11x)$.

Answer

$$\int \cos(6x)\cos(5x)\,dx = \frac{1}{2}\sin x + \frac{1}{22}\sin(11x) + C$$

Integrating Products and Powers of tan x and sec x

Before discussing the integration of products and powers of $\tan x$ and $\sec x$, it is useful to recall the integrals involving $\tan x$ and $\sec x$ we have already learned:

1. $\int \sec^2 x\,dx = \tan x + C$
2. $\int \sec x \tan x\,dx = \sec x + C$
3. $\int \tan x\,dx = \ln|\sec x| + C$
4. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$.

For most integrals of products and powers of $\tan x$ and $\sec x$, we rewrite the expression we wish to integrate as the sum or difference of integrals of the form $\int \tan^j x \sec^2 x\,dx$ or $\int \sec^j x \tan x\,dx$. As we see in the following example, we can evaluate these new integrals by using u-substitution.

Example 7.2.7: Evaluating $\int \sec^j x \tan x\,dx$

Evaluate $\int \sec^5 x \tan x\,dx$.

Solution: Start by rewriting $\sec^5 x \tan x$ as $\sec^4 x \sec x \tan x$.

$$\begin{aligned}
\int \sec^5 x \tan x\,dx &= \int \sec^4 x \sec x \tan x\,dx \\
&= \int u^4\,du && \text{Let } u = \sec x;\ \text{then, } du = \sec x \tan x\,dx. \\
&= \tfrac{1}{5}u^5 + C && \text{Evaluate the integral.} \\
&= \tfrac{1}{5}\sec^5 x + C && \text{Substitute } \sec x = u.
\end{aligned}$$

You can read some interesting information at this website to learn about a common integral involving the secant.

Exercise 7.2.7

Evaluate $\int \tan^5 x \sec^2 x\,dx$.

Hint

Let $u = \tan x$ and $du = \sec^2 x$.

Answer

$$\int \tan^5 x \sec^2 x\,dx = \tfrac{1}{6}\tan^6 x + C$$

We now take a look at the various strategies for integrating products and powers of $\sec x$ and $\tan x$.

Problem-Solving Strategy: Integrating $\int \tan^k x \sec^j x\, dx$

To integrate $\int \tan^k x \sec^j x\, dx$, use the following strategies:

1. If j is even and $j \geq 2$, rewrite $\sec^j x = \sec^{j-2} x \sec^2 x$ and use $\sec^2 x = \tan^2 x + 1$ to rewrite $\sec^{j-2} x$ in terms of $\tan x$. Let $u = \tan x$ and $du = \sec^2 x$.

2. If k is odd and $j \geq 1$, rewrite $\tan^k x \sec^j x = \tan^{k-1} x \sec^{j-1} x \sec x \tan x$ and use $\tan^2 x = \sec^2 x - 1$ to rewrite $\tan^{k-1} x$ in terms of $\sec x$. Let $u = \sec x$ and $du = \sec x \tan x\, dx$. (Note: If j is even and k is odd, then either strategy 1 or strategy 2 may be used.)

3. If k is odd where $k \geq 3$ and $j = 0$, rewrite $\tan^k x = \tan^{k-2} x \tan^2 x = \tan^{k-2} x(\sec^2 x - 1) = \tan^{k-2} x \sec^2 x - \tan^{k-2} x$. It may be necessary to repeat this process on the $\tan^{k-2} x$ term.

4. If k is even and j is odd, then use $\tan^2 x = \sec^2 x - 1$ to express $\tan^k x$ in terms of $\sec x$. Use integration by parts to integrate odd powers of $\sec x$.

Example 7.2.8: Integrating $\int \tan^k x \sec^j x\, dx$ when j is Even

Evaluate $\int \tan^6 x \sec^4 x\, dx$.

Solution

Since the power on $\sec x$ is even, rewrite $\sec^4 x = \sec^2 x \sec^2 x$ and use $\sec^2 x = \tan^2 x + 1$ to rewrite the first $\sec^2 x$ in terms of $\tan x$. Thus,

$$\begin{aligned}
\int \tan^6 x \sec^4 x\, dx &= \int \tan^6 x(\tan^2 x + 1) \sec^2 x\, dx \\
&= \int u^6(u^2 + 1)\, du && \text{Let } u = \tan x \text{ and } du = \sec^2 x. \\
&= \int (u^8 + u^6)\, du && \text{Expand.} \\
&= \frac{1}{9}u^9 + \frac{1}{7}u^7 + C && \text{Evaluate the integral.} \\
&= \frac{1}{9}\tan^9 x + \frac{1}{7}\tan^7 x + C. && \text{Substitute } \tan x = u.
\end{aligned}$$

Example 7.2.9: Integrating $\int \tan^k x \sec^j x\, dx$ when k is Odd

Evaluate $\int \tan^5 x \sec^3 x\, dx$.

Solution

Since the power on $\tan x$ is odd, begin by rewriting $\tan^5 x \sec^3 x = \tan^4 x \sec^2 x \sec x \tan x$. Thus,

$$\int \tan^5 x \sec^3 x\,dx = \int \tan^4 x \sec^2 x \sec x \tan x\,dx.$$

$$= \int (\tan^2 x)^2 \sec^2 x \sec x \tan x\,dx \qquad \text{Write } \tan^4 x = (\tan^2 x)^2.$$

$$= \int (\sec^2 x - 1)^2 \sec^2 x \sec x \tan x\,dx \qquad \text{Use } \tan^2 x = \sec^2 x - 1.$$

$$= \int (u^2 - 1)^2 u^2 du \qquad \text{Let } u = \sec x \text{ and } du = \sec x \tan x\,dx$$

$$= \int (u^6 - 2u^4 + u^2) du \qquad \text{Expand.}$$

$$= \frac{1}{7}u^7 - \frac{2}{5}u^5 + \frac{1}{3}u^3 + C \qquad \text{Integrate.}$$

$$= \frac{1}{7}\sec^7 x - \frac{2}{5}\sec^5 x + \frac{1}{3}\sec^3 x + C \qquad \text{Substitute } \sec x = u.$$

Example 7.2.10: Integrating $\int \tan^k x\,dx$ where k is Odd and $k \geq 3$

Evaluate $\int \tan^3 x\,dx$.

Solution

Begin by rewriting $\tan^3 x = \tan x \tan^2 x = \tan x(\sec^2 x - 1) = \tan x \sec^2 x - \tan x$. Thus,

$$\int \tan^3 x\,dx = \int (\tan x \sec^2 x - \tan x)\,dx$$

$$= \int \tan x \sec^2 x\,dx - \int \tan x\,dx$$

$$= \frac{1}{2}\tan^2 x - \ln|\sec x| + C.$$

For the first integral, use the substitution $u = \tan x$. For the second integral, use the formula.

Example 7.2.11: Integrating $\int \sec^3 x\,dx$

Integrate $\int \sec^3 x\,dx$.

Solution

This integral requires integration by parts. To begin, let $u = \sec x$ and $dv = \sec^2 x$. These choices make $du = \sec x \tan x$ and $v = \tan x$. Thus,

$$\int \sec^3 x\,dx = \sec x \tan x - \int \tan x \sec x \tan x\,dx$$

$$= \sec x \tan x - \int \tan^2 x \sec x\,dx \qquad \text{Simplify.}$$

$$= \sec x \tan x - \int (\sec^2 x - 1)\sec x\,dx \qquad \text{Substitute } \tan^2 x = \sec^2 x - 1.$$

$$= \sec x \tan x + \int \sec x\,dx - \int \sec^3 x\,dx \qquad \text{Rewrite.}$$

$$= \sec x \tan x + \ln|\sec x + \tan x| - \int \sec^3 x\,dx. \qquad \text{Evaluate } \int \sec x\,dx.$$

We now have

$$\int \sec^3 x\,dx = \sec x \tan x + \ln|\sec x + \tan x| - \int \sec^3 x\,dx.$$

Since the integral $\int \sec^3 x\,dx$ has reappeared on the right-hand side, we can solve for $\int \sec^3 x\,dx$ by adding it to both sides. In doing so, we obtain

$$2\int \sec^3 x\,dx = \sec x \tan x + \ln|\sec x + \tan x|.$$

Dividing by 2, we arrive at

$$\int \sec^3 x\,dx = \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$$

? Exercise 7.2.8

Evaluate $\int \tan^3 x \sec^7 x\,dx$.

Hint

Use Example 7.2.9 as a guide.

Answer

$$\int \tan^3 x \sec^7 x\,dx = \frac{1}{9}\sec^9 x - \frac{1}{7}\sec^7 x + C$$

Reduction Formulas

Evaluating $\int \sec^n x\,dx$ for values of n where n is odd requires integration by parts. In addition, we must also know the value of $\int \sec^{n-2} x\,dx$ to evaluate $\int \sec^n x\,dx$. The evaluation of $\int \tan^n x\,dx$ also requires being able to integrate $\int \tan^{n-2} x\,dx$. To make the process easier, we can derive and apply the following power reduction formulas. These rules allow us to replace the integral of a power of $\sec x$ or $\tan x$ with the integral of a lower power of $\sec x$ or $\tan x$.

Rule: Reduction Formulas for $\int \sec^n x\,dx$ and $\int \tan^n x\,dx$

$$\int \sec^n x\,dx = \frac{1}{n-1}\sec^{n-2} x \tan x + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx$$

$$\int \tan^n x\,dx = \frac{1}{n-1}\tan^{n-1} x - \int \tan^{n-2} x\,dx$$

The first power reduction rule may be verified by applying integration by parts. The second may be verified by following the strategy outlined for integrating odd powers of $\tan x$.

✓ Example 7.2.12: Revisiting $\int \sec^3 x\,dx$

Apply a reduction formula to evaluate $\int \sec^3 x\,dx$.

Solution: By applying the first reduction formula, we obtain

$$\begin{aligned}\int \sec^3 x\,dx &= \frac{1}{2}\sec x \tan x + \frac{1}{2}\int \sec x\,dx \\ &= \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C.\end{aligned}$$

✓ Example 7.2.13: Using a Reduction Formula

Evaluate $\int \tan^4 x\,dx$.

Solution: Applying the reduction formula for $\int \tan^4 x\,dx$ we have

$$\begin{aligned}\int \tan^4 x\,dx &= \frac{1}{3}\tan^3 x - \int \tan^2 x\,dx \\ &= \frac{1}{3}\tan^3 x - (\tan x - \int \tan^0 x\,dx) && \text{Apply the reduction formula to } \int \tan^2 x\,dx. \\ &= \frac{1}{3}\tan^3 x - \tan x + \int 1\,dx && \text{Simplify.} \\ &= \frac{1}{3}\tan^3 x - \tan x + x + C && \text{Evaluate } \int 1\,dx\end{aligned}$$

? Exercise 7.2.9

Apply the reduction formula to $\int \sec^5 x\,dx$.

Hint

Use reduction formula 1 and let $n = 5$.

Answer

$$\int \sec^5 x\,dx = \frac{1}{4}\sec^3 x \tan x + \frac{3}{4}\int \sec^3 x$$

Key Concepts

Integrals of trigonometric functions can be evaluated by the use of various strategies. These strategies include

1. Applying trigonometric identities to rewrite the integral so that it may be evaluated by u-substitution
2. Using integration by parts
3. Applying trigonometric identities to rewrite products of sines and cosines with different arguments as the sum of individual sine and cosine functions
4. Applying reduction formulas

Key Equations

To integrate products involving $\sin(ax)$, $\sin(bx)$, $\cos(ax)$, and $\cos(bx)$, use the substitutions.

- **Sine Products**

  $\sin(ax)\sin(bx) = \frac{1}{2}\cos((a-b)x) - \frac{1}{2}\cos((a+b)x)$

- **Sine and Cosine Products**

  $\sin(ax)\cos(bx) = \frac{1}{2}\sin((a-b)x) + \frac{1}{2}\sin((a+b)x)$

- **Cosine Products**

  $\cos(ax)\cos(bx) = \frac{1}{2}\cos((a-b)x) + \frac{1}{2}\cos((a+b)x)$

- **Power Reduction Formula**

  $$\int \sec^n x\,dx = \frac{1}{n-1}\sec^{n-2} x \tan x + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx$$

- **Power Reduction Formula**

  $$\int \tan^n x\,dx = \frac{1}{n-1}\tan^{n-1} x - \int \tan^{n-2} x\,dx$$

Glossary

power reduction formula

a rule that allows an integral of a power of a trigonometric function to be exchanged for an integral involving a lower power

trigonometric integral

an integral involving powers and products of trigonometric functions

This page titled 7.2: Trigonometric Integrals is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.